Course overview and introduction

PSTAT 131/231

Week 1

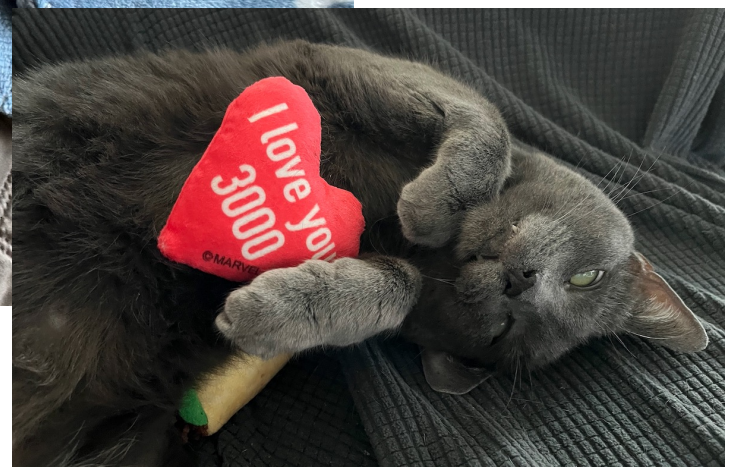
Introductions

Ph.D. Quantitative Psychology

B.S. Psychology



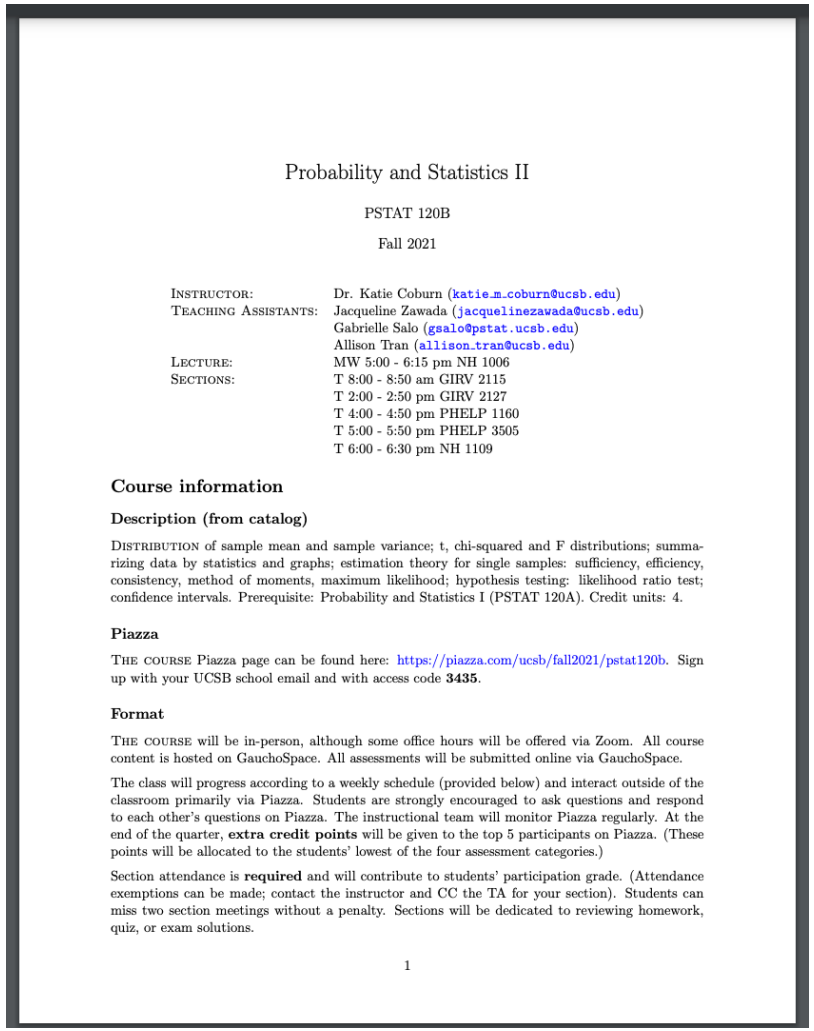




UNIVERSITY OF CALIFORNIA
MERCED



Syllabus



Probability and Statistics II

PSTAT 120B

Fall 2021

Instructor:	Dr. Katie Coburn (katie_m_coburn@ucsb.edu)
Teaching Assistants:	Jacqueline Zawada (jacquelinezawada@ucsb.edu)
	Gabrielle Salo (gsalo@pstat.ucsb.edu)
	Allison Tran (allison_tran@ucsb.edu)
Lecture:	MW 5:00 - 6:15 pm NH 1006
Sections:	T 8:00 - 8:50 am GIRV 2115
	T 2:00 - 2:50 pm GIRV 2127
	T 4:00 - 4:50 pm PHELP 1160
	T 5:00 - 5:50 pm PHELP 3505
	T 6:00 - 6:30 pm NH 1109

Course information

Description (from catalog)

Distribution of sample mean and sample variance; t, chi-squared and F distributions; summarizing data by statistics and graphs; estimation theory for single samples: sufficiency, efficiency, consistency, method of moments, maximum likelihood; hypothesis testing: likelihood ratio test; confidence intervals. Prerequisite: Probability and Statistics I (PSTAT 120A). Credit units: 4.

Piazza

The course Piazza page can be found here: https://piazza.com/ucsb/fall2021/pstat120b. Sign up with your UCSB school email and with access code **3435**.

Format

The course will be in-person, although some office hours will be offered via Zoom. All course content is hosted on GauchoSpace. All assessments will be submitted online via GauchoSpace.

The class will progress according to a weekly schedule (provided below) and interact outside of the classroom primarily via Piazza. Students are strongly encouraged to ask questions and respond to each other's questions on Piazza. The instructional team will monitor Piazza regularly. At the end of the quarter, **extra credit points** will be given to the top 5 participants on Piazza. (These points will be allocated to the students' lowest of the four assessment categories.)

Section attendance is **required** and will contribute to students' participation grade. (Attendance exemptions can be made; contact the instructor and CC the TA for your section). Students can miss two section meetings without a penalty. Sections will be dedicated to reviewing homework, quiz, or exam solutions.

1

R Setup

Introduction

Required reading:

- For **undergraduates**, ch. 1 of ISLR

- For **graduate students**, ch. 1 of ESL

What is machine learning?

- Application of learning algorithms and statistical methods to real-world data sets:

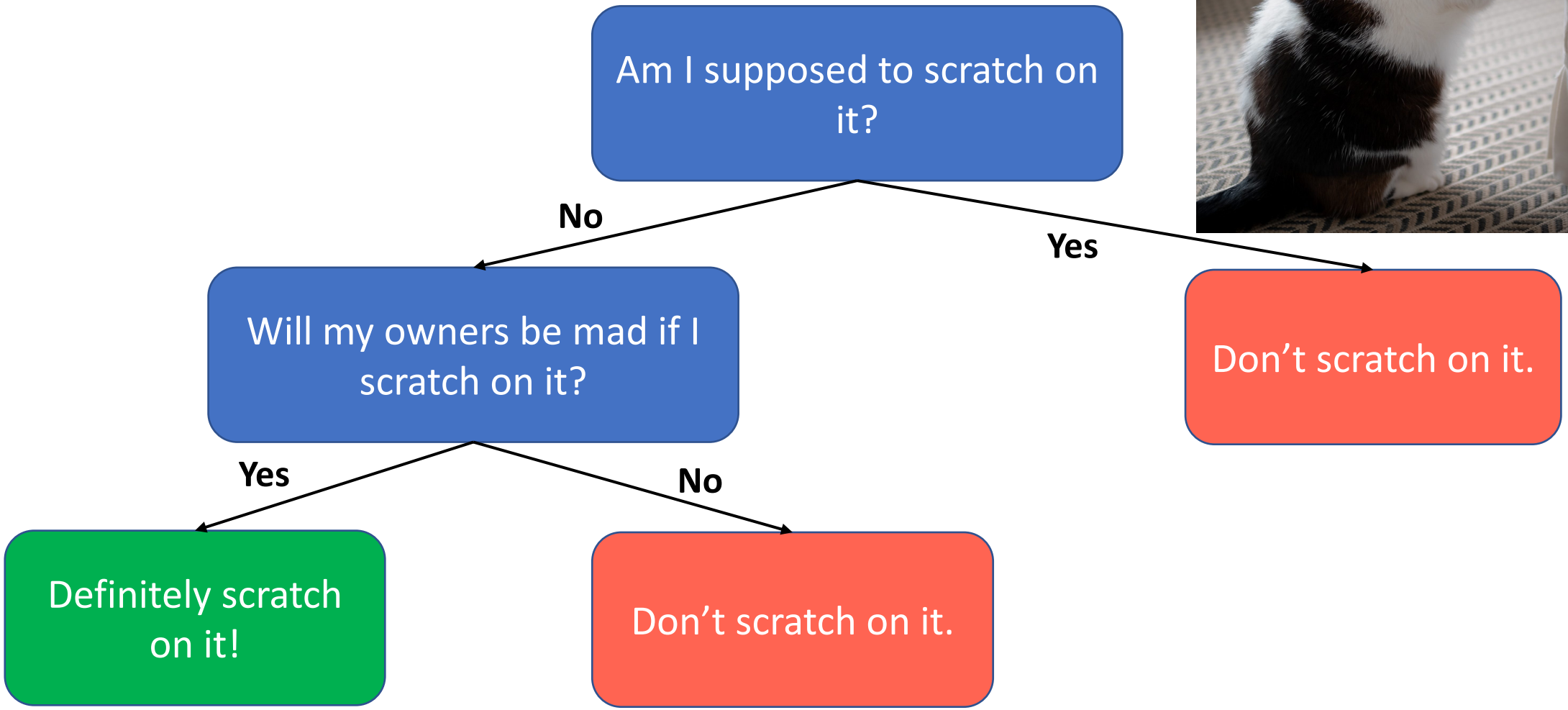
- i.e., how can a **machine** use **data** to **learn**

- This is a non-trivial problem!

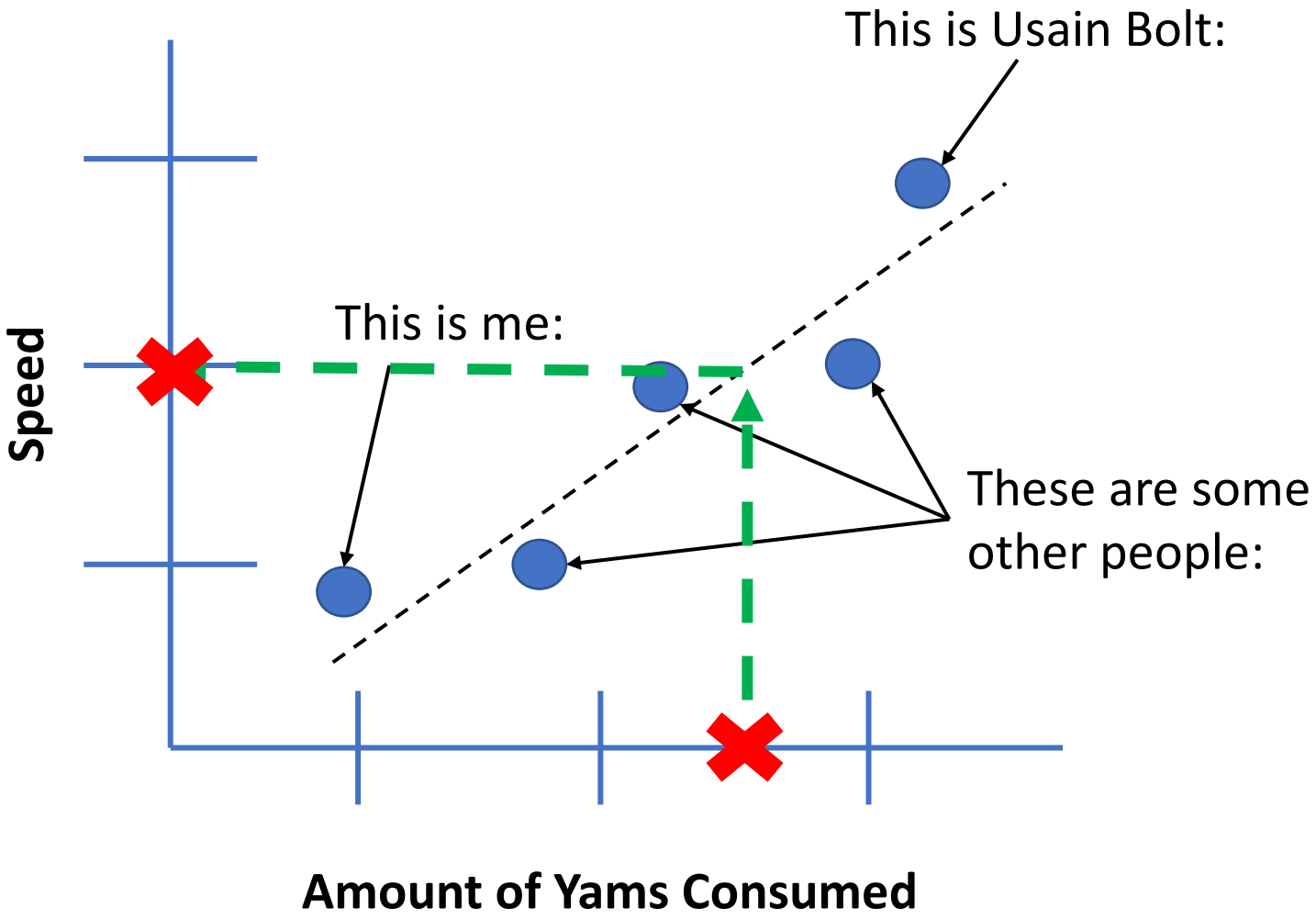
A silly introduction

How a cat decides whether to scratch on something:





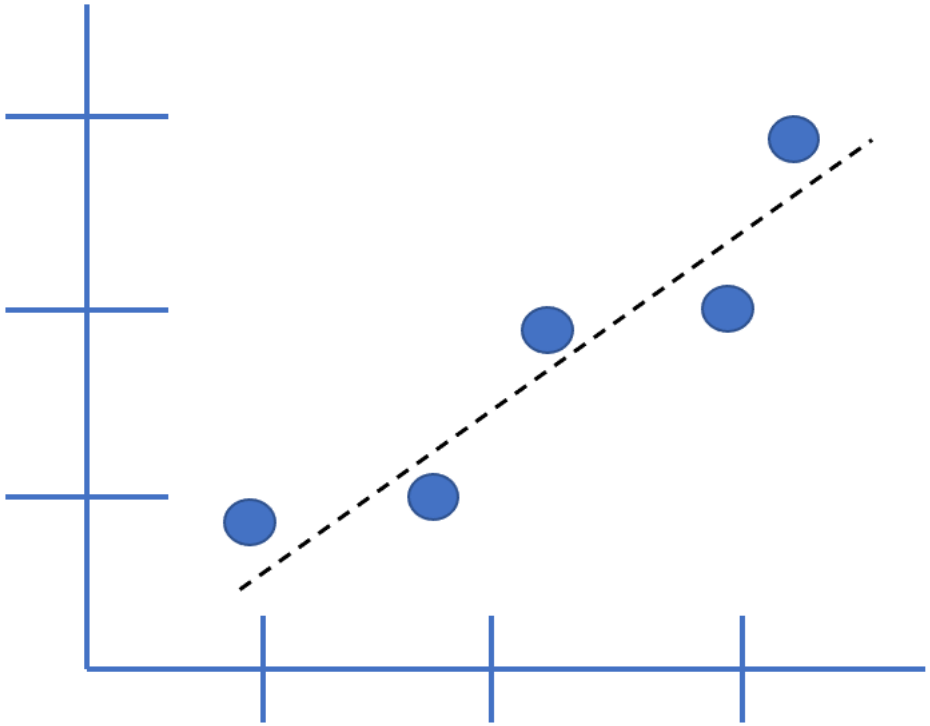
[Jamaica's sprinting glory: Is it the yams or the genes?]

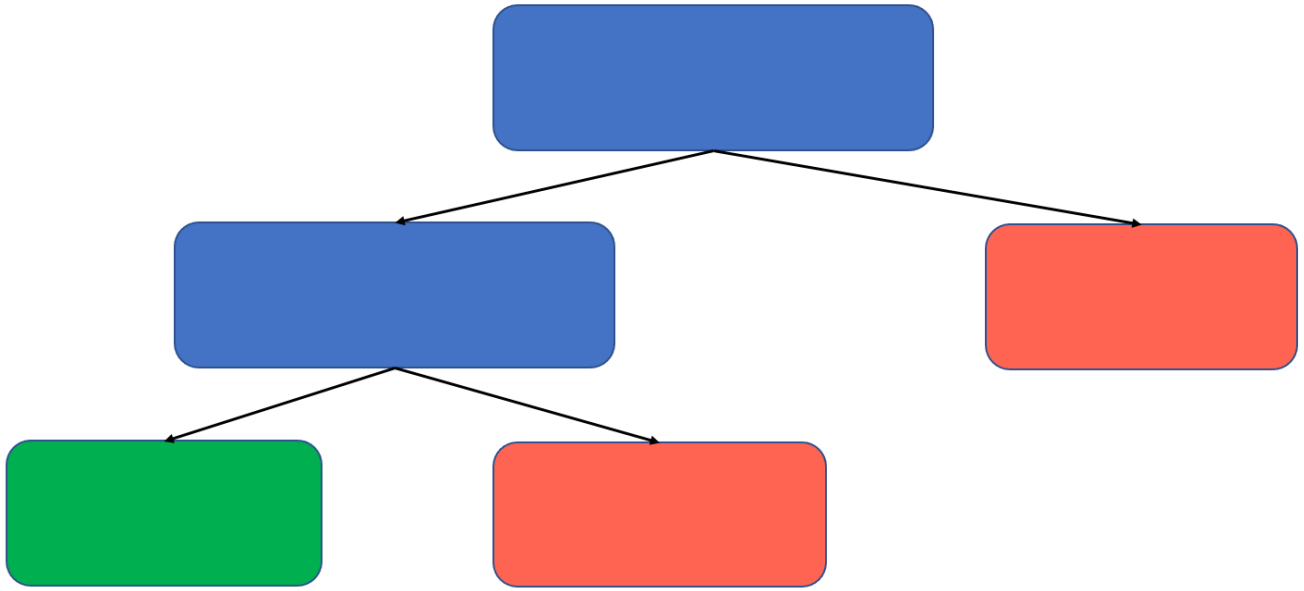






- In general, machine learning is all about making **predictions** or **classifications**.





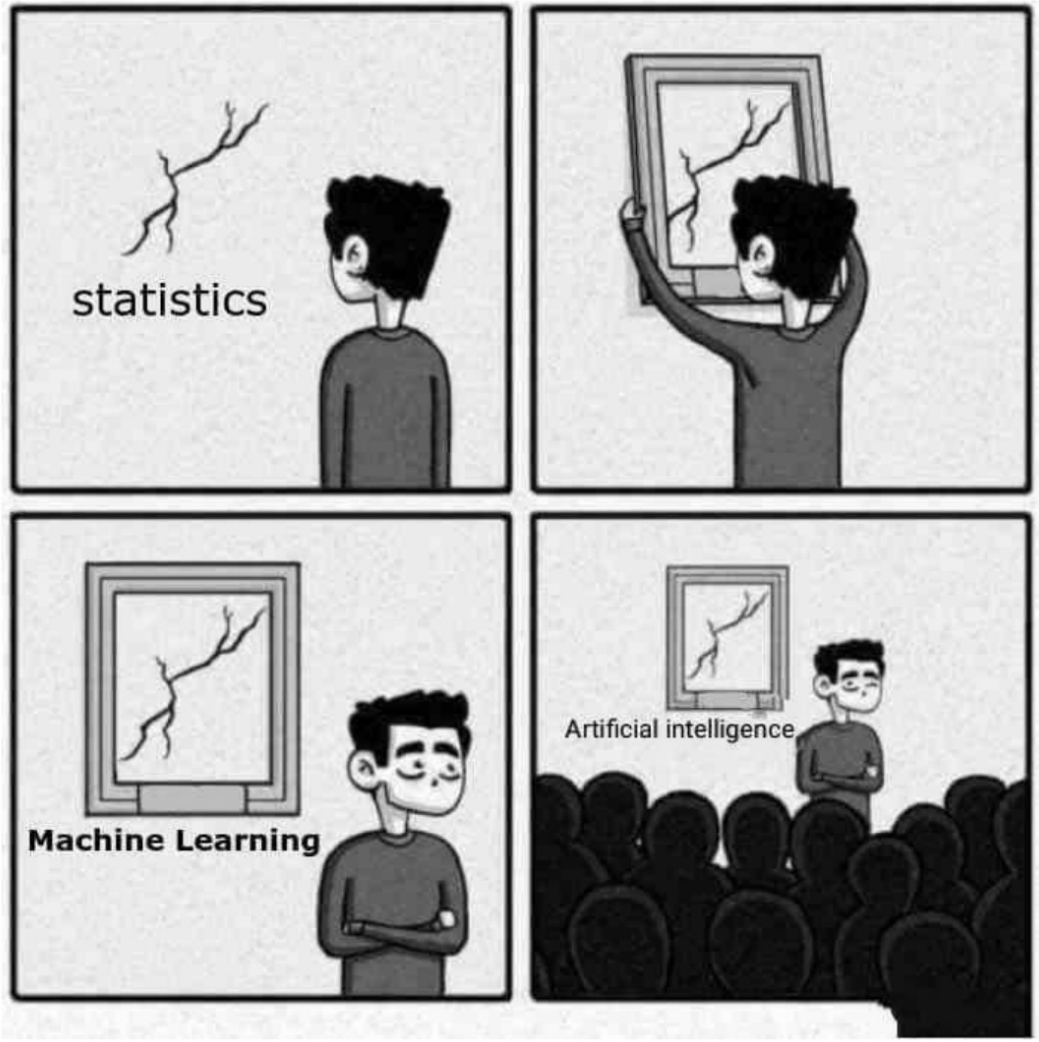


https://xkcd.com/1838/



https://xkcd.com/1425/

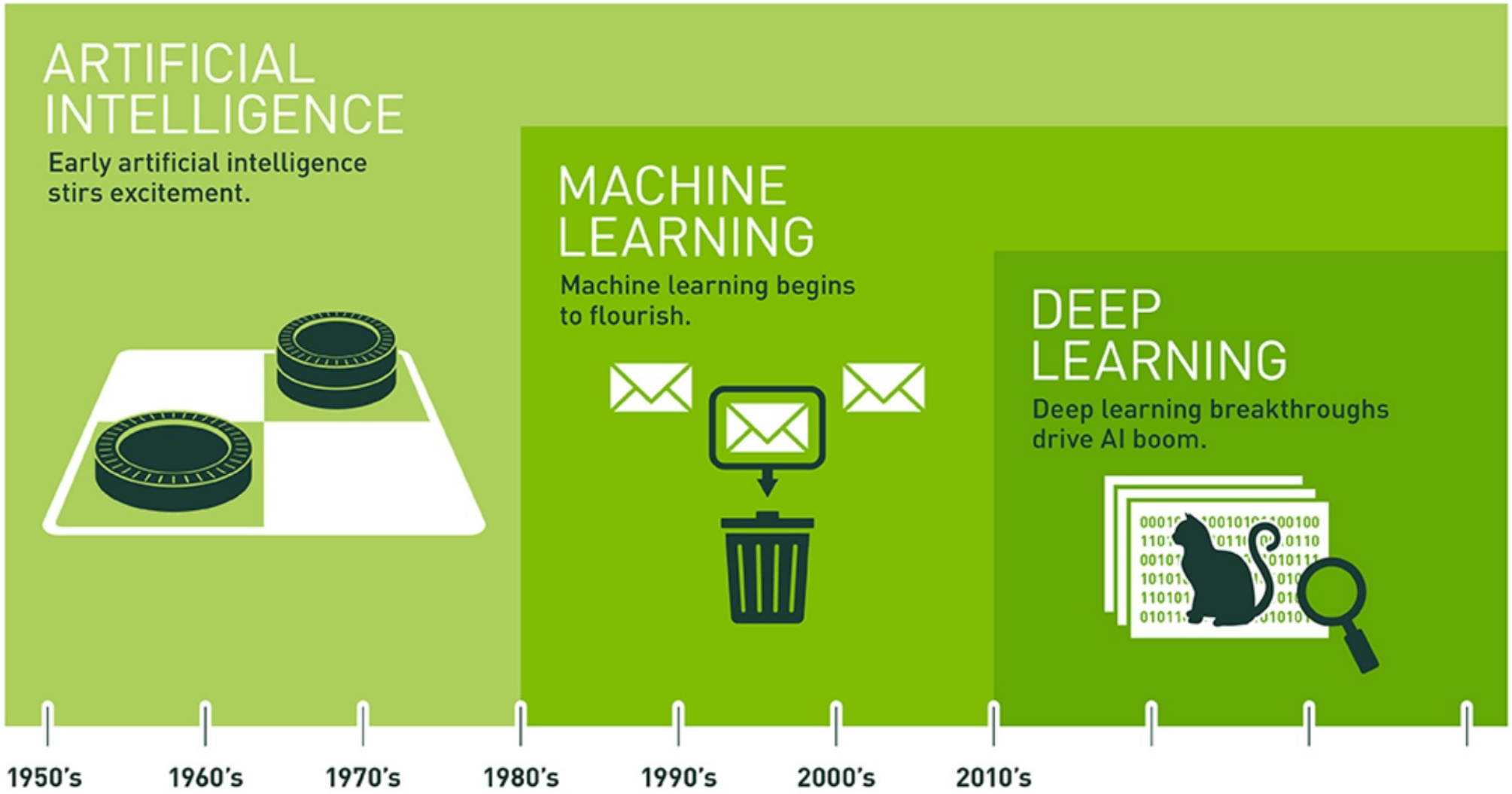
Machine learning vs. statistical learning



Machine learning vs. statistical learning?

- arose as a subfield of AI in CS
- algorithmic perspective
- focus on AI and prediction accuracy

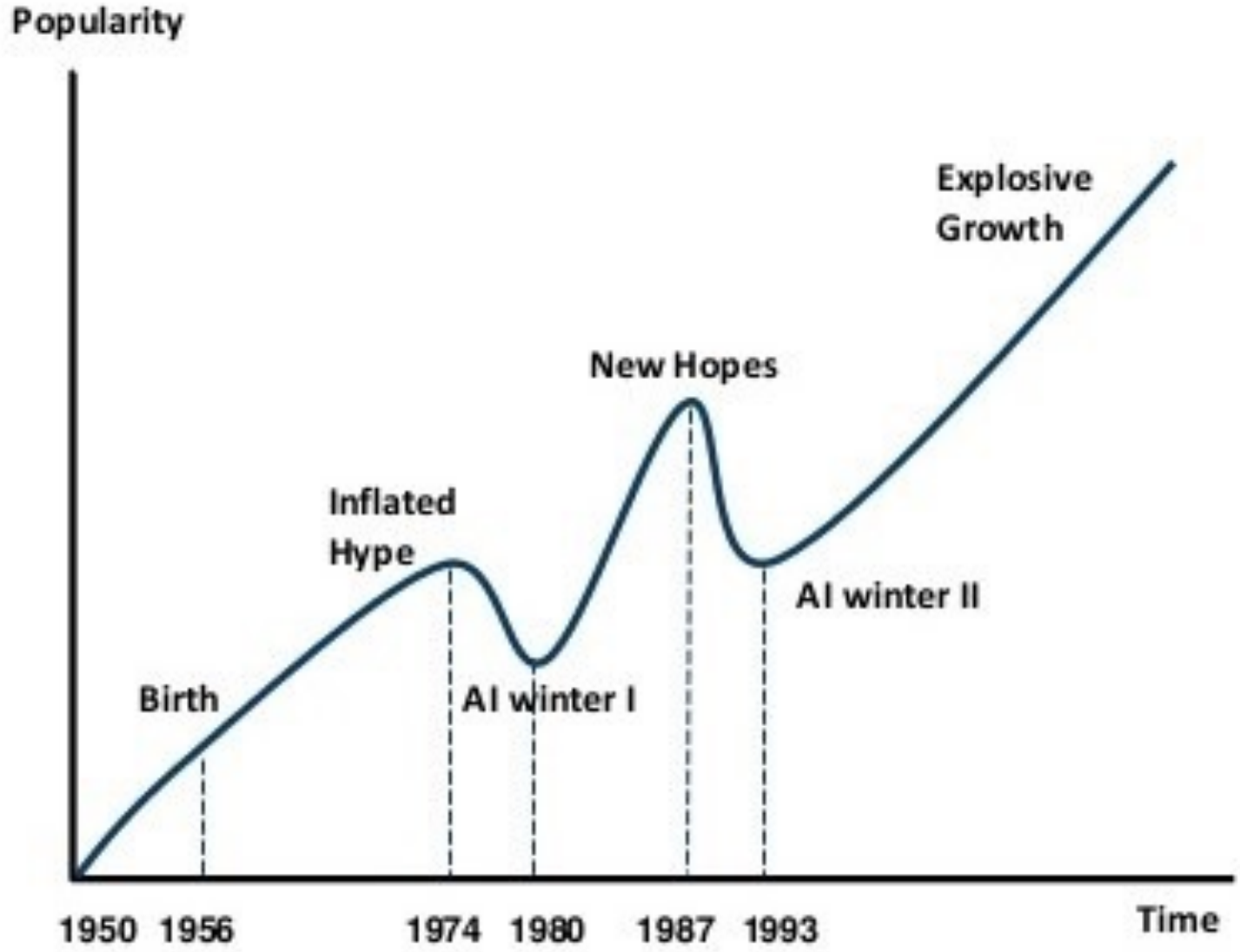
- arose as a subfield of statistics
- statistical perspective
- focus on interpretability and inference



Since an early flush of optimism in the 1950s, smaller subsets of artificial intelligence – first machine learning, then deep learning, a subset of machine learning – have created ever larger disruptions.

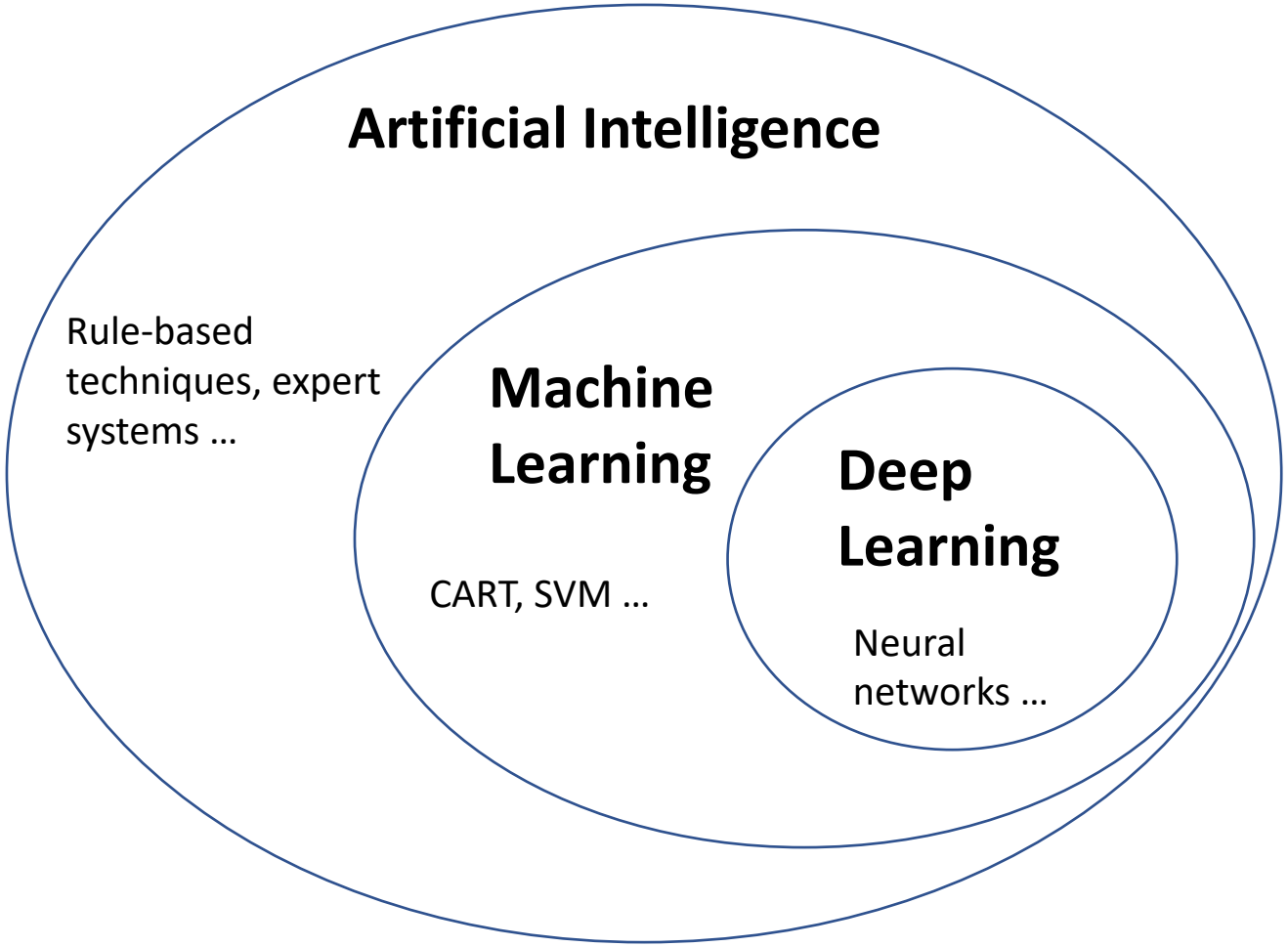
Source: Nvidia

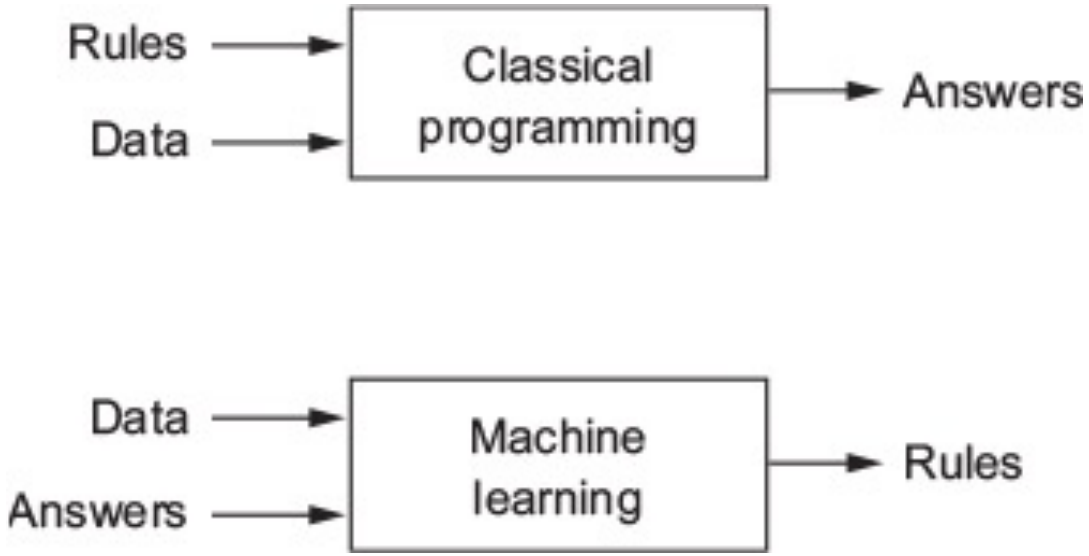
AI HAS A LONG HISTORY OF BEING "THE NEXT BIG THING"…



Timeline of AI Development

- **1950s-1960s**: First AI boom - the age of reasoning, prototype AI developed
- **1970s**: AI winter I
- **1980s-1990s**: Second AI boom: the age of Knowledge representation (appearance of expert systems capable of reproducing human decision-making)
- **1990s**: AI winter II
- **1997**: Deep Blue beats Gary Kasparov
- **2006**: University of Toronto develops Deep Learning
- **2011**: IBM's Watson won Jeopardy
- **2016**: Go software based on Deep Learning beats world's champions


Artificial Intelligence
Rule-based techniques, expert systems …
Machine Learning
CART, SVM …
Deep Learning
Neural networks …


Rules
Data
Classical programming
Answers
Data
Answers
Machine learning
Rules

"Imagine an old-fashioned program to identify a dog," he said. "A software engineer would write a thousand if-then-else statements: *if* it has ears, and a snout, *and* has hair, and *is not* a rat . . . and so forth, ad infinitum. But that's not how a child learns to identify a dog, of course. At first, she learns by seeing dogs and being told that they are dogs. She makes mistakes, and corrects herself. She thinks that a wolf is a dog—but is told that it belongs to an altogether different category. And so she shifts her understanding bit by bit: this is 'dog,' that is 'wolf.' The machine-learning algorithm, like the child, pulls information from a training set that has been classified. Here's a dog, and here's not a dog. It then extracts features from one set versus another. And, by testing itself against hundreds and thousands of classified images, it begins to create its own way to recognize a dog—again, the way a child does." It just knows *how* to do it.

https://www.newyorker.com/magazine/2017/04/03/ai-versus-md







LENSA



DALL·E





Welcome to ChatGPT

Log in with your OpenAI account to continue

Log in | Sign up

Jukebox

We're introducing Jukebox, a neural net that generates music, including rudimentary singing, as raw audio in a variety of genres and artist styles. We're releasing the model weights and code, along with a tool to explore the generated samples.

READ PAPER | VIEW CODE

April 30, 2020
12 minute read, 10 day listen





Lee Sedol vs. AlphaGo in 2016, Source: The New Yorker

https://www.wired.com/2016/03/two-moves-alphago-lee-sedol-redefined-future/



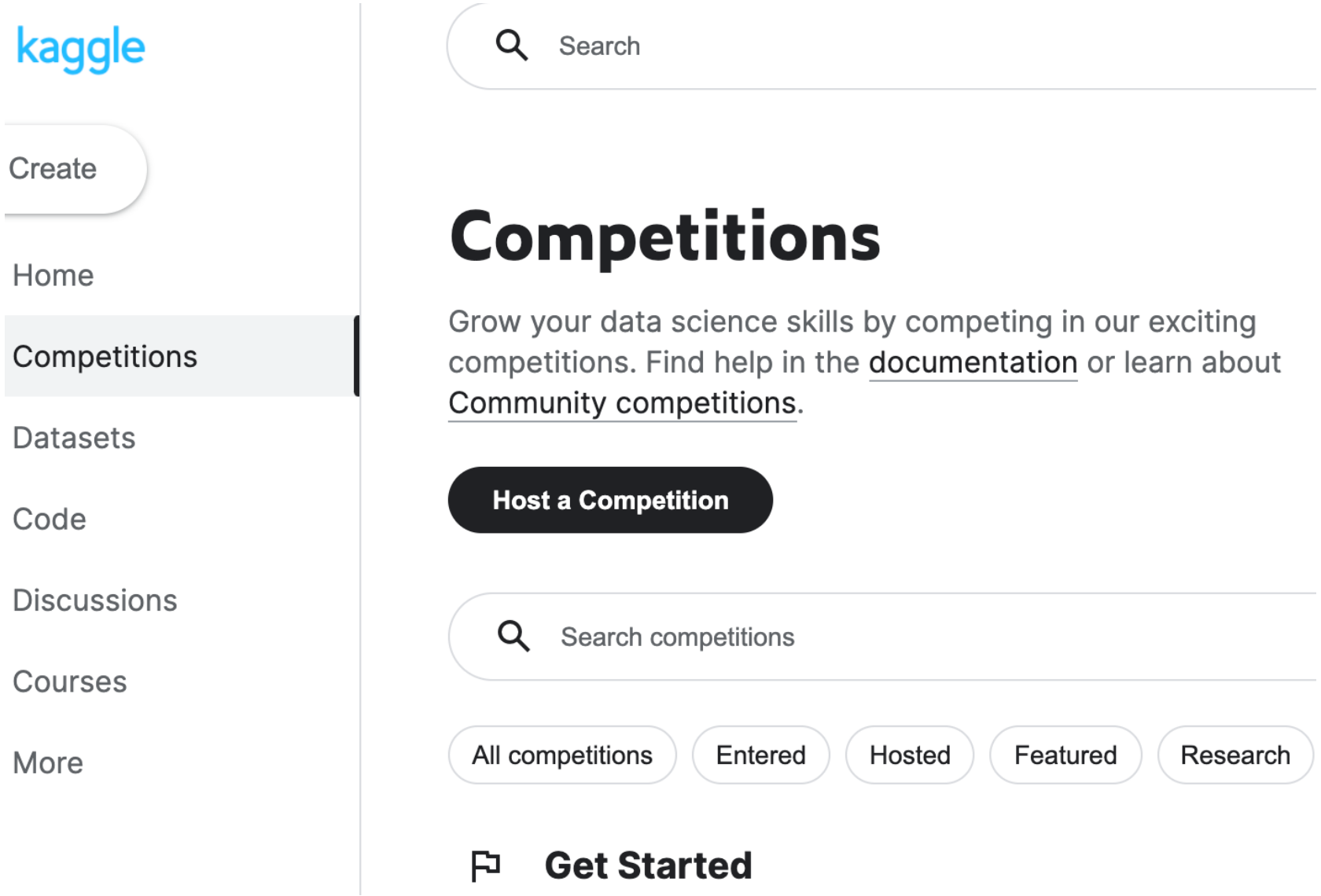
Image courtesy of Google

Google's artificial intelligence becomes first non-human to qualify as a driver

Machine learning competitions

- Netflix Challenge
- 100,000,000+ ratings from 400,000+ users of 17,000+ movies
- Predict user rating for a film, winner got $1,000,000!





https://www.kaggle.com/competitions

Machine learning in research: Examples

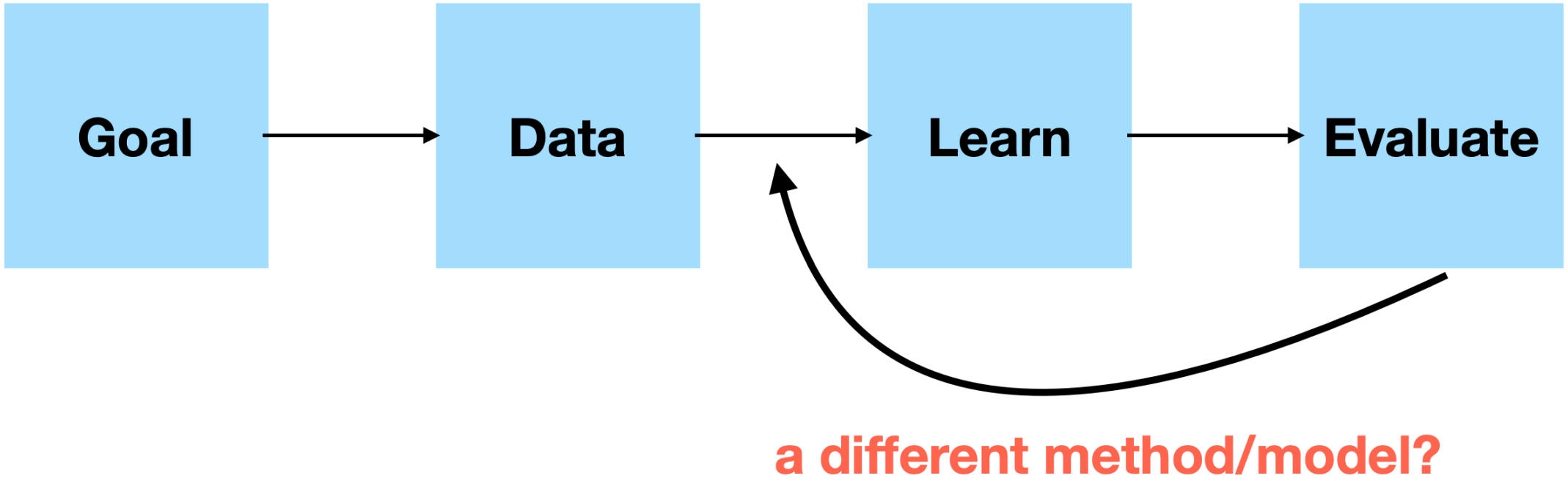
- High-energy physics experiments
- Microarrays for gene expression (20,000+ features)
- Genome-wide association studies (millions of features)

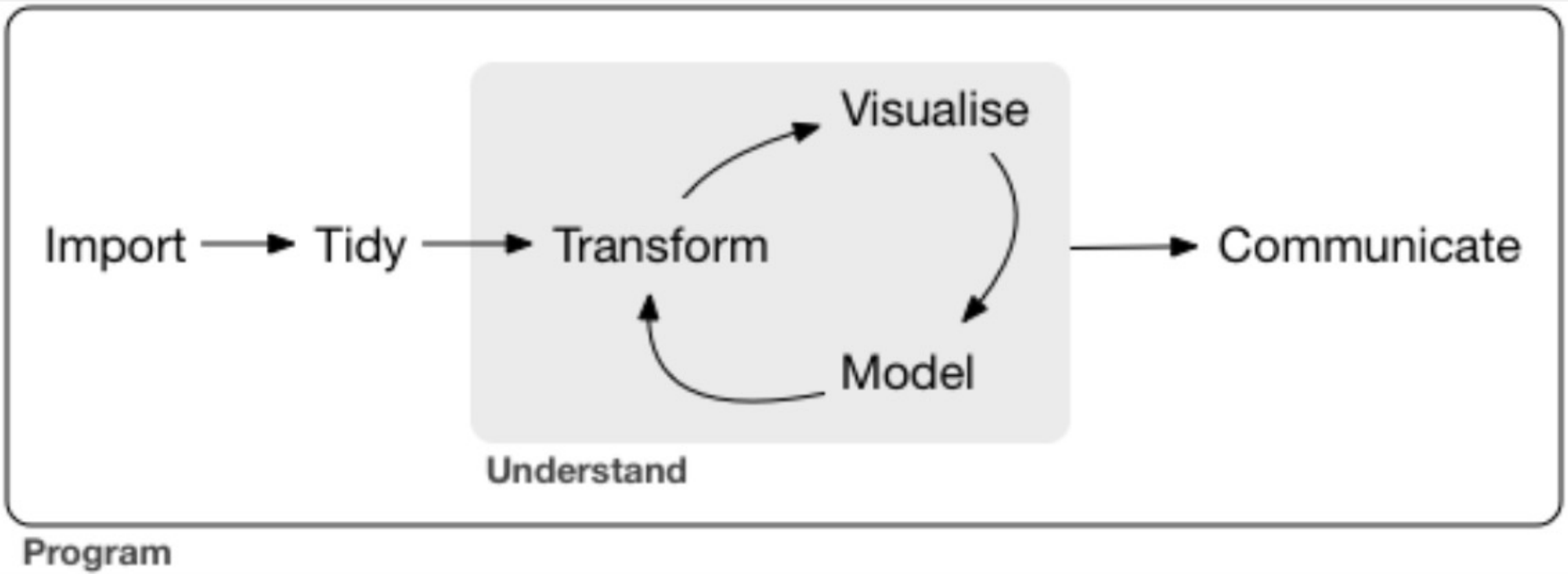
Unprecedented large scale of data

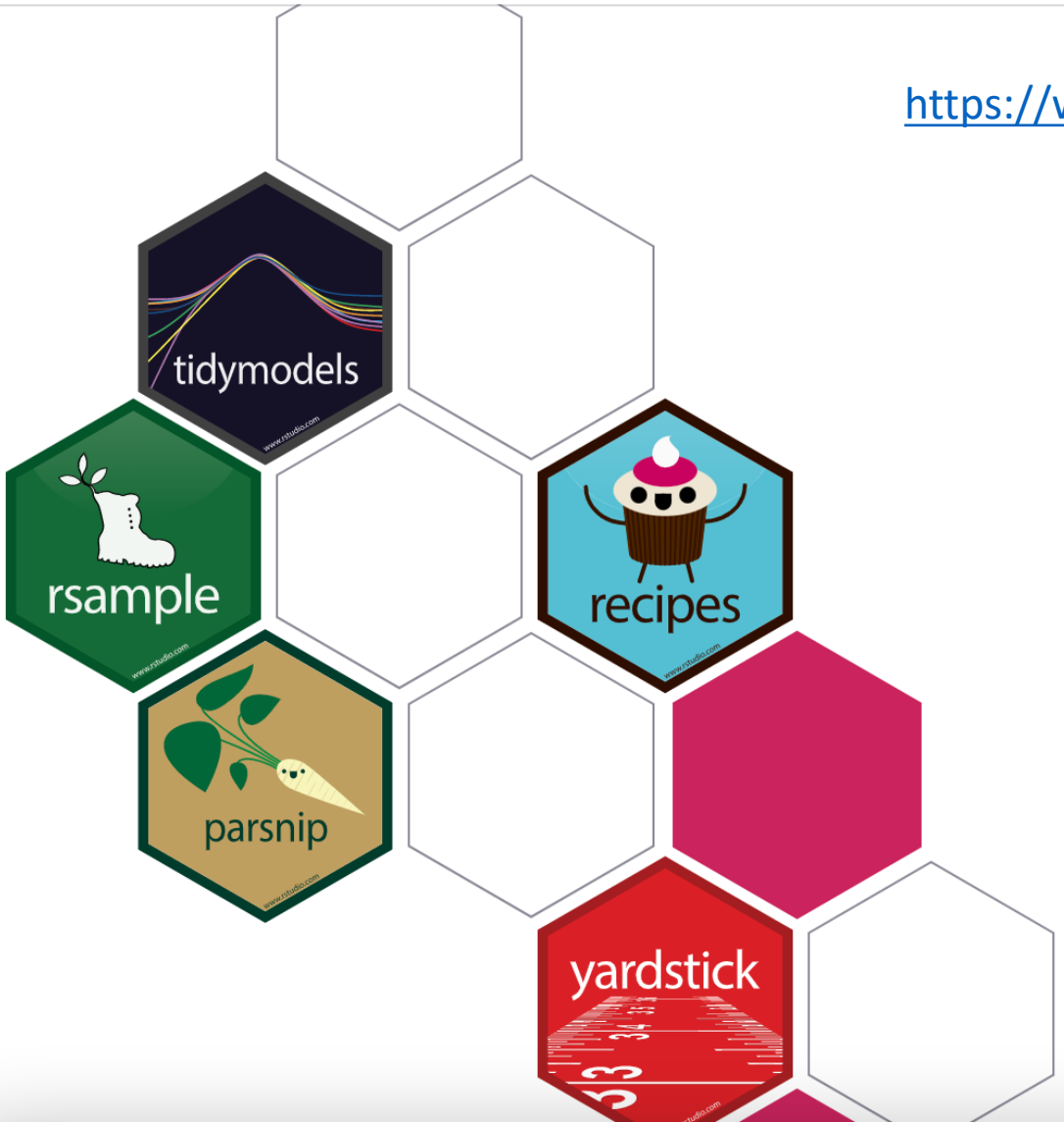
+

Highly complicated underlying structure

A (very) rough ML workflow:




Visualise
Import
Tidy
Transform
Communicate
Model
Understand
Program



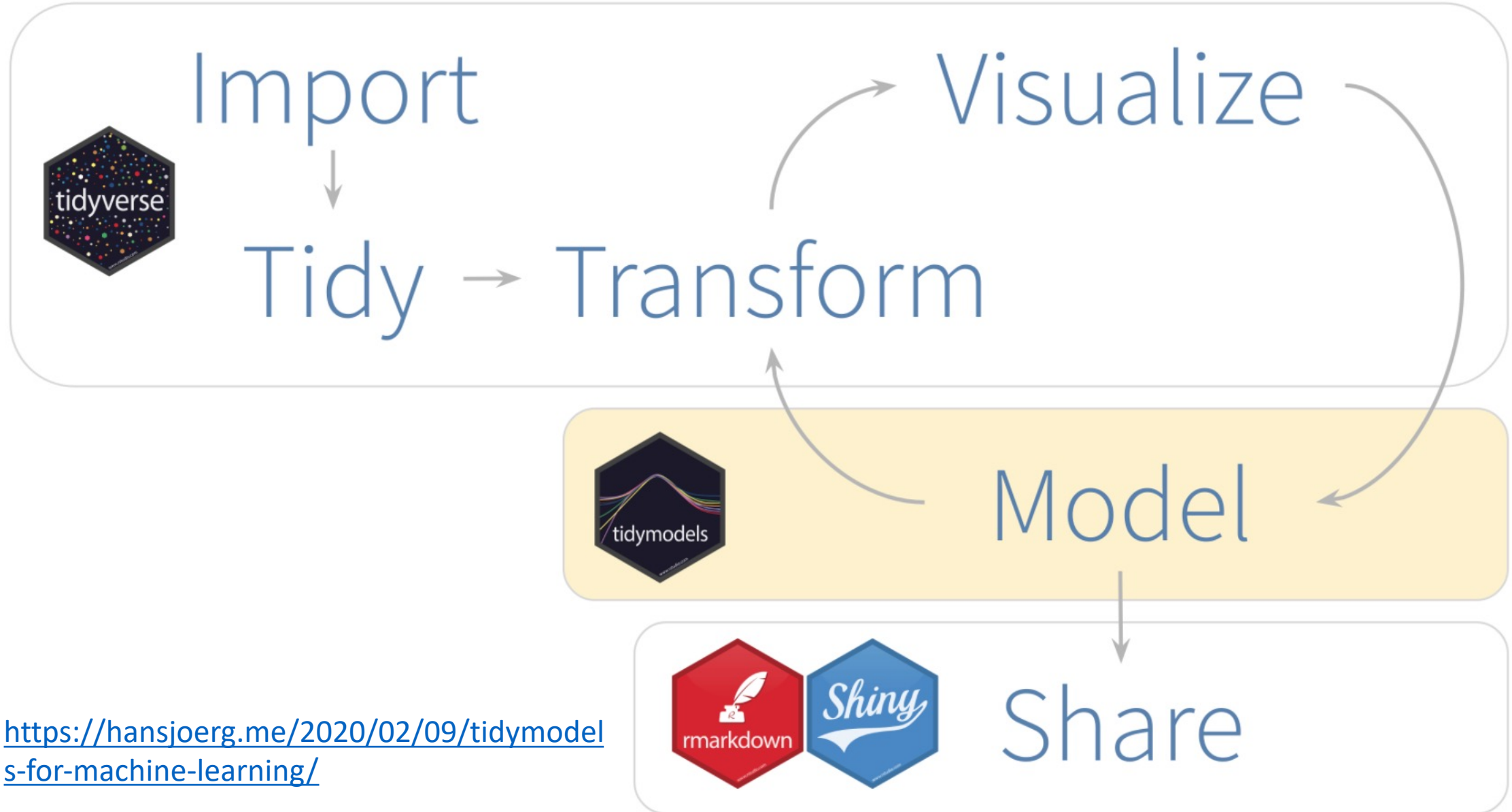
https://www.tidymodels.org/

TIDYMODELS

The tidymodels framework is a collection of packages for modeling and machine learning using tidyverse principles.

Install tidymodels with:

```
install.packages("tidymodels")
```



https://hansjoerg.me/2020/02/09/tidymodels-for-machine-learning/

https://rviews.rstudio.com/2019/06/19/a-gentle-intro-to-tidymodels/

An example:

- At the population level:

$$Y$$

Response

(output, target, dependent variable)

Y = **wage**

$$(X_1, \ldots, X_p)$$

Predictors

(input, features, covariates, independent variables)

X1 = employee's **age**
X2 = **education** level
X3 = calendar **year**

Goal: Predict an employee's **wage** based on their **age**, **education** level, and the calendar **year**

Example data:

- At the observation level:

wage

$$\begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_9 \end{pmatrix}$$

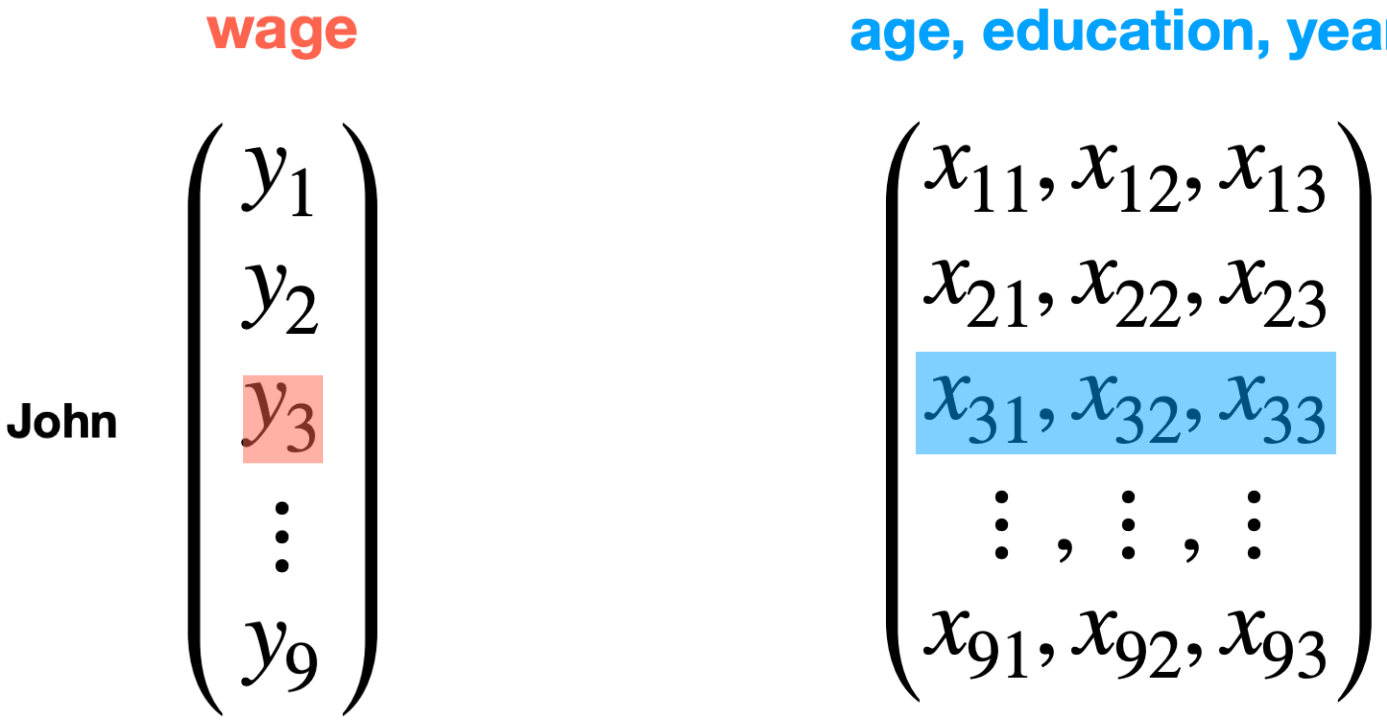
age, education, year

$$\begin{pmatrix} x_{11}, x_{12}, x_{13} \\ x_{21}, x_{22}, x_{23} \\ x_{31}, x_{32}, x_{33} \\ \vdots\ ,\ \vdots\ ,\ \vdots \\ x_{91}, x_{92}, x_{93} \end{pmatrix}$$

Goal: Predict an employee's **wage** based on their **age**, **education** level, and the calendar **year**

Example data:

- At the observation level:



Goal: Predict an employee's **wage** based on their **age**, **education** level, and the calendar **year**

Example data:

- At the observation level:

wage

$$\begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_9 \end{pmatrix}$$

age, education, year

$$\begin{pmatrix} x_{11}, x_{12}, x_{13} \\ x_{21}, x_{22}, x_{23} \\ x_{31}, x_{32}, x_{33} \\ \vdots, \vdots, \vdots \\ x_{91}, x_{92}, x_{93} \end{pmatrix}$$

all 9 employees' education levels

Goal: Predict an employee's **wage** based on their **age**, **education** level, and the calendar **year**

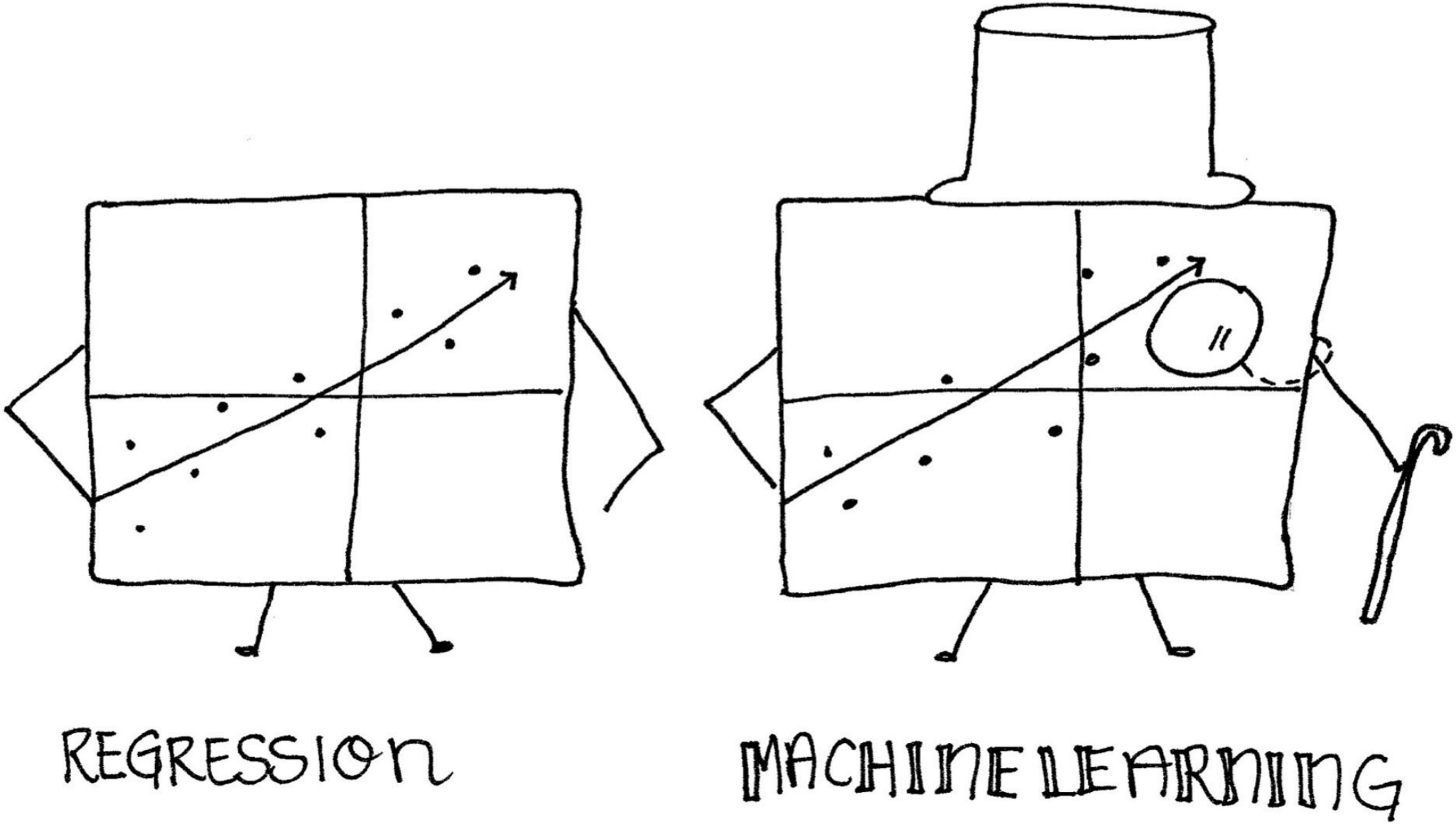
A very basic learning model

$$Y = f(X_1, \ldots, X_p) + \varepsilon$$

Response **is a function of** **Predictors** **and** **Error ("noise")**

- Goal: **learn** f from observations of data

Twitter

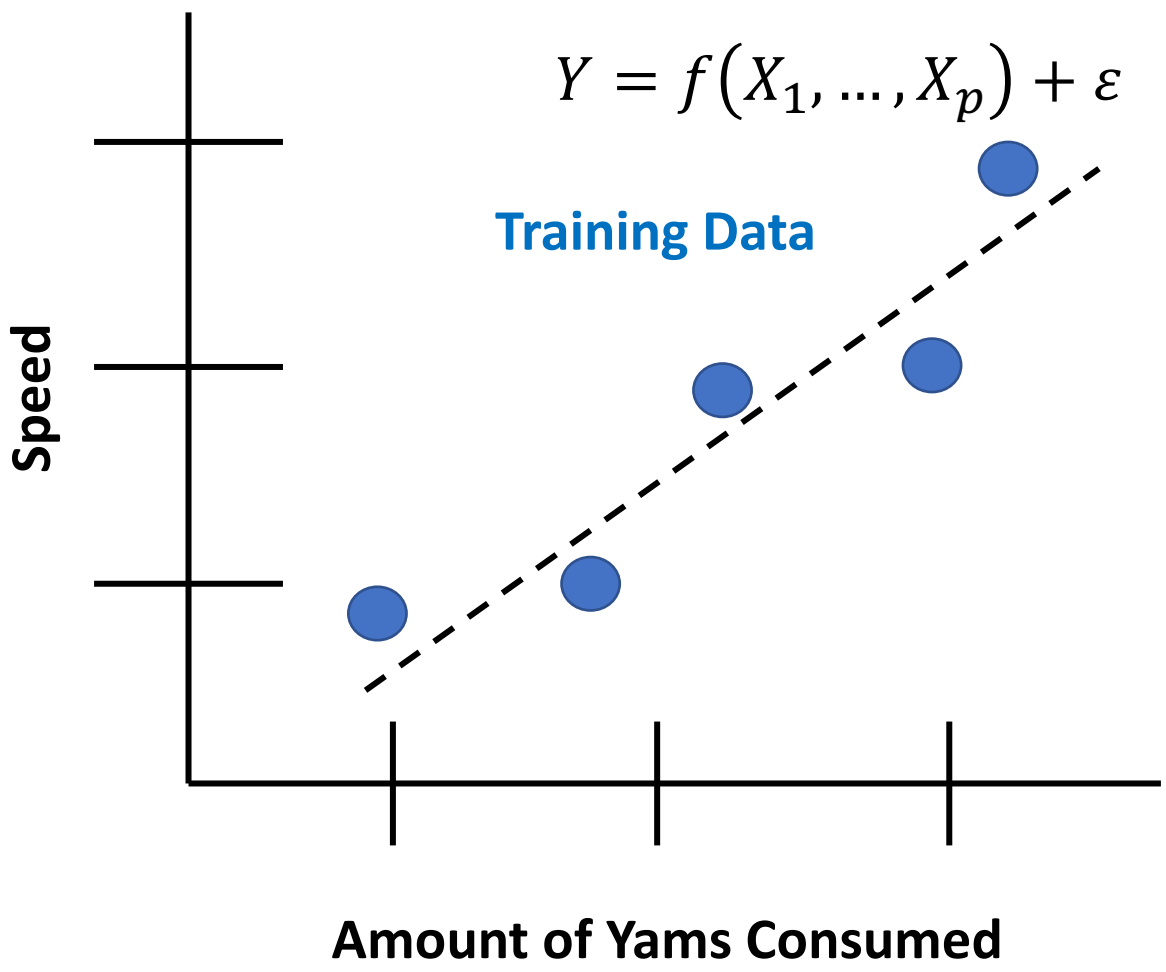


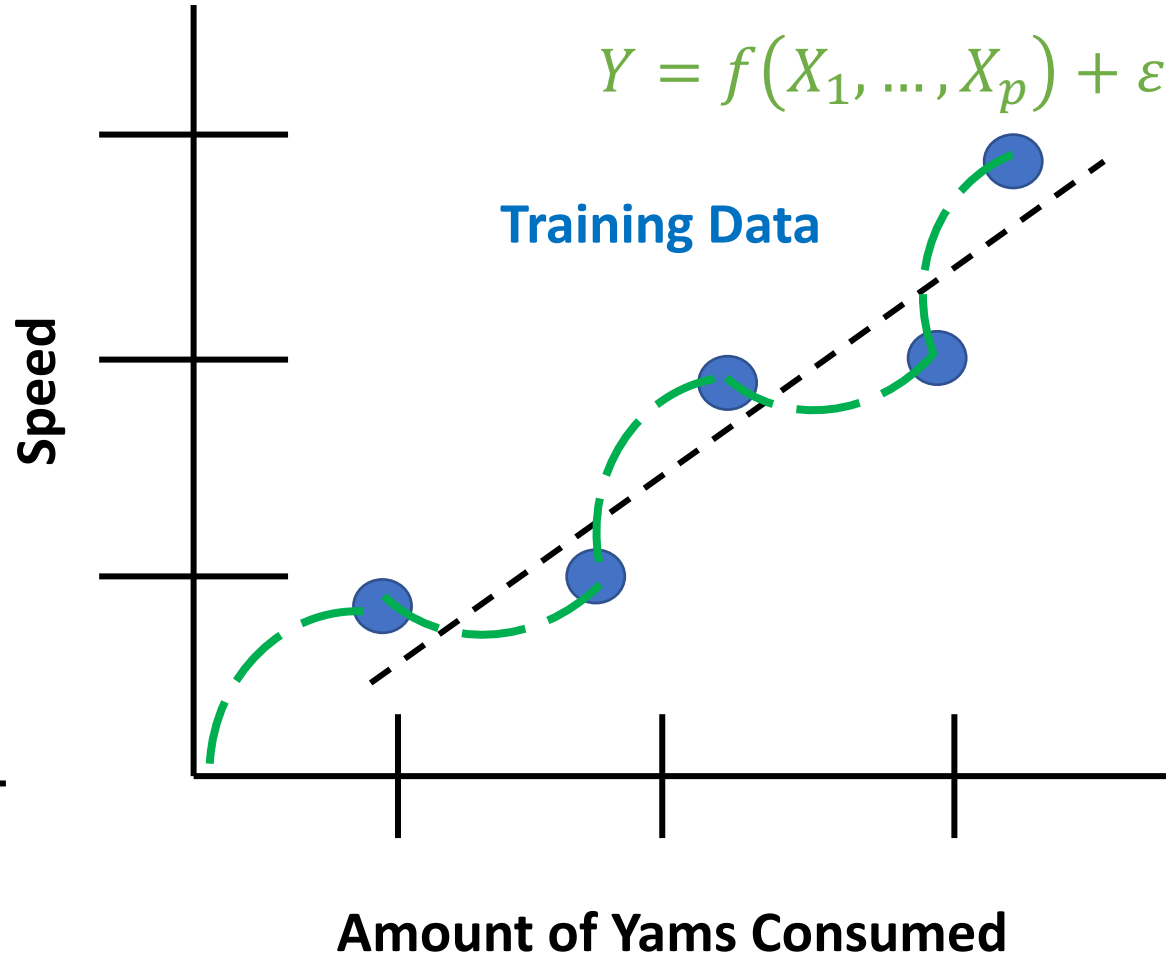
credit: Chelsea Parlett Pelleriti

Evaluate: How well did the machine learn?

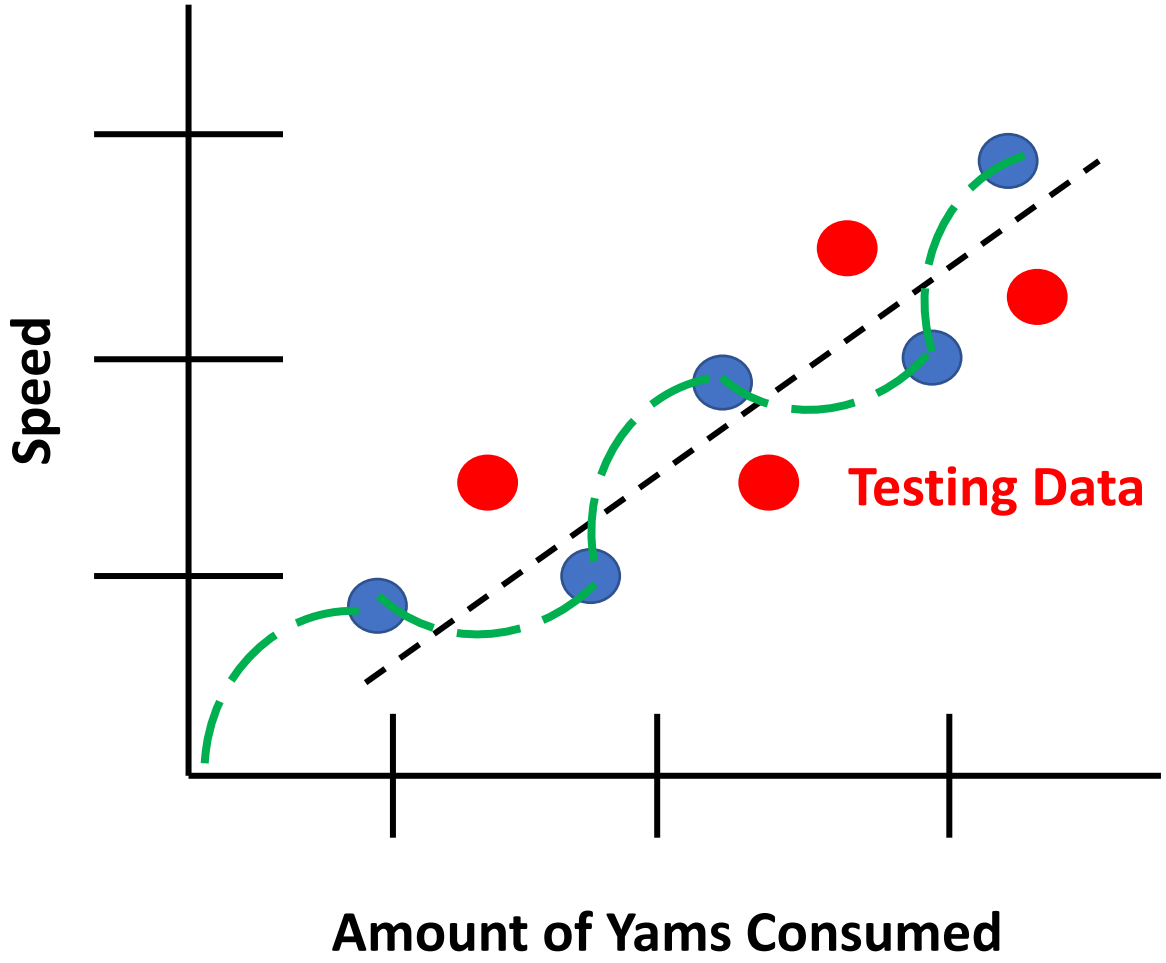
- In **supervised learning**, the actual data Y is the **supervisor**
- Need to give the model observed output and input

Learner	**Supervisor**
College student	Answer key
Machine	Raw observed output





The **green squiggle** seems to fit the **training data** better than the **black line** but remember – the goal of machine learning is to make predictions!

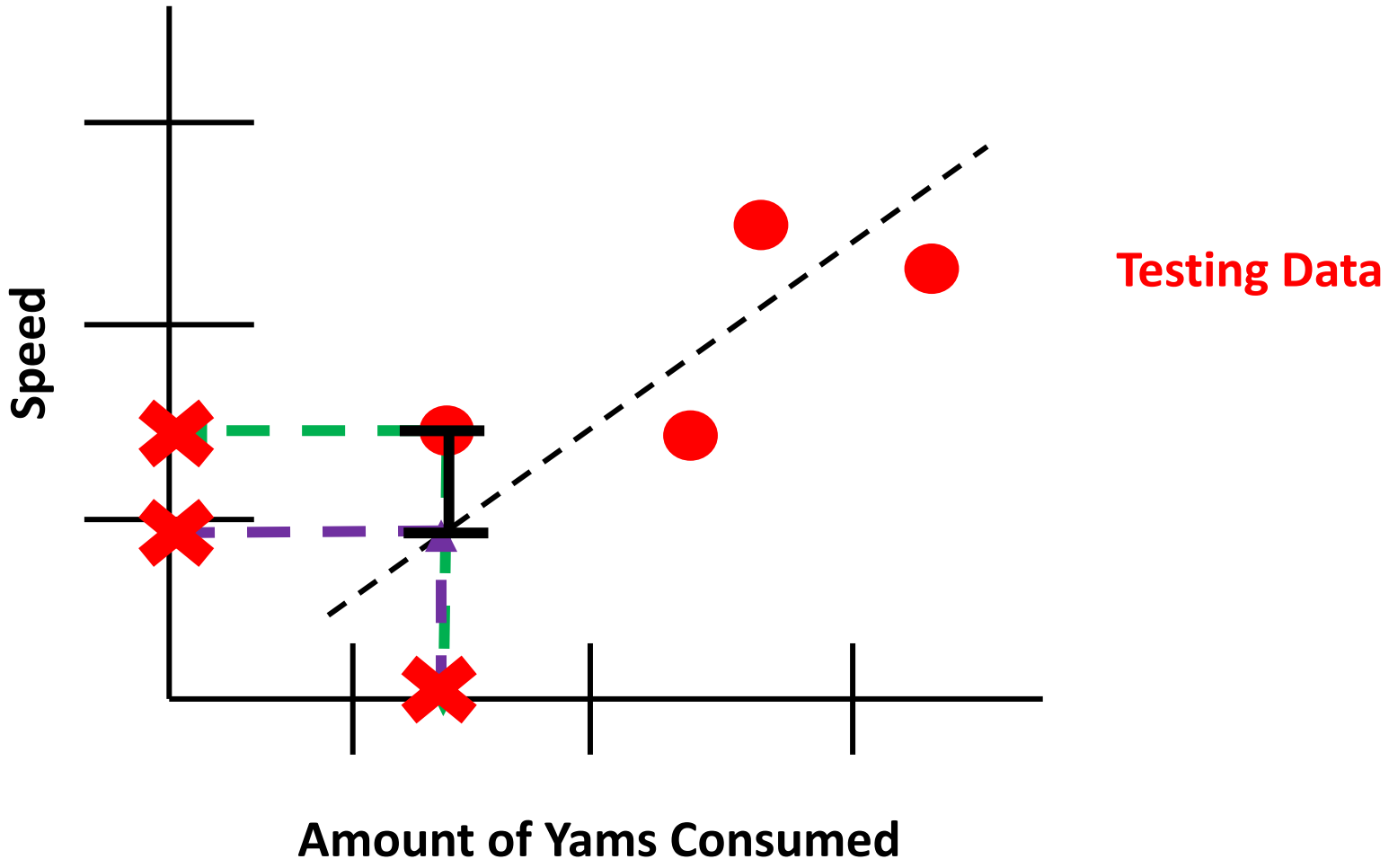


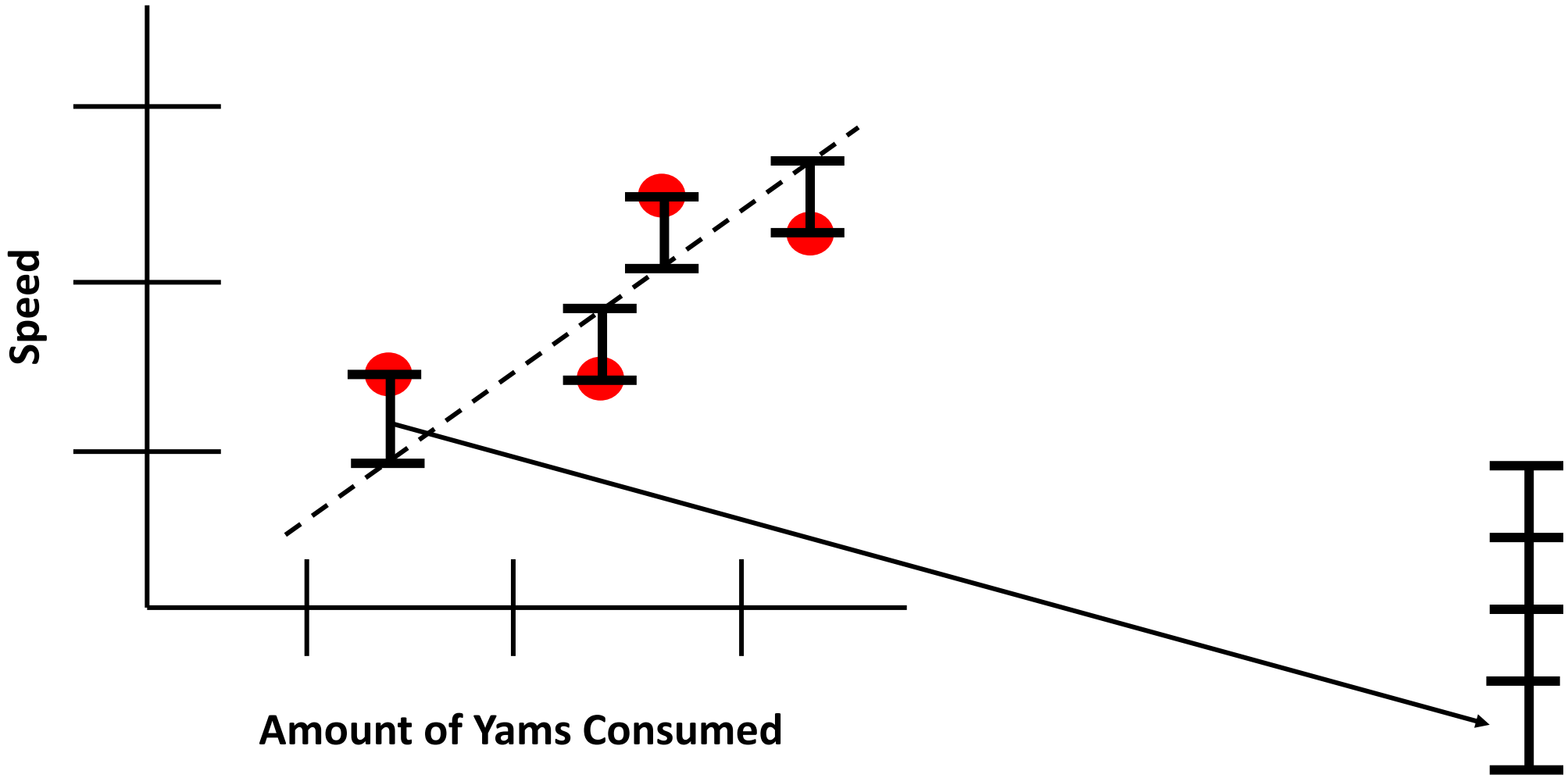
speed yams

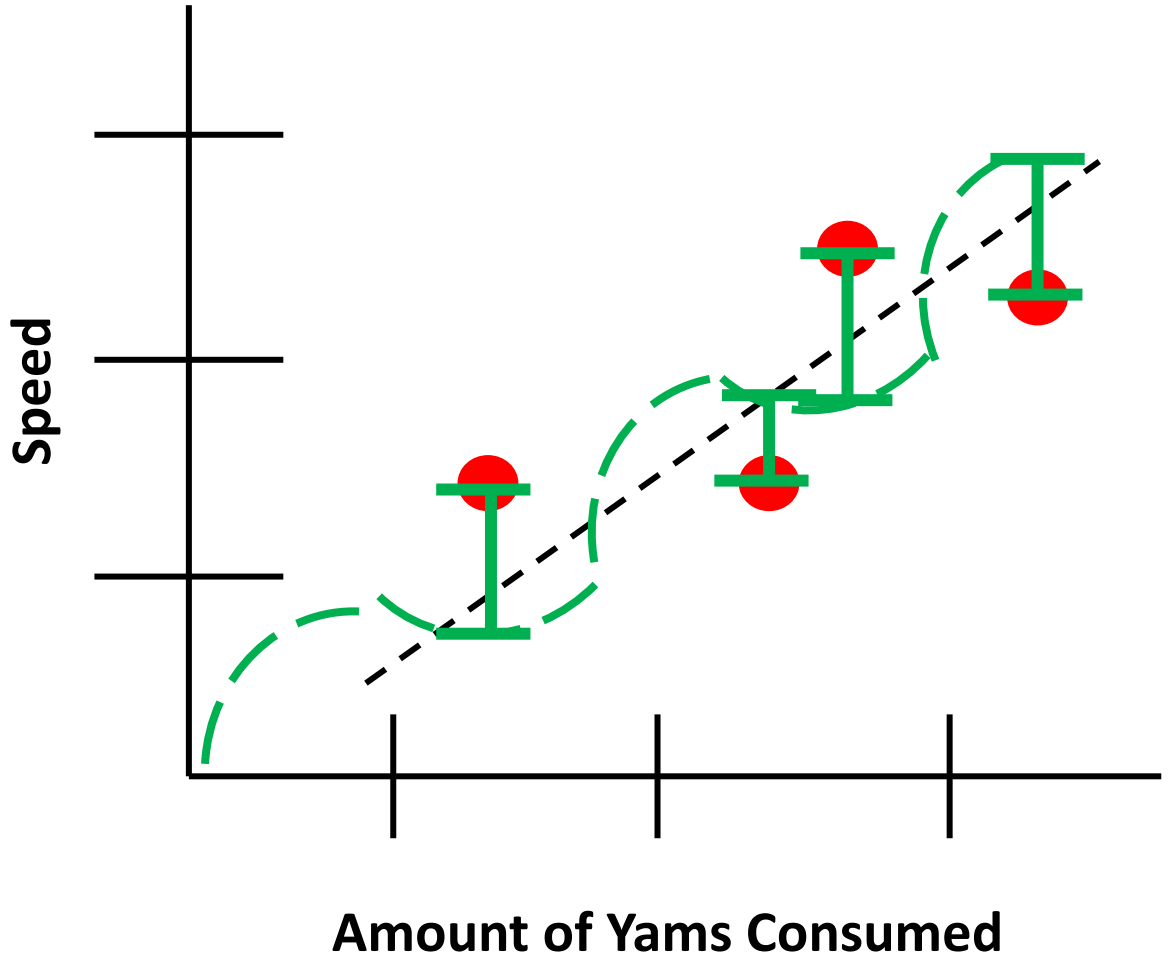
$$\begin{pmatrix} y_1 & x_1 \\ y_2 & x_2 \\ y_3 & x_3 \\ y_4 & x_4 \\ y_5 & x_5 \\ y_6 & x_6 \\ y_7 & x_7 \\ y_8 & x_8 \\ y_9 & x_9 \end{pmatrix}$$

$$\{(\boldsymbol{x_1}, y_1), (\boldsymbol{x_2}, y_2), \dots, (\boldsymbol{x_5}, y_5)\}$$

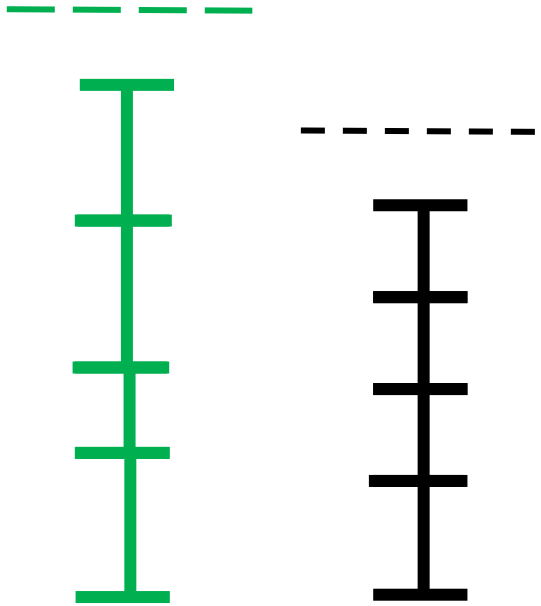
$$\{(\tilde{\boldsymbol{x}}_{\boldsymbol{6}}, \tilde{y}_6), (\tilde{\boldsymbol{x}}_{\boldsymbol{7}}, \tilde{y}_7), \dots, (\tilde{\boldsymbol{x}}_{\boldsymbol{9}}, \tilde{y}_9)\}$$


Speed
Amount of Yams Consumed
Testing Data


Speed
Amount of Yams Consumed



$$\sum_{i=6}^{9} \left| \tilde{y}_i - \hat{f}(\tilde{x}_i) \right|$$



Supervised learning

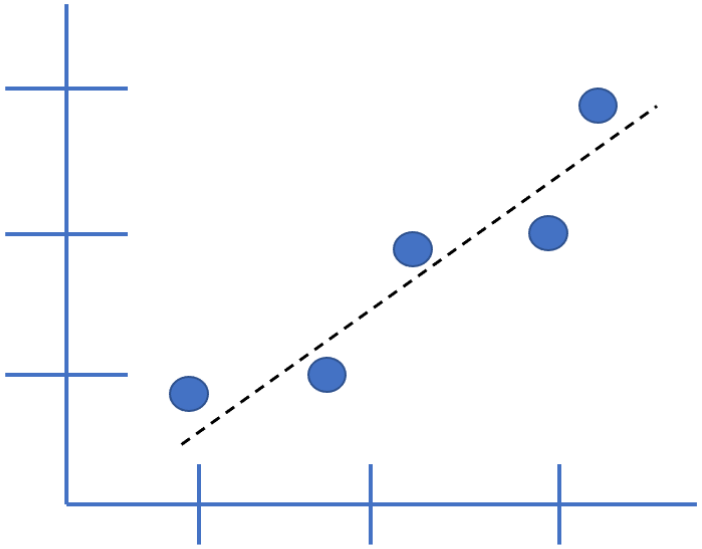
- Prediction:
  - accurately predict future response given predictors
- Estimation:
  - understand how predictors affect response
- Model selection:
  - find the “best” model for response given predictors
- Inference:
  - assess the quality of our predictions and (or) estimation

Regression models vs. classification models

Regression

Y is **quantitative**

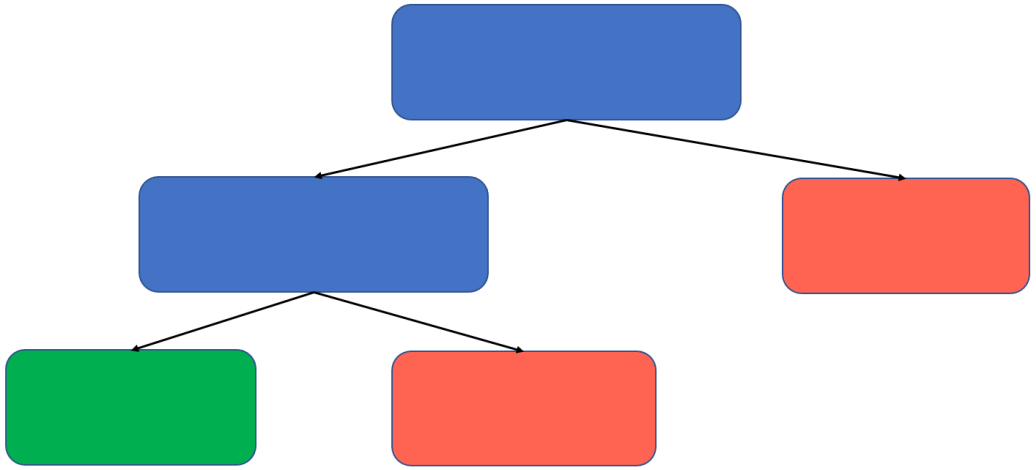
Numerical values (e.g., price, blood pressure, speed)



Classification

Y is **qualitative**

Categorical values (e.g., survived/died, spam/not spam, scratch/don't scratch)



Common error metrics

Regression models

Mean squared error

Root mean squared error

Mean absolute error

R-squared

Classification models

Accuracy

Error rate

Precision

Area under ROC curve

Unsupervised learning

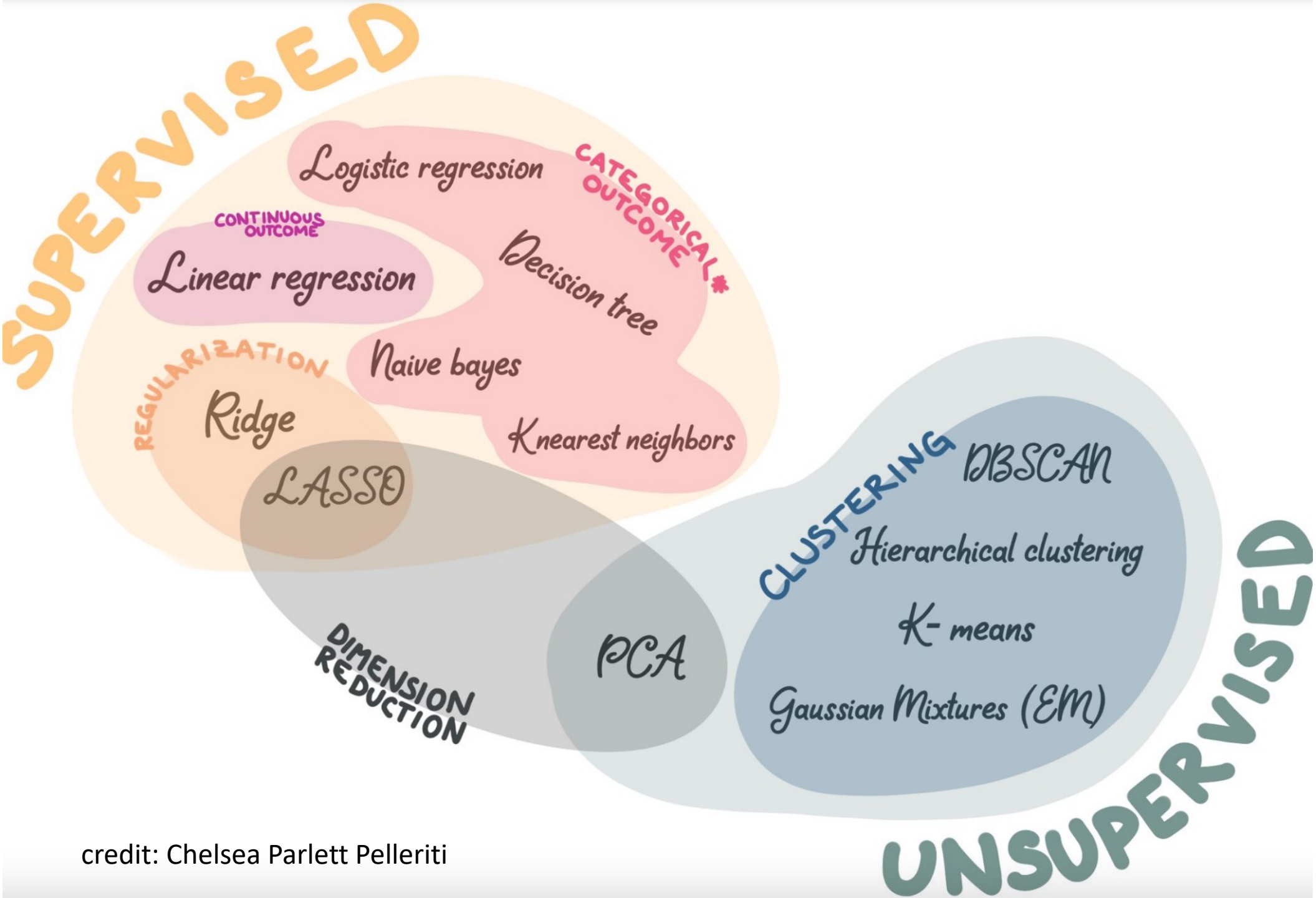
- Learn without a supervisor!
- You never see the answer key

NO Response

$(X_1, \ldots, X_p)$

Predictors

(input, features, covariates, independent variables)

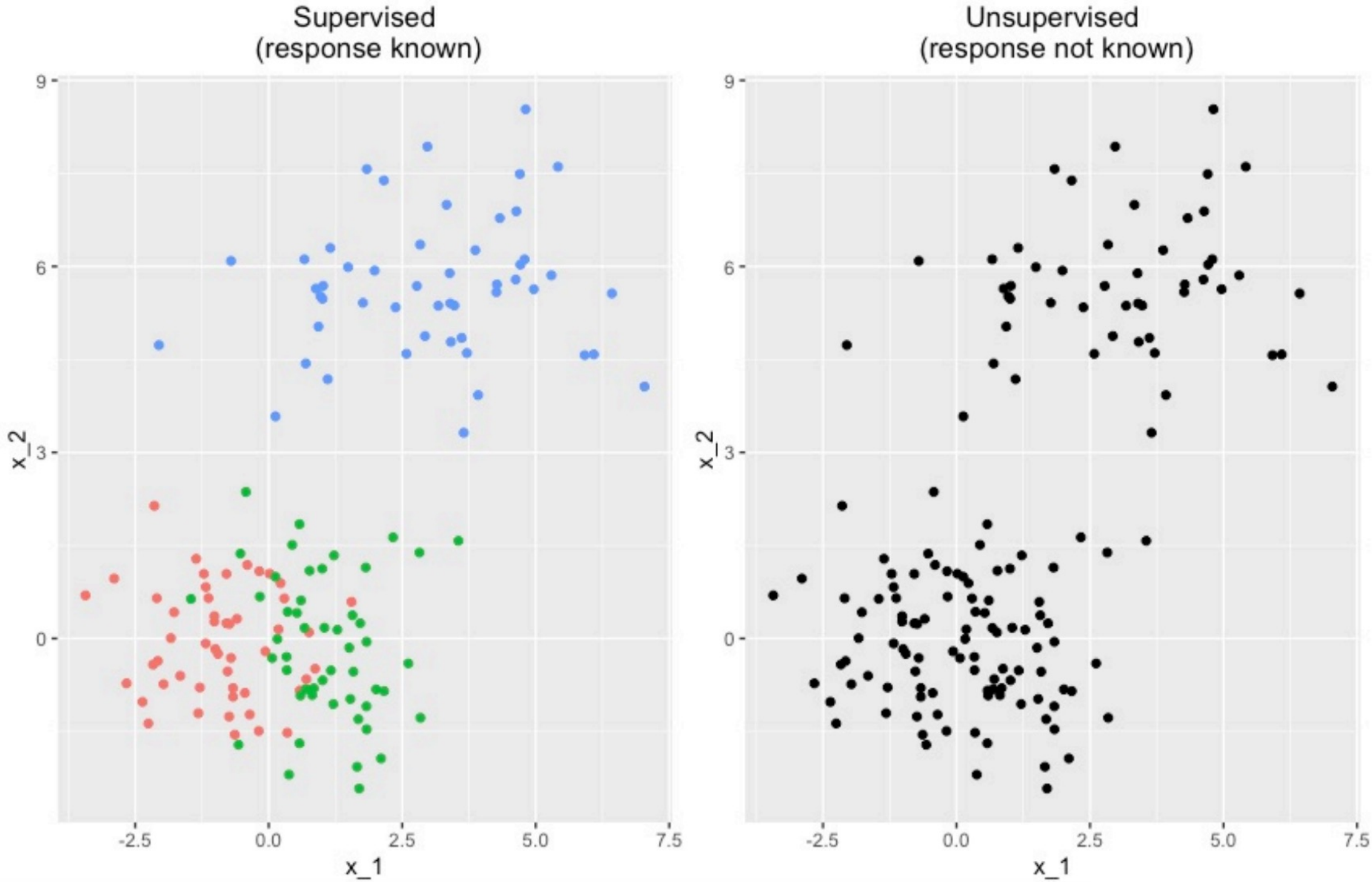


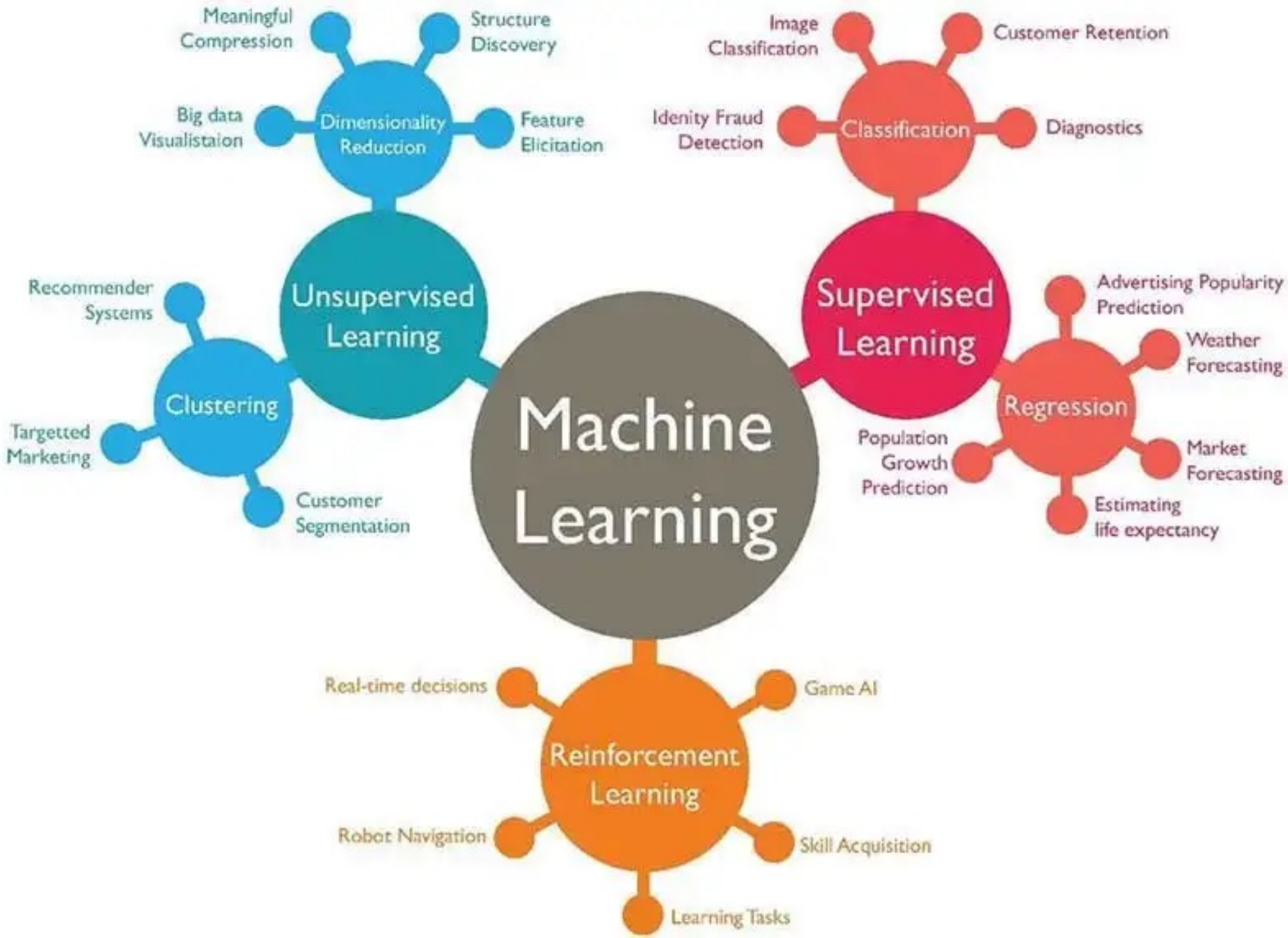
credit: Chelsea Parlett Pelleriti

Supervised learning vs. unsupervised learning

- Linear regression
- Logistic regression
- *k*-nearest neighbors
- Decision trees
- Random forests
- Support vector machine(s)
- Neural networks

- Principal Component Analysis (PCA)
- *k*-means clustering
- Hierarchical clustering
- Neural networks


Supervised
(response known)
Unsupervised
(response not known)
x_2
x_1
9
6
3
0
-2.5
0.0
2.5
5.0
7.5


Machine Learning
Unsupervised Learning
Supervised Learning
Reinforcement Learning
Dimensionality Reduction
Meaningful Compression
Structure Discovery
Big data Visualistaion
Feature Elicitation
Clustering
Recommender Systems
Targetted Marketing
Customer Segmentation
Classification
Image Classification
Customer Retention
Idenity Fraud Detection
Diagnostics
Regression
Advertising Popularity Prediction
Weather Forecasting
Population Growth Prediction
Market Forecasting
Estimating life expectancy
Real-time decisions
Game AI
Robot Navigation
Skill Acquisition
Learning Tasks

Mechanistic vs. empirically-driven

Parametric vs. non-parametric

- Assume a parametric form for f (i.e., $\beta_0 + \beta_1 X_1 + \cdots$)
  - Won't match true unknown f
  - Can add parameters = more flexibility
  - Too many = overfitting

- No/few assumptions about f
  - Require **a larger** # of observations
  - Much more flexible by default
  - Can also overfit

$$Y = f(X_1, \ldots, X_p) + \varepsilon$$

Uses of statistical models:

PREDICTIVE:

- What combo of features fits best?
- Aim is to predict *Y* with minimum reducible error
- Not focused on hypothesis tests

DESCRIPTIVE:

- Choose model to best visually emphasize a trend in data
- i.e., using a line on a scatterplot

INFERENTIAL:

- What features are significant?
- Aim is to test theories
- (Possibly) causal claims
- State relationship between outcome & predictor(s)

General comments

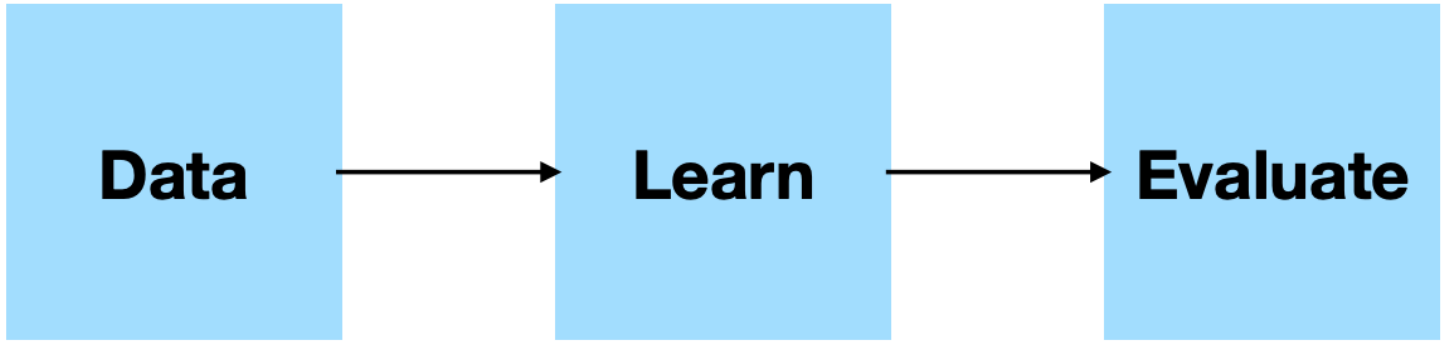
- It’s important to:

- understand the ideas behind these methods
- understand the simpler methods first
- accurately evaluate the performance of a method

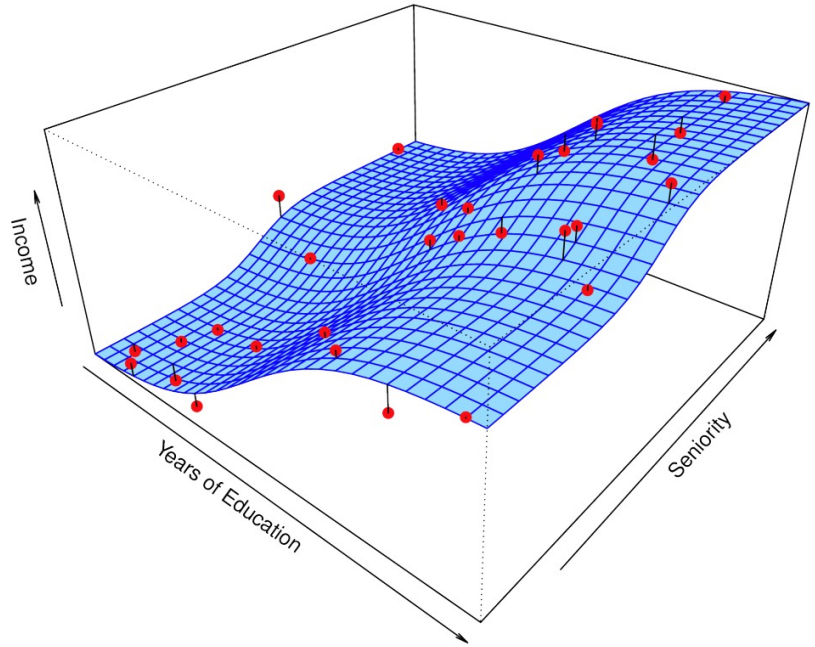
Bias-variance tradeoff

Regression setting

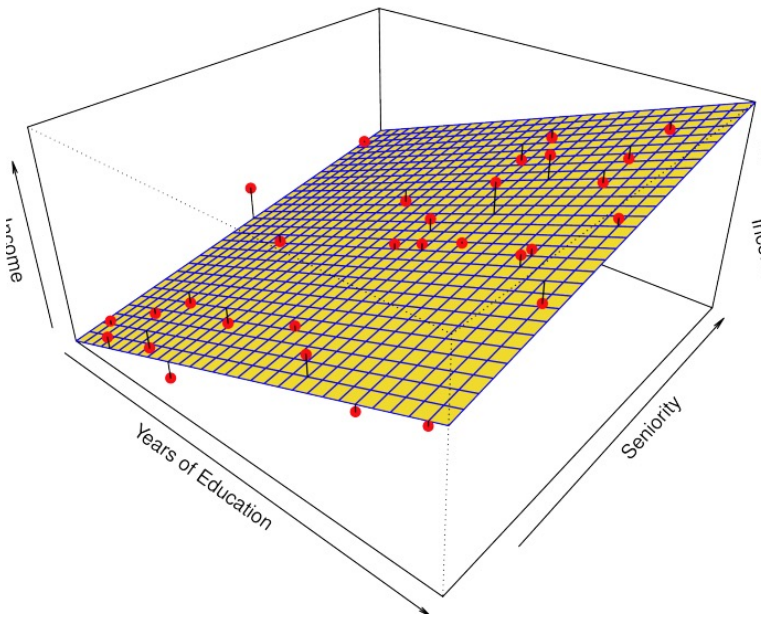
$$Y = f(X_1, \ldots, X_p) + \varepsilon$$

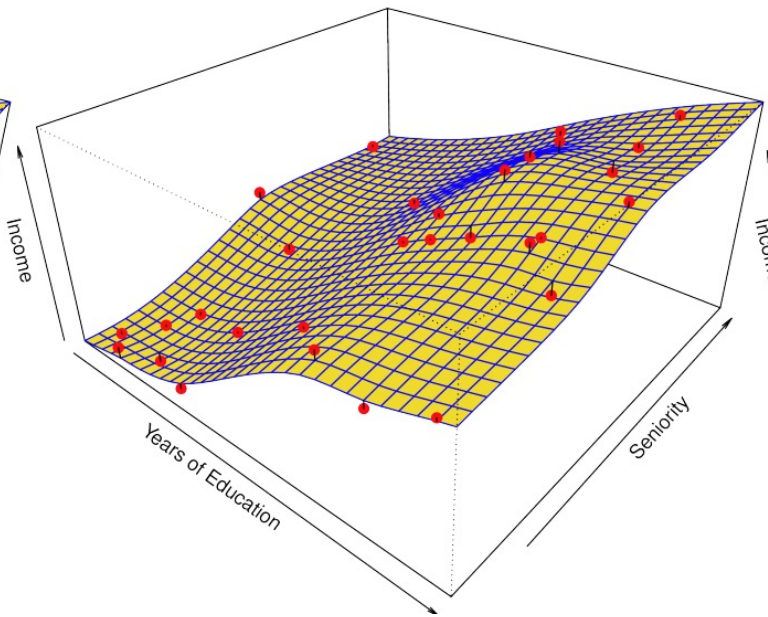


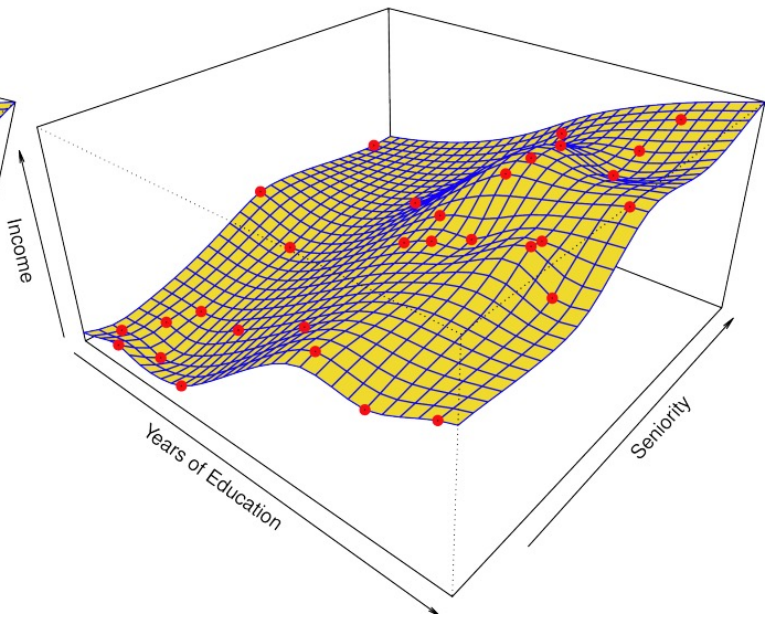
- Different statistical methods give different f()
- Each function/model has different flexibility & complexity



model flexibility/complexity

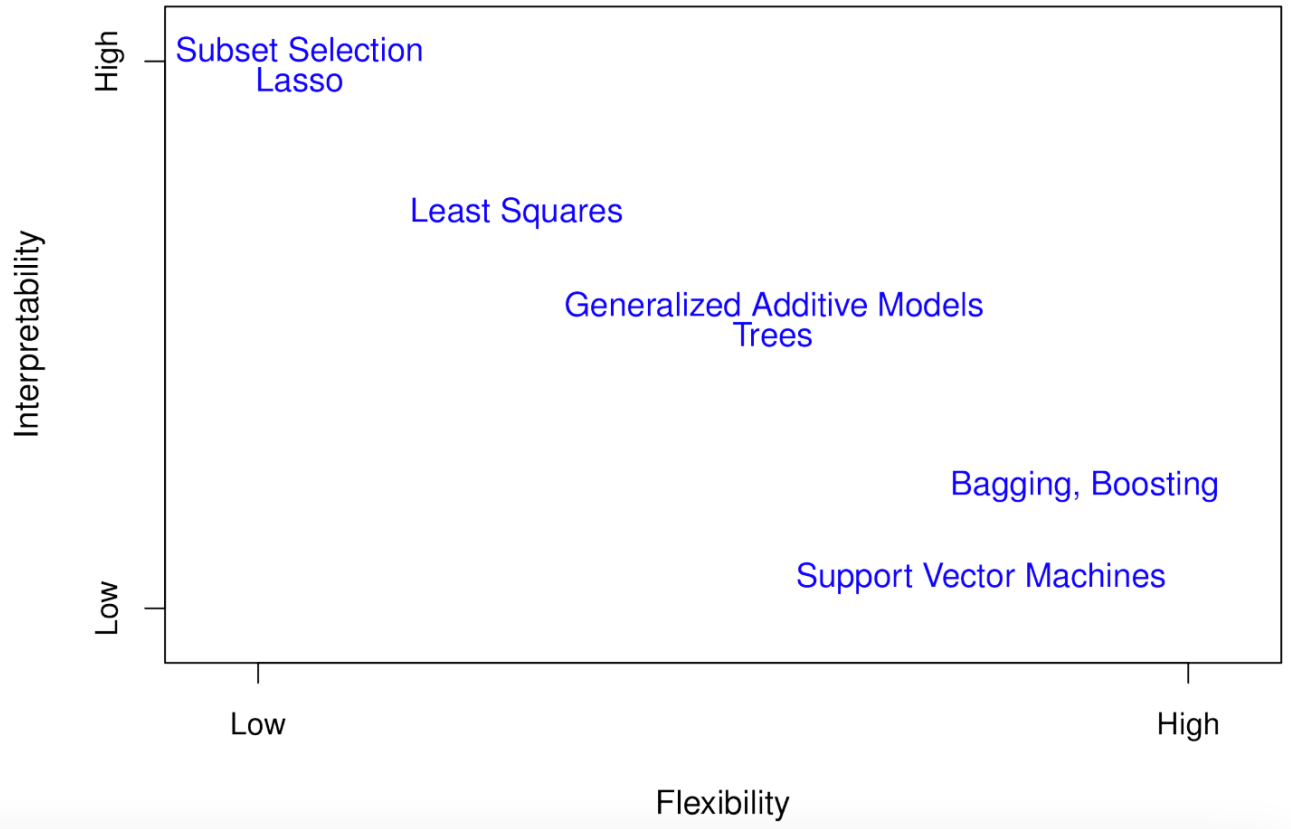






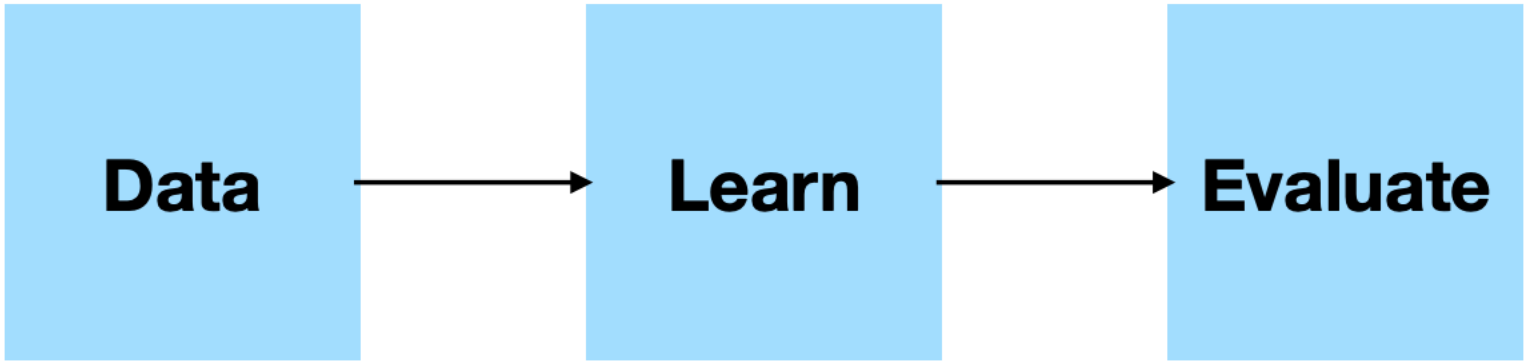
Model flexibility vs. interpretability

- Why should we ever choose a less flexible method over a more flexible one?
- Better **interpretability** (this slide) and, sometimes, smaller **error** (next)



Regression setting

$$Y = f(X_1, \ldots, X_p) + \varepsilon$$



Training set

$\{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \ldots, (\mathbf{x}_n, y_n)\}$

$\hat{f}$

how well does $\hat{f}$ learn?

- No single method is the best choice for all data sets
- Selecting a good statistical learning method depends on **evaluating** its performance

$$Y = f(X_1, \ldots, X_p) + \varepsilon$$



Training set

$\{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \ldots, (\mathbf{x}_n, y_n)\}$

$\hat{f}$

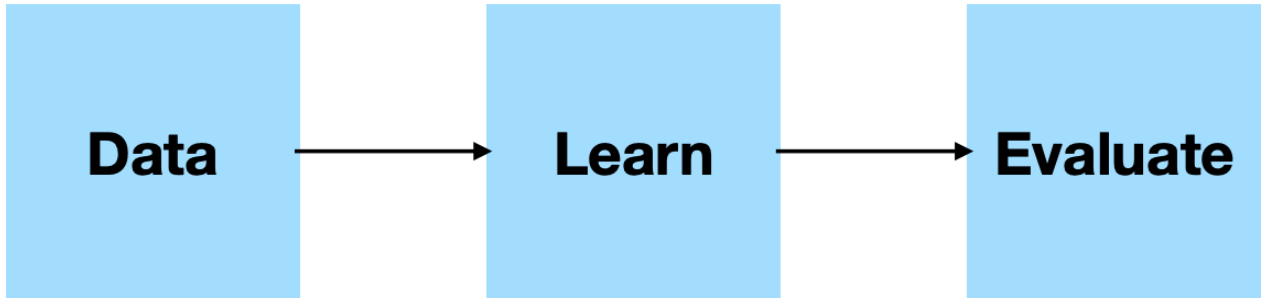
how well does $\hat{f}$ learn?

$$\text{Training MSE} = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{f}(\mathbf{x}_i)\right)^2$$

prediction that $\hat{f}$ gives for the ith training observation

Training Mean Squared Error

$$Y = f(X_1, \ldots, X_p) + \varepsilon$$



Training set

$\{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \ldots, (\mathbf{x}_n, y_n)\}$

$\hat{f}$

how well does $\hat{f}$ learn?

We are less interested in how $\hat{f}$ performs on training set

Consider test set $\{(\tilde{\mathbf{x}}_1, \tilde{y}_1), (\tilde{\mathbf{x}}_2, \tilde{y}_2), \ldots, (\tilde{\mathbf{x}}_m, \tilde{y}_m)\}$, not seen or used to train $\hat{f}$

$$\text{Test MSE} = \frac{1}{m} \sum_{i=1}^{m} \left(\tilde{y}_i - \hat{f}(\tilde{\mathbf{x}}_i) \right)^2$$

prediction that $\hat{f}$ gives for the ith test data

Test Mean Squared Error

Training MSE vs. test MSE

Training set

$$\{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \ldots, (\mathbf{x}_n, y_n)\}$$

Test set

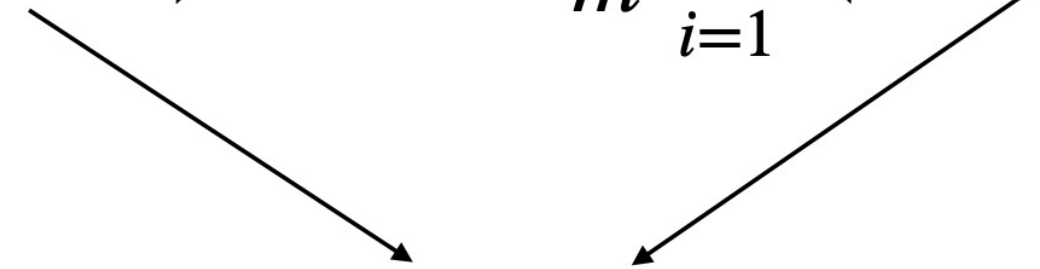
$$\{(\tilde{\mathbf{x}}_1, \tilde{y}_1), (\tilde{\mathbf{x}}_2, \tilde{y}_2), \ldots, (\tilde{\mathbf{x}}_m, y_m)\}$$

Training MSE

$$\frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{f}(\mathbf{x}_i)\right)^2$$

Test MSE

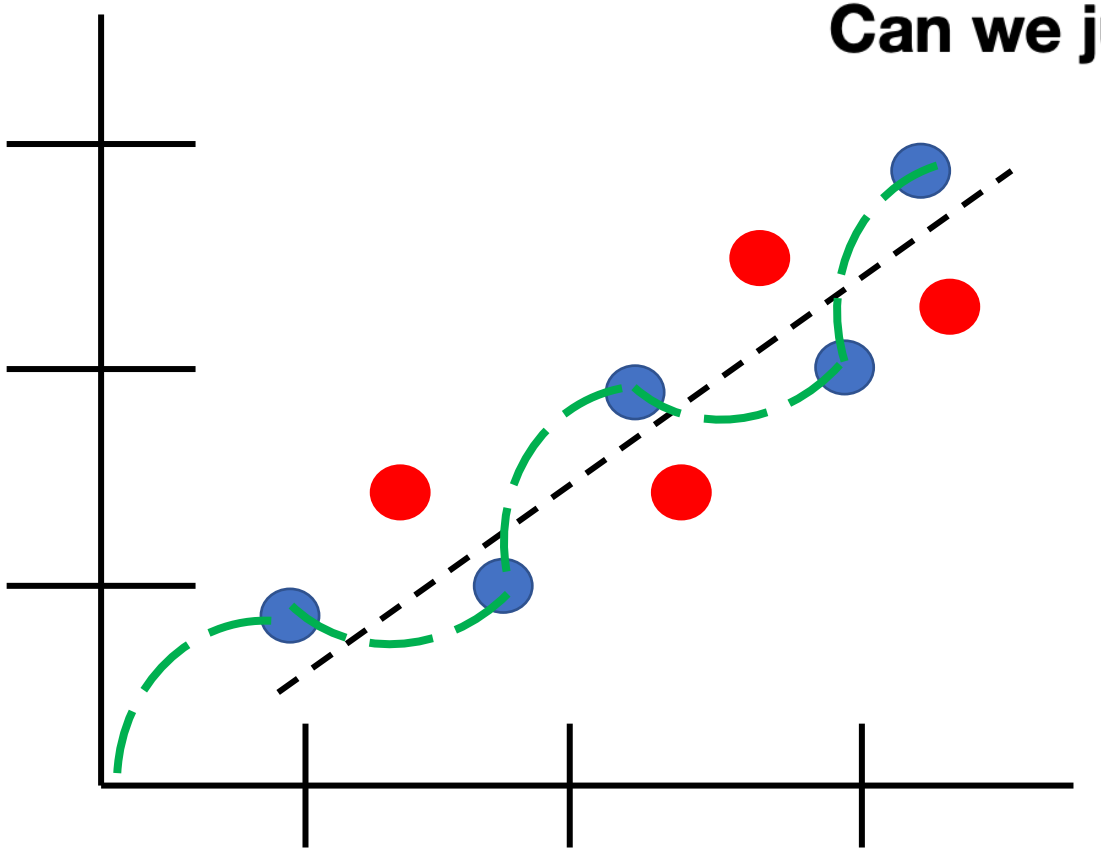
$$\frac{1}{m}\sum_{i=1}^{m}\left(\tilde{y}_i - \hat{f}(\tilde{\mathbf{x}}_i)\right)^2$$



$\hat{f}$ **is obtained on the training set!**

We want $\hat{f}$ for which the test MSE is as small as possible

Can we just select $\hat{f}$ that minimizes Training MSE



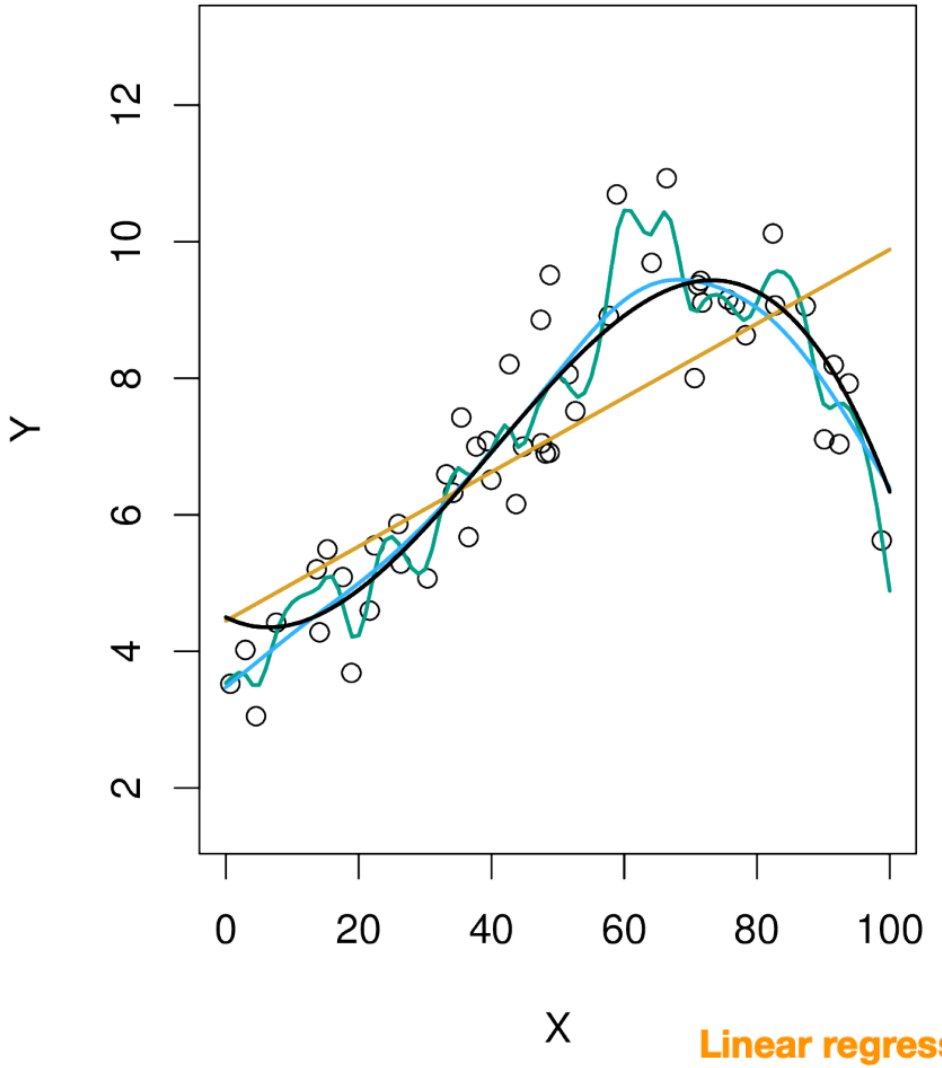
$$\frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{f}(\mathbf{x}_i)\right)^2$$

?

NO!

A method that has lowest training MSE does not necessarily imply that it also has the lowest test MSE

Lowest training MSE ≠ lowest test MSE



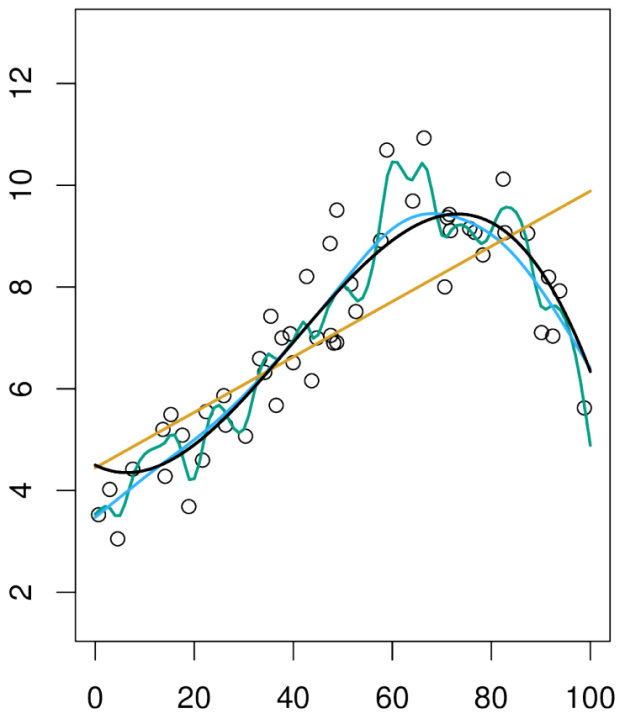
Higher model flexibility →
matches data points better

Higher model flexibility →
more wiggly curve

Higher model flexibility ↛
better fit to the true f

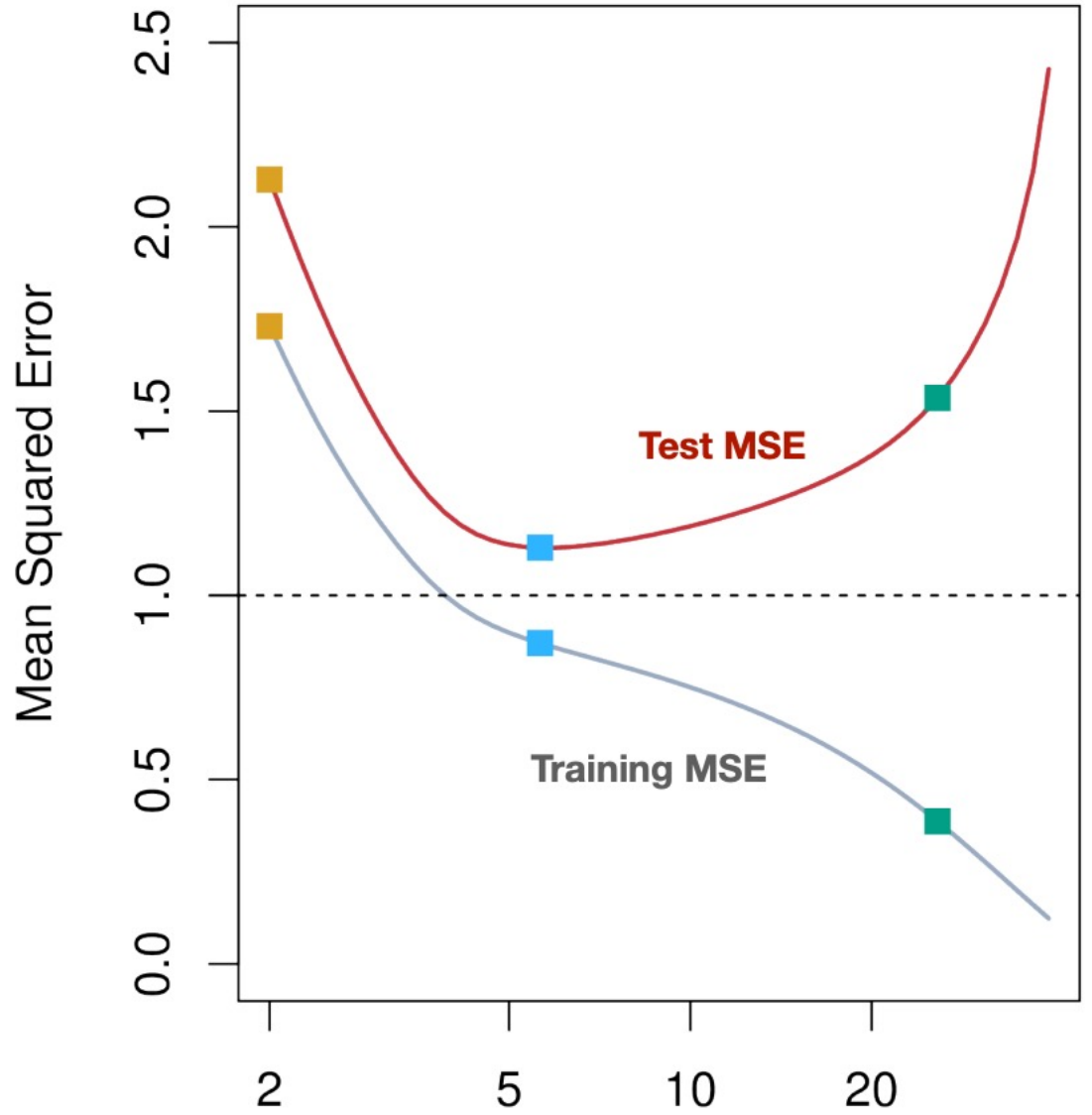
Linear regression **True f** **smoothing spline (simple)** **smoothing spline (complicated)**

level of model flexibility



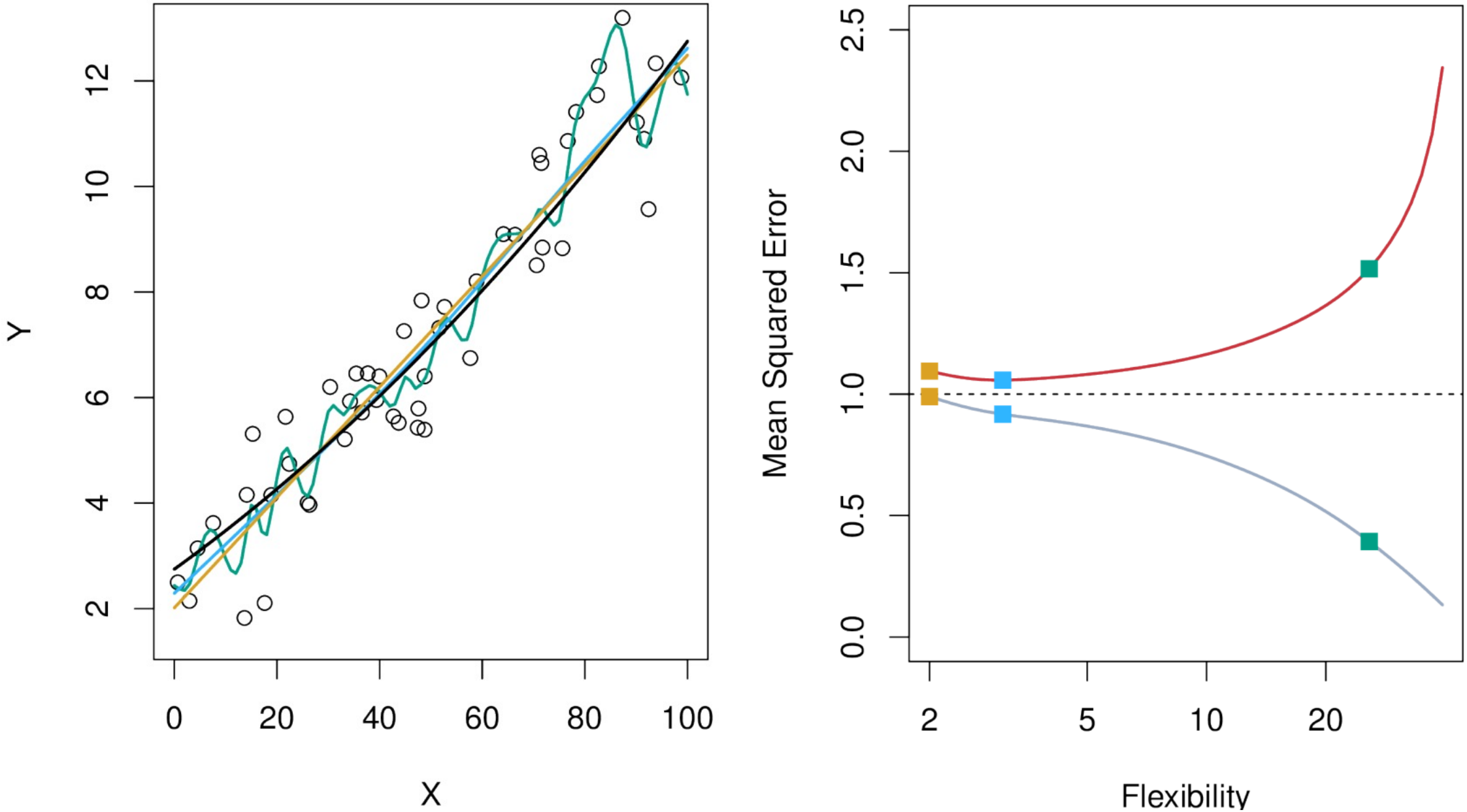
Higher model flexibility → matches data points better (≈lower training MSE)

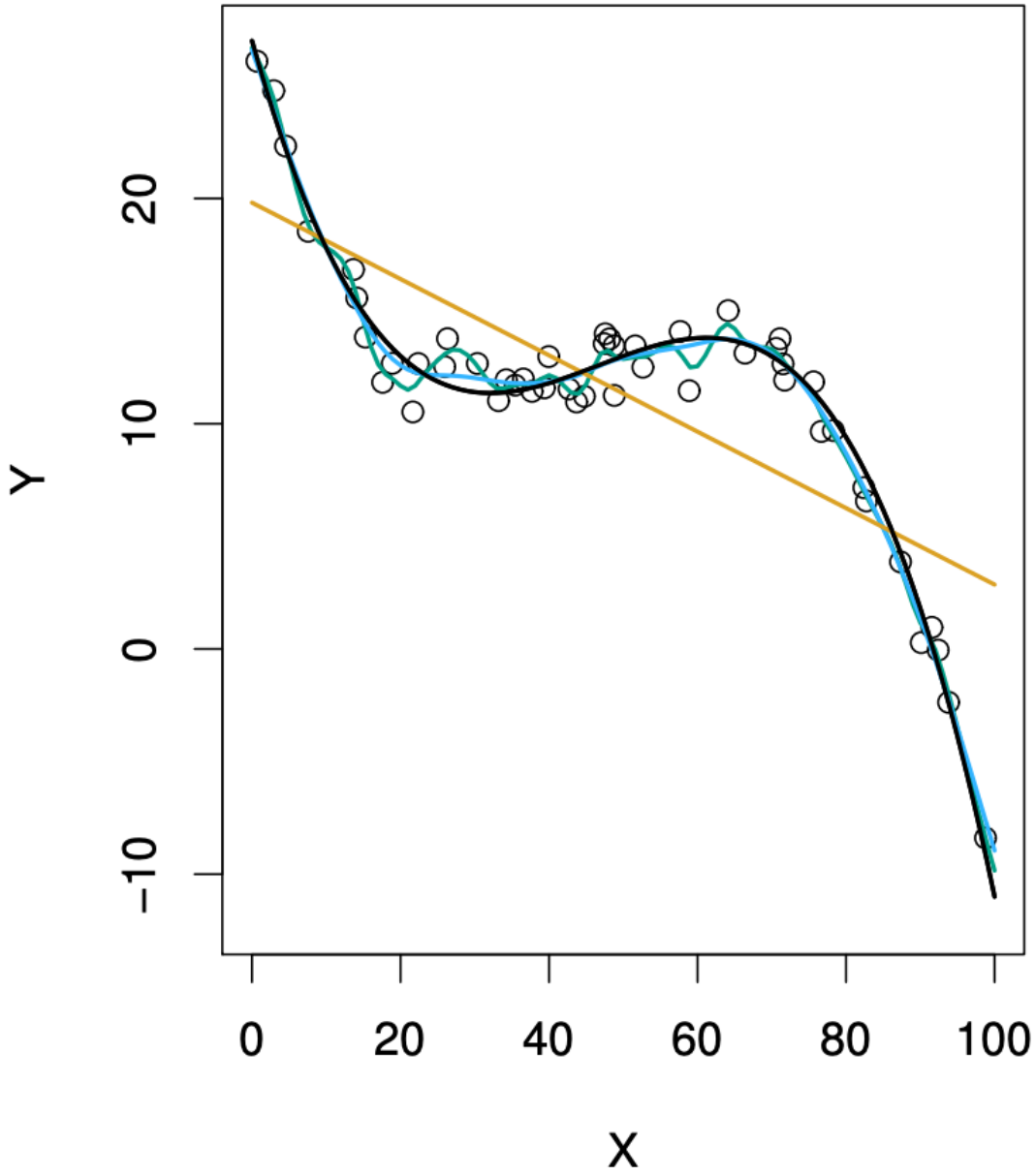
Higher model flexibility ↛ better fit to the true f (≈lower test MSE)

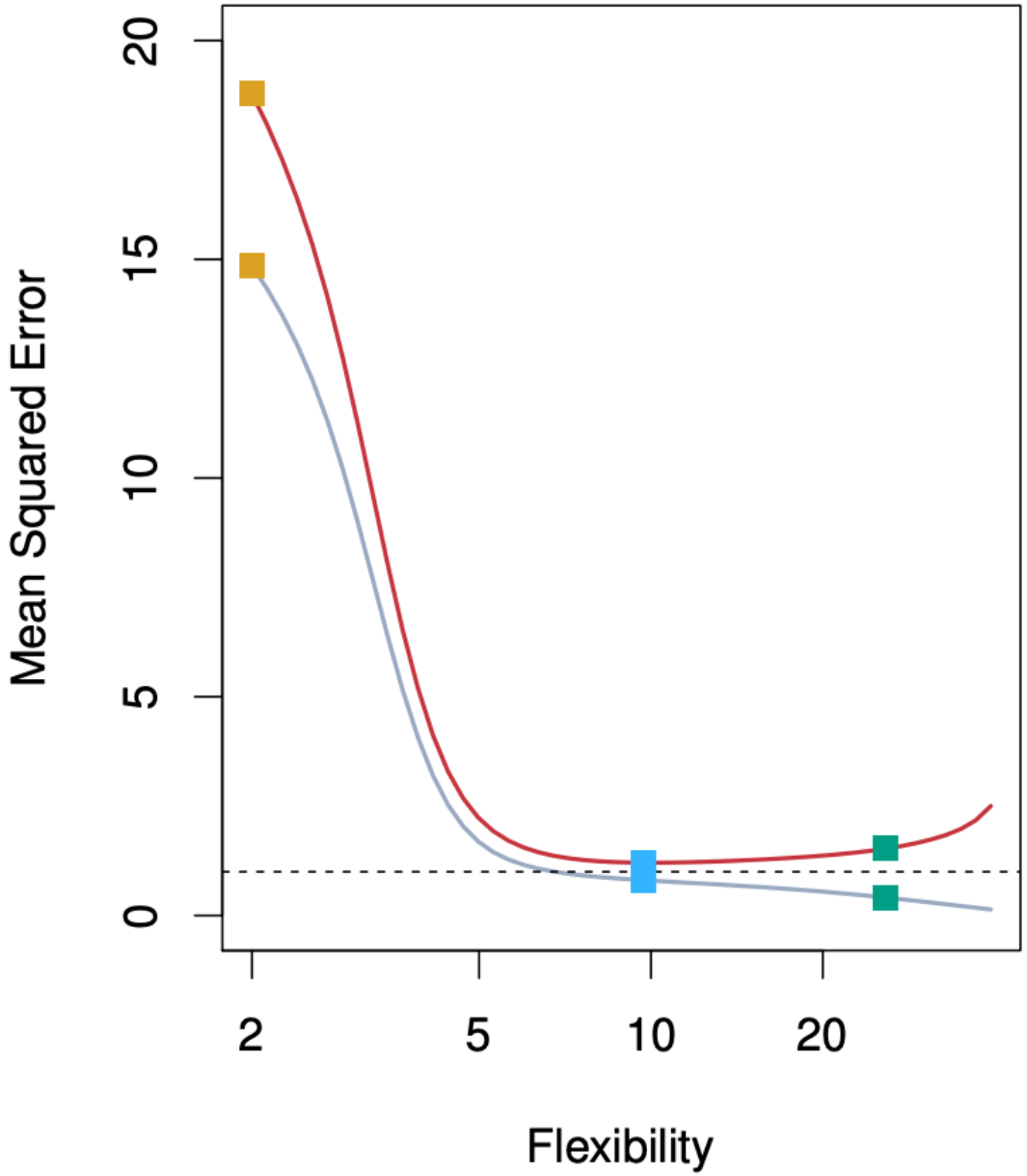


Linear regression True f smoothing spline (simple) smoothing spline (complicated)

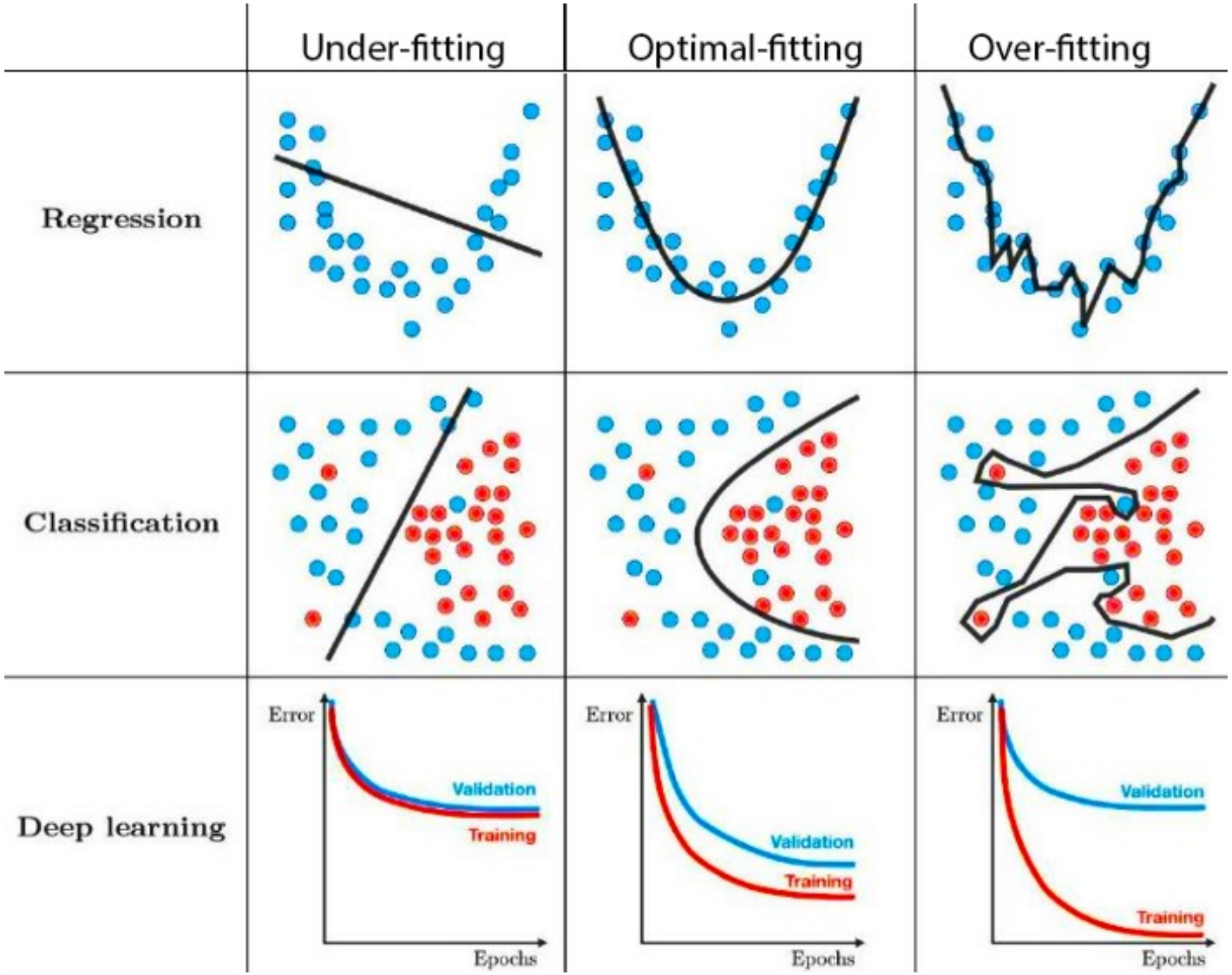
level of model flexibility

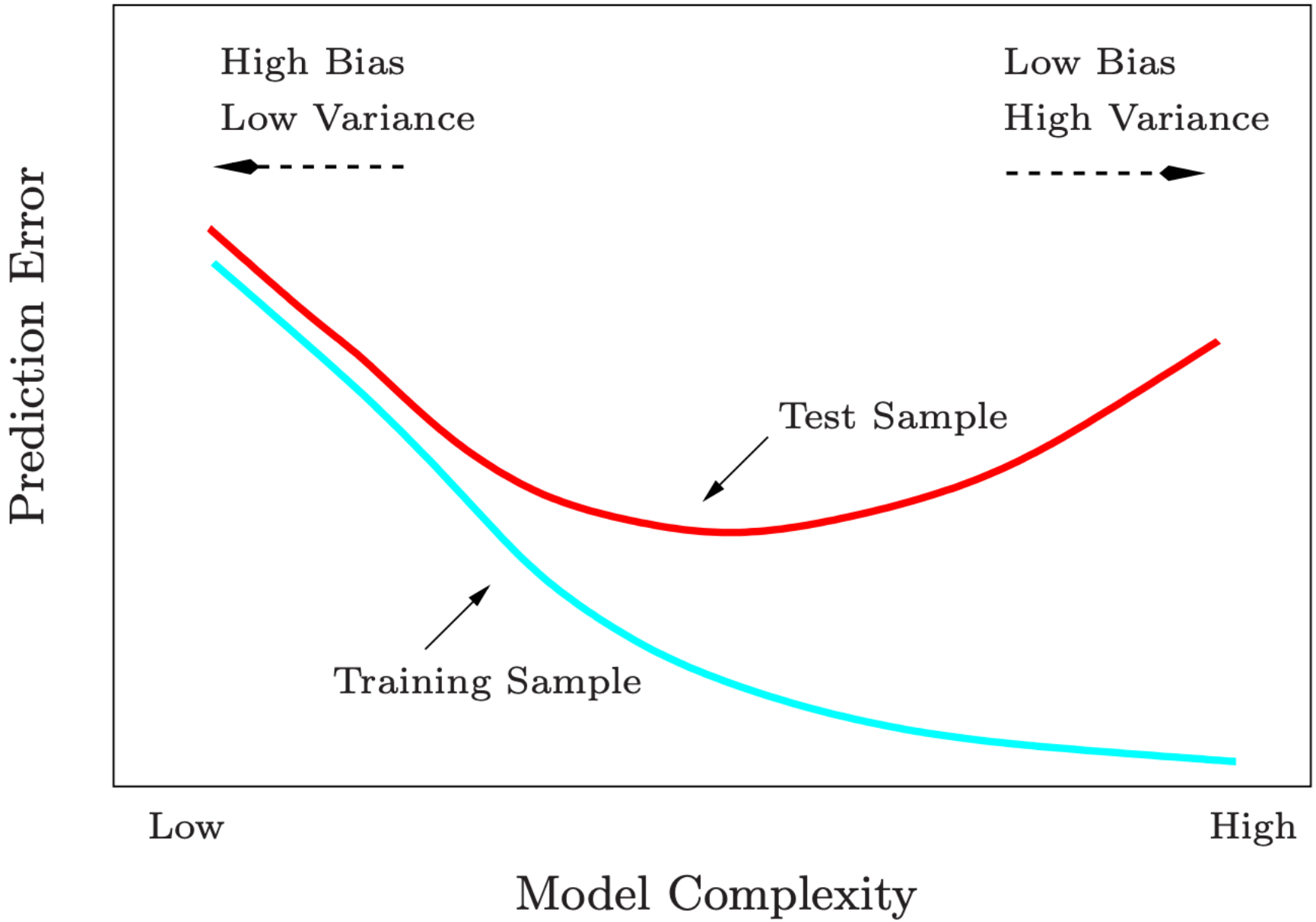

Y
X
0
20
40
60
80
100
2
4
6
8
10
12
Mean Squared Error
Flexibility
0.0
0.5
1.0
1.5
2.0
2.5
2
5
10
20
Linear regression
True f
smoothing spline (simple)
smoothing spline (complicated)
level of model flexibility


Y
20
10
0
−10
0
20
40
60
80
100
X

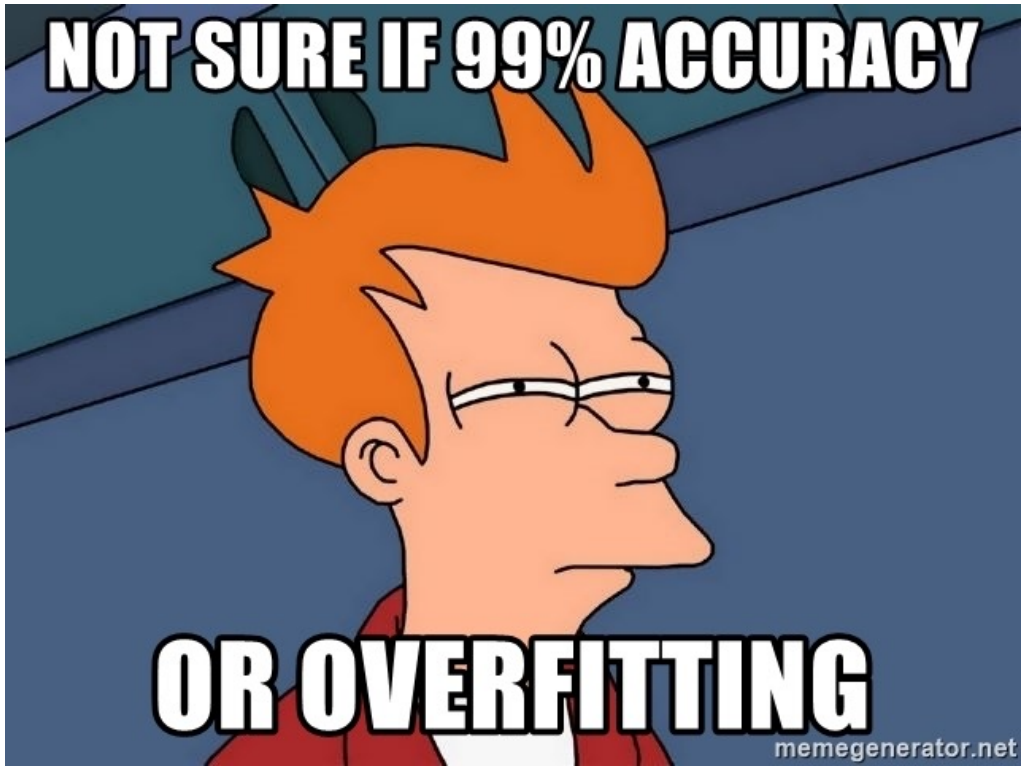

Mean Squared Error
20
15
10
5
0
2
5
10
20
Flexibility

Overfitting = Small train MSE, large test MSE




High Bias
Low Variance
Low Bias
High Variance
Prediction Error
Test Sample
Training Sample
Low
High
Model Complexity


THE BEST WAY TO
EXPLAIN OVERFITTING


NOT SURE IF 99% ACCURACY
OR OVERFITTING
memegenerator.net

Estimating test MSE

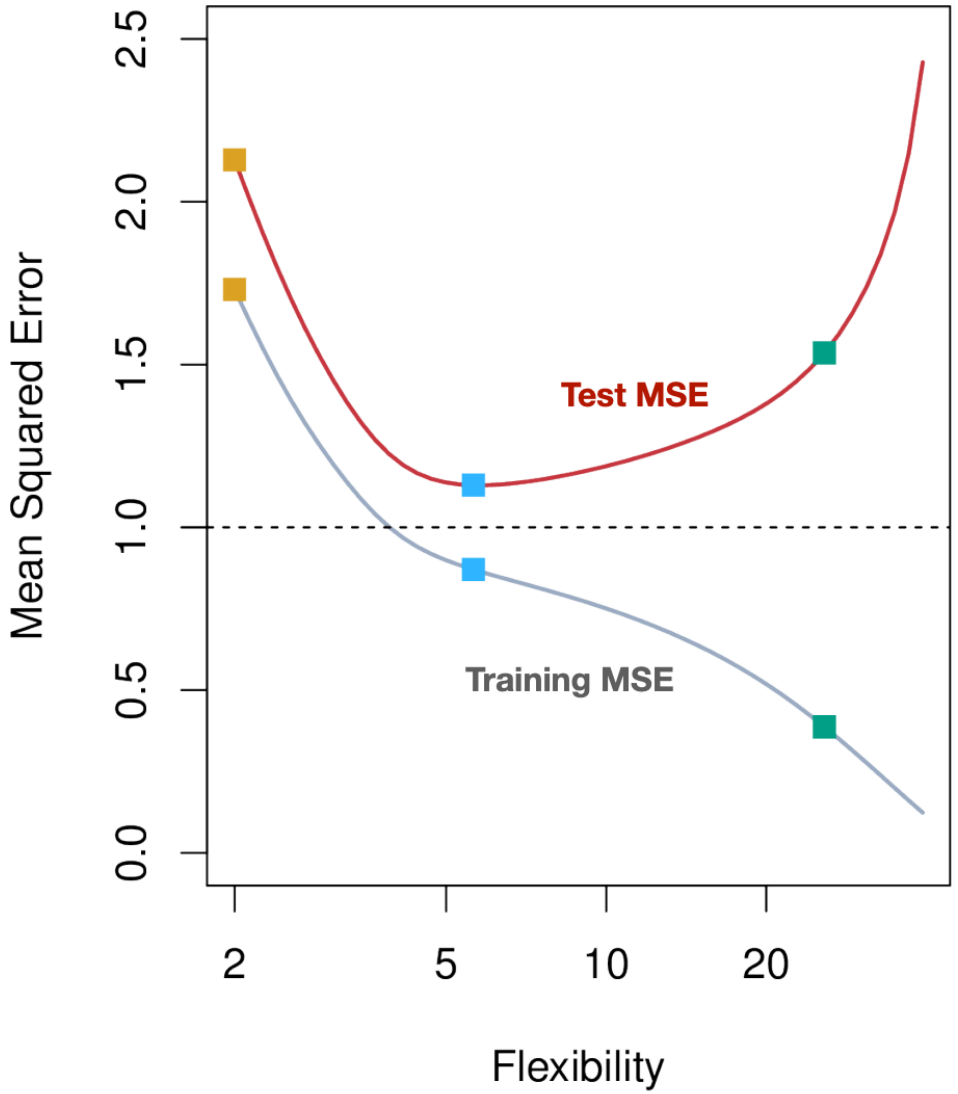
- Computing training MSE is easy – test MSE, not so much
  - Either testing data is not available or it is, but we don't want to use it too much
  - It would just become another training set!

- What do we do?

Cross-validation:

a method for estimating test MSE using only training data!

Bias-variance decomposition



Bias-variance decomposition

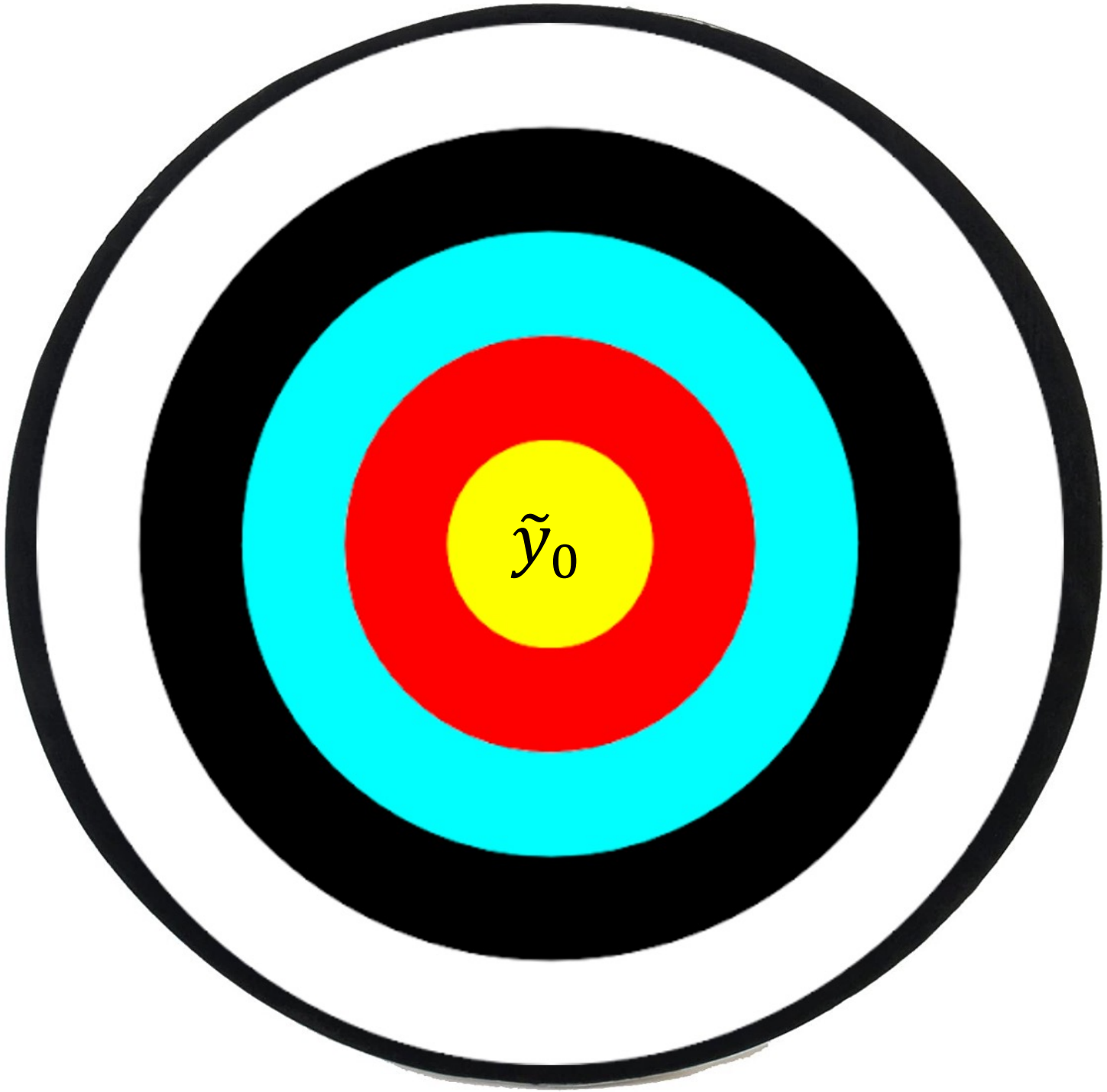
$$E\left[\left(\tilde{y}_0 - \hat{f}(\tilde{\boldsymbol{x}}_{\boldsymbol{0}})\right)^2\right] = Var\left(\hat{f}(\tilde{\boldsymbol{x}}_{\boldsymbol{0}})\right) + \left(Bias(\hat{f}(\tilde{\boldsymbol{x}}_{\boldsymbol{0}})\right)^2 + Var(\varepsilon)$$

expected test MSE **variance on test data** **bias on test data** **irreducible error**

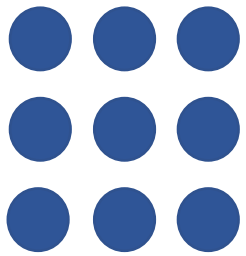
- $\hat{f}$ is learned from a training set
- $(\tilde{\boldsymbol{x}}_{\boldsymbol{0}}, \tilde{y}_0)$ is a test observation, independent of the training set, randomly sampled from the model

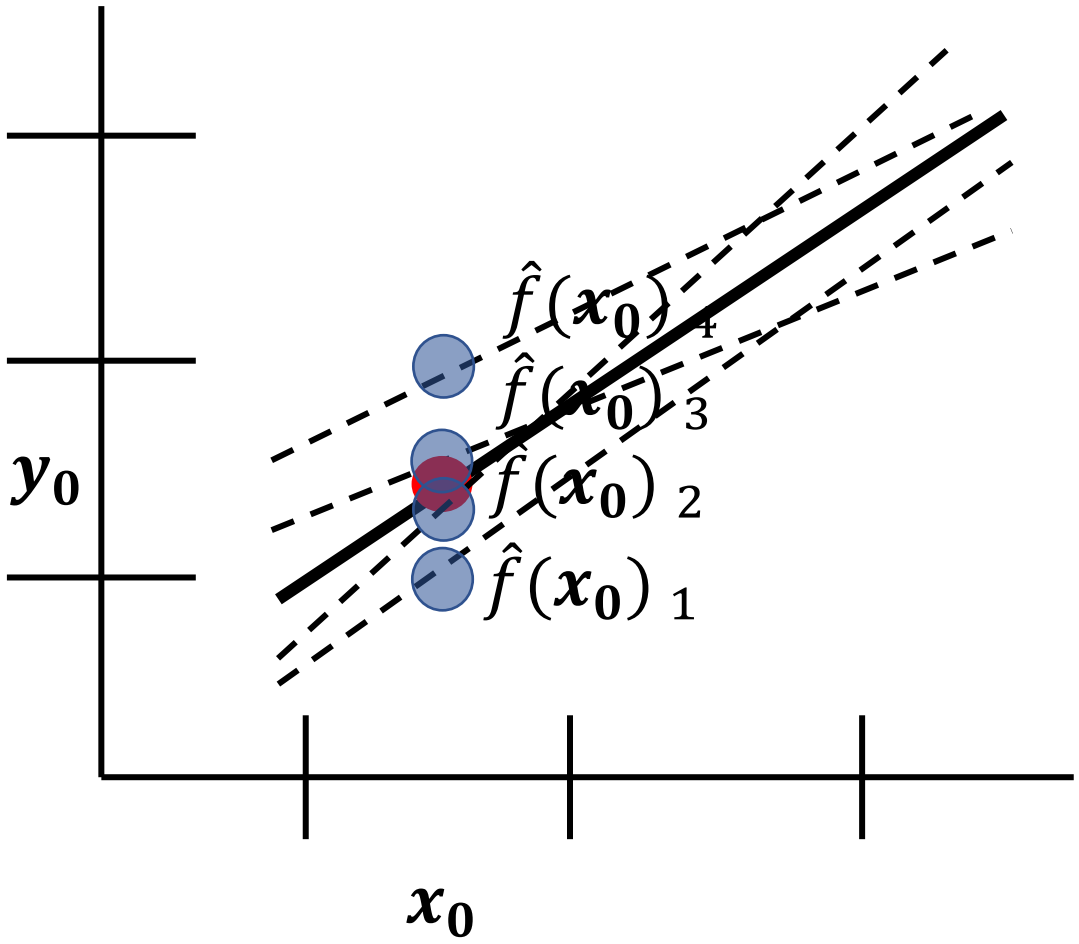
$$Y = f(X) + \varepsilon$$

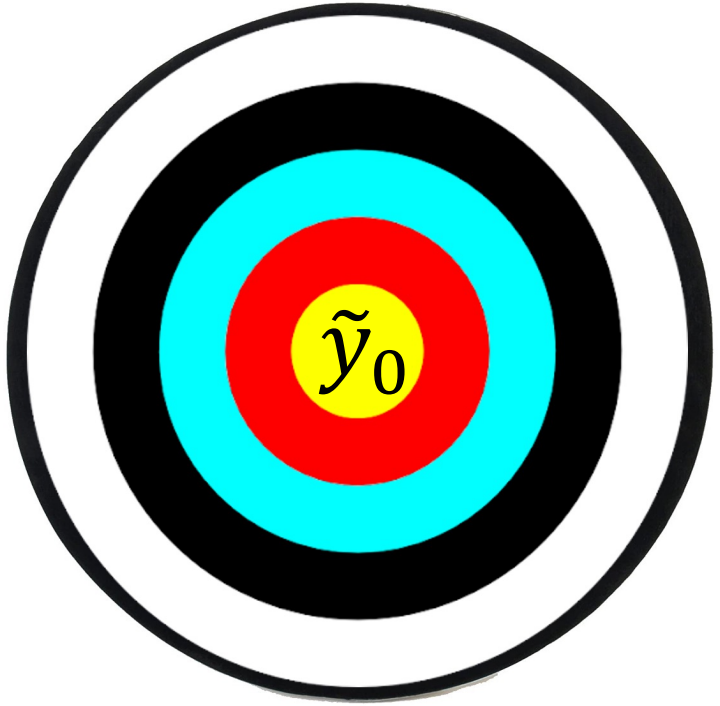
non-random **zero-mean noise**



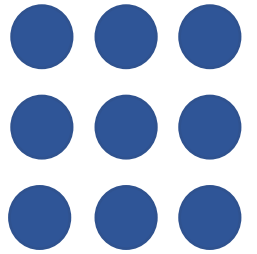
$\hat{f}(\tilde{\boldsymbol{x}}_0)$



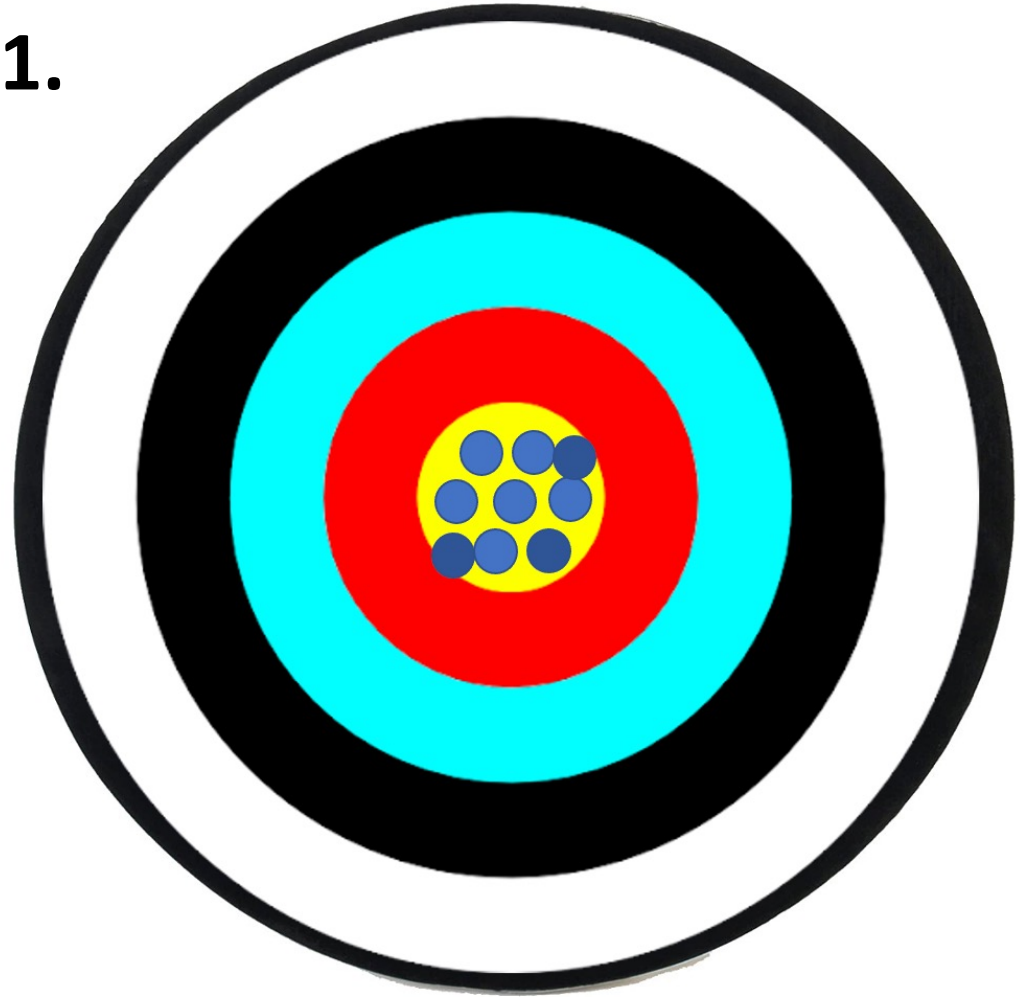




$\hat{f}(\tilde{x}_0)$



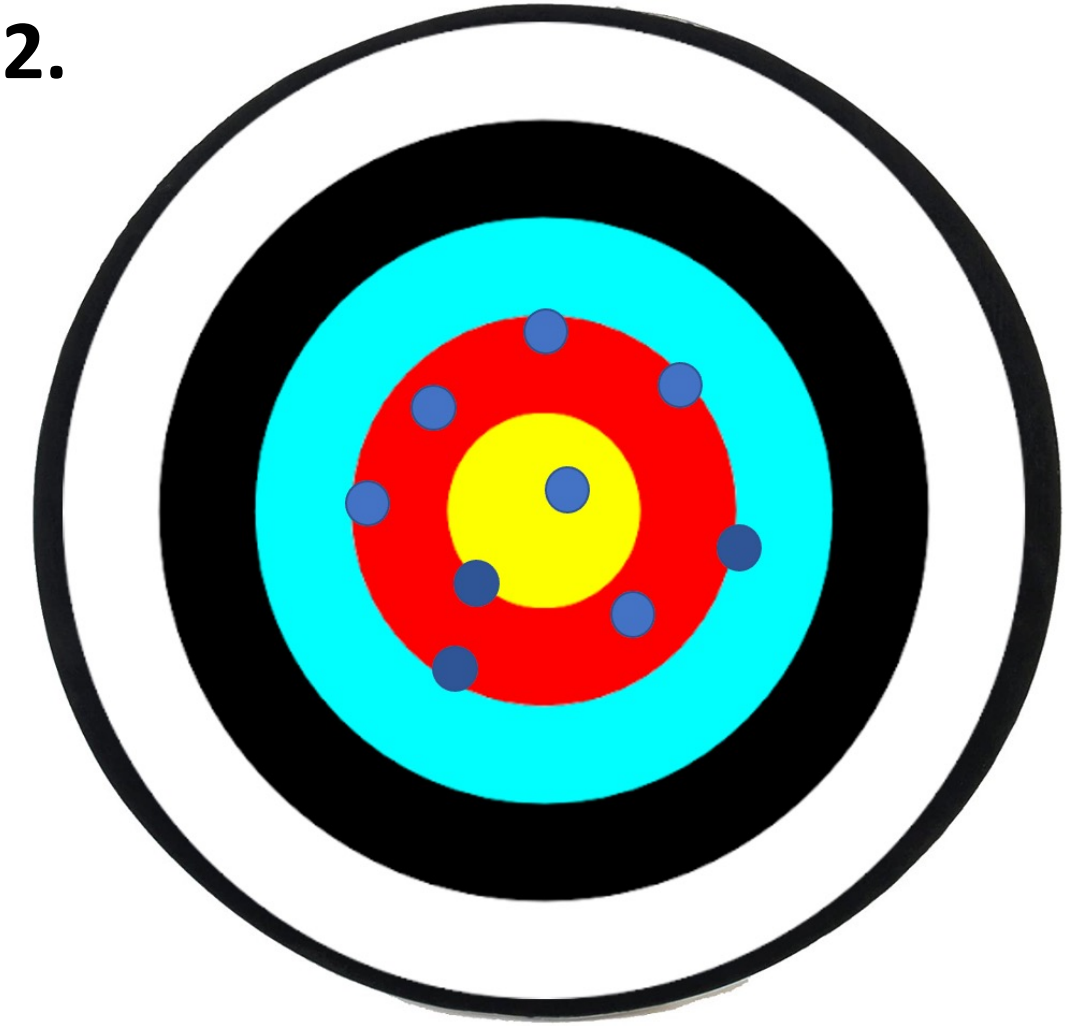
1.



$$E\left[\left(\tilde{y}_0 - \hat{f}(\tilde{\mathbf{x}}_\mathbf{0})\right)^2\right] = Var\left(\hat{f}(\tilde{\mathbf{x}}_\mathbf{0})\right) + \left(Bias(\hat{f}(\tilde{\mathbf{x}}_\mathbf{0}))\right)^2 + Var(\varepsilon)$$

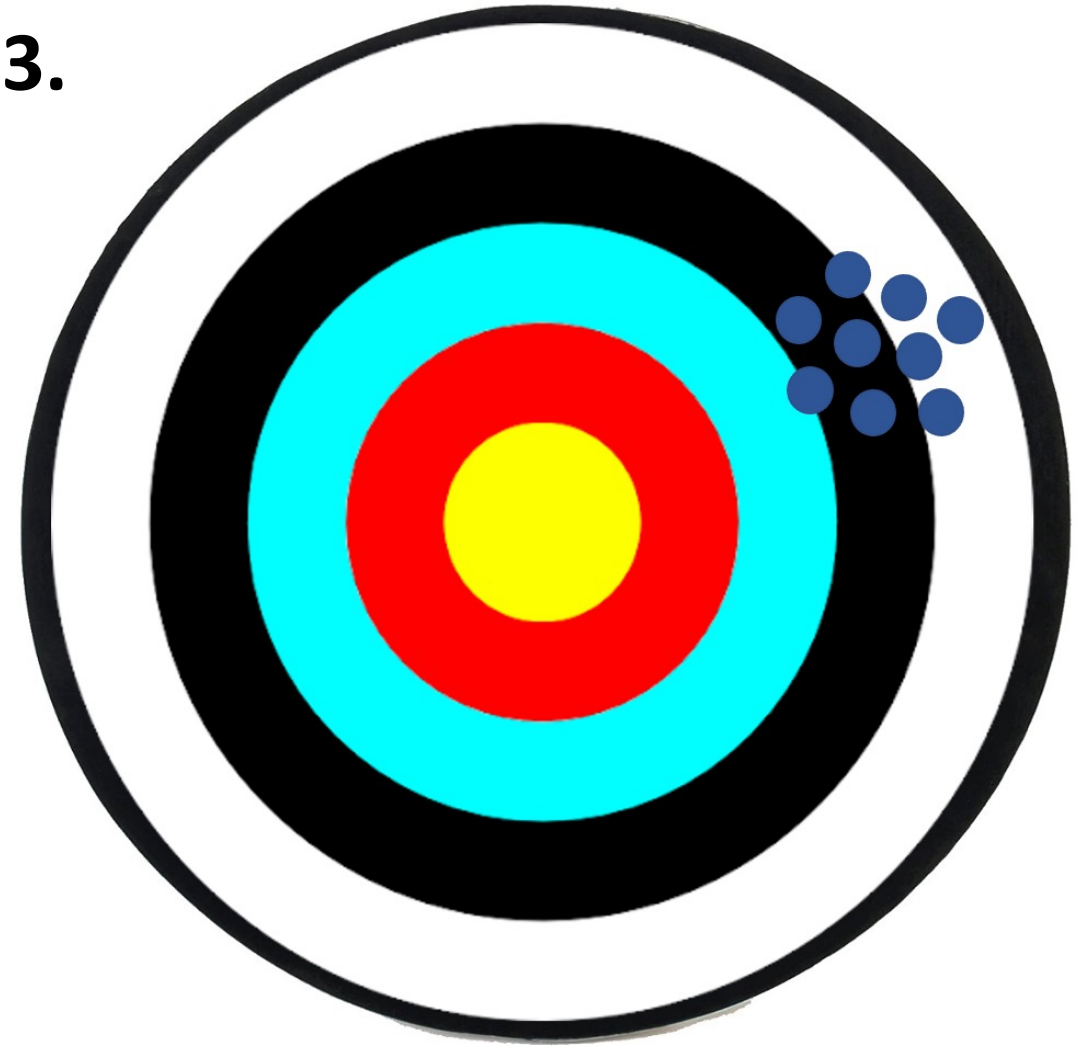
expected test MSE | **variance on test data** | **bias on test data** | **irreducible error**

2.



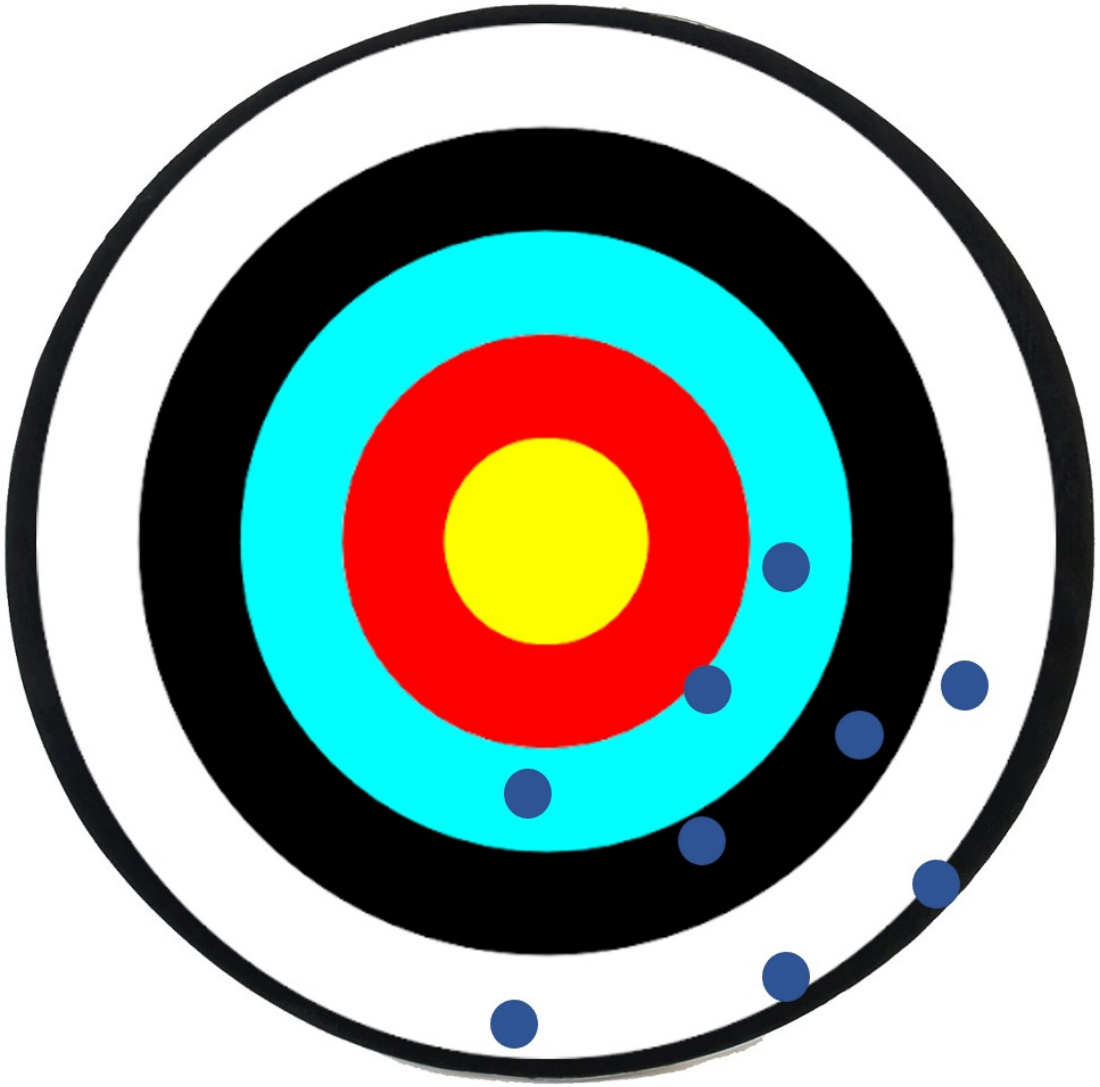
$$E\left[\left(\tilde{y}_0 - \hat{f}(\tilde{\boldsymbol{x}}_{\boldsymbol{0}})\right)^2\right] = Var\left(\hat{f}(\tilde{\boldsymbol{x}}_{\boldsymbol{0}})\right) + \left(Bias(\hat{f}(\tilde{\boldsymbol{x}}_{\boldsymbol{0}})\right)^2 + Var(\varepsilon)$$

3.



$$E\left[\left(\tilde{y}_0 - \hat{f}(\tilde{\mathbf{x}}_\mathbf{0})\right)^2\right] = \cancel{Var\left(\hat{f}(\tilde{\mathbf{x}}_\mathbf{0})\right)} + \left(Bias(\hat{f}(\tilde{\mathbf{x}}_\mathbf{0})\right)^2 + Var(\varepsilon)$$

4.



$$E\left[\left(\tilde{y}_0 - \hat{f}(\tilde{\boldsymbol{x}}_{\boldsymbol{0}})\right)^2\right] = Var\left(\hat{f}(\tilde{\boldsymbol{x}}_{\boldsymbol{0}})\right) + \left(Bias(\hat{f}(\tilde{\boldsymbol{x}}_{\boldsymbol{0}})\right)^2 + Var(\varepsilon)$$

Bias-variance decomposition

$$E\left[\left(\tilde{y}_0 - \hat{f}(\tilde{\mathbf{x}}_\mathbf{0})\right)^2\right] = Var\left(\hat{f}(\tilde{\mathbf{x}}_\mathbf{0})\right) + \left(Bias(\hat{f}(\tilde{\mathbf{x}}_\mathbf{0})\right)^2 + Var(\varepsilon)$$

Expected test MSE

Irreducible error

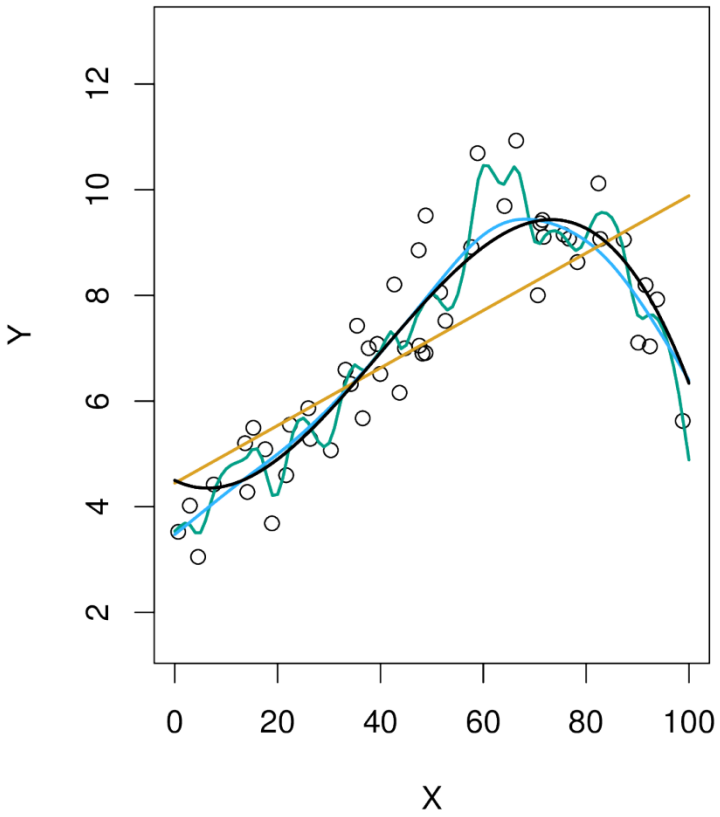
$$Y = f(X) + \varepsilon$$

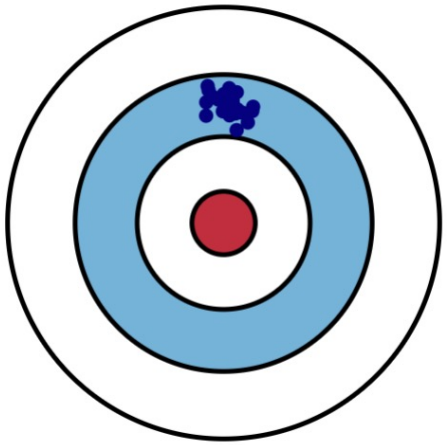
non-random **zero-mean noise**

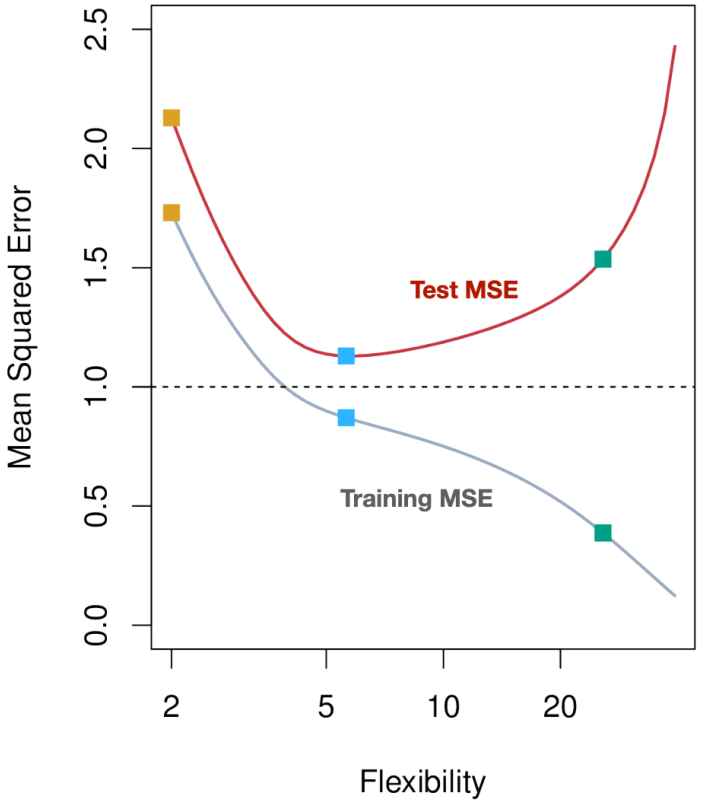
Bias

$$E\left[\left(\tilde{y}_0 - \hat{f}(\tilde{\boldsymbol{x}}_0)\right)^2\right] = Var\left(\hat{f}(\tilde{\boldsymbol{x}}_0)\right) + \left(Bias(\hat{f}(\tilde{\boldsymbol{x}}_0)\right)^2 + Var(\varepsilon)$$

$$Bias(\hat{f}(\tilde{\boldsymbol{x}}_0) = E\left[\hat{f}(\tilde{\boldsymbol{x}}_0)\right] - f(\tilde{\boldsymbol{x}}_0)$$



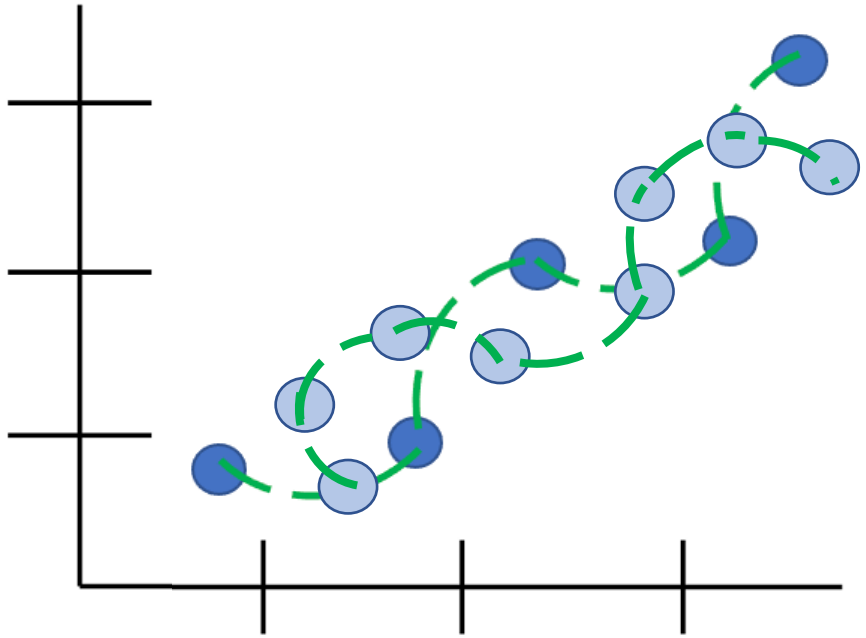


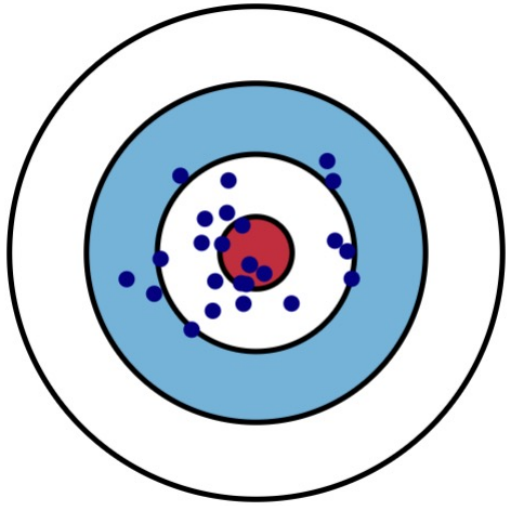


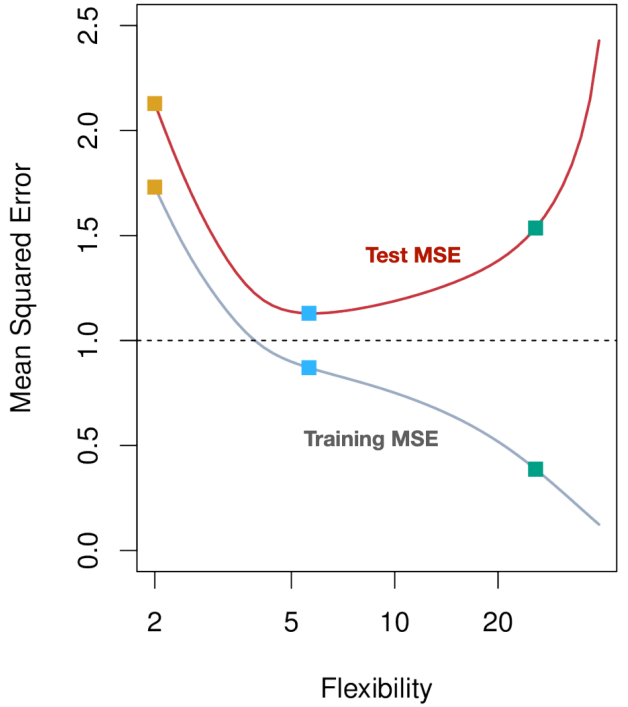
Variance

$$E\left[\left(\tilde{y}_0 - \hat{f}(\tilde{\boldsymbol{x}}_0)\right)^2\right] = Var\left(\hat{f}(\tilde{\boldsymbol{x}}_0)\right) + \left(Bias(\hat{f}(\tilde{\boldsymbol{x}}_0)\right)^2 + Var(\varepsilon)$$

$$Var\left(\hat{f}(\boldsymbol{x}_0)\right) = E\left[\left(\hat{f}(\boldsymbol{x}_0) - E[\hat{f}(\boldsymbol{x}_0)]\right)^2\right]$$

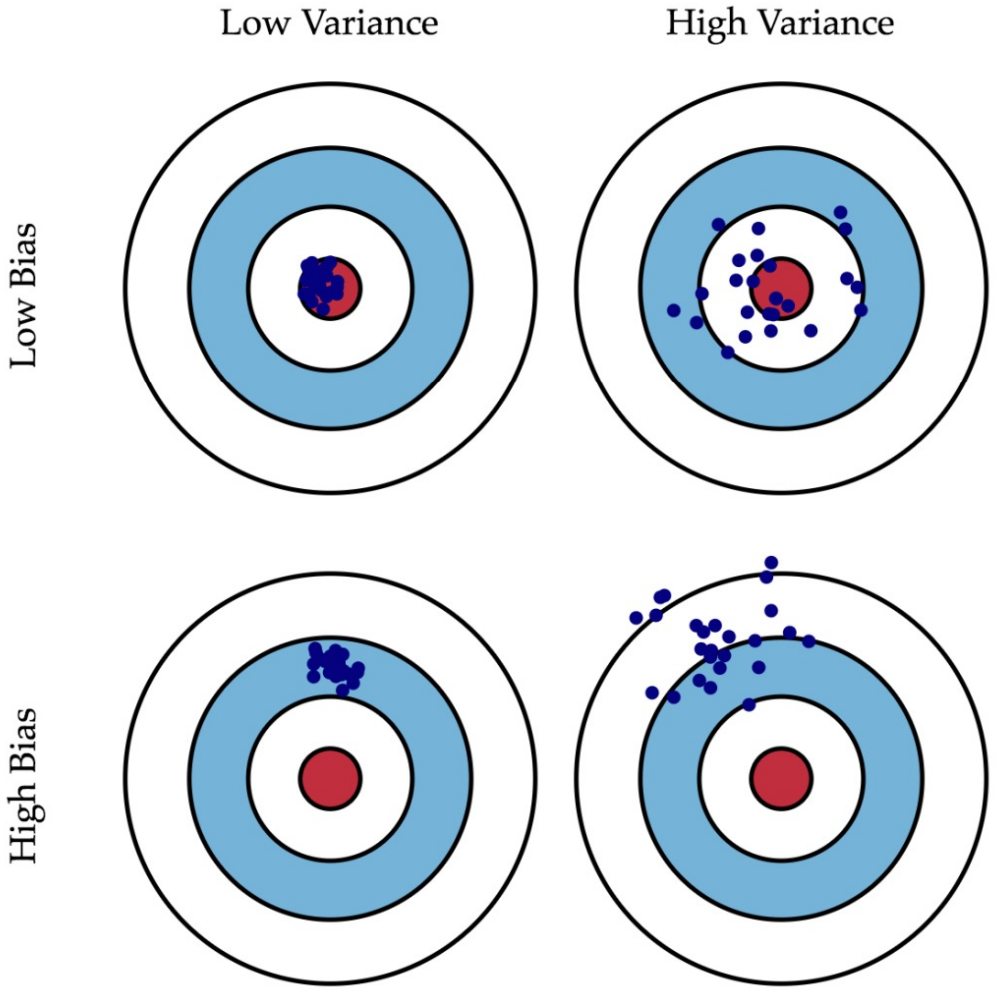






Bias-variance tradeoff

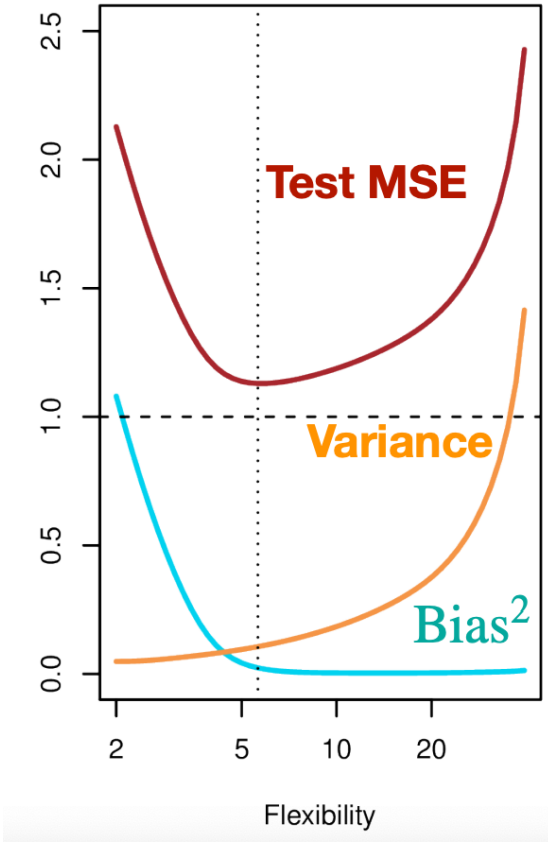
$$E\left[\left(\tilde{y}_0 - \hat{f}(\tilde{\boldsymbol{x}}_{\boldsymbol{0}})\right)^2\right] = Var\left(\hat{f}(\tilde{\boldsymbol{x}}_{\boldsymbol{0}})\right) + \left(Bias(\hat{f}(\tilde{\boldsymbol{x}}_{\boldsymbol{0}})\right)^2 + Var(\varepsilon)$$



Bias-variance tradeoff

$$E\left[\left(\tilde{y}_0 - \hat{f}(\tilde{\boldsymbol{x}}_0)\right)^2\right] = Var\left(\hat{f}(\tilde{\boldsymbol{x}}_0)\right) + \left(Bias(\hat{f}(\tilde{\boldsymbol{x}}_0)\right)^2 + Var(\varepsilon)$$

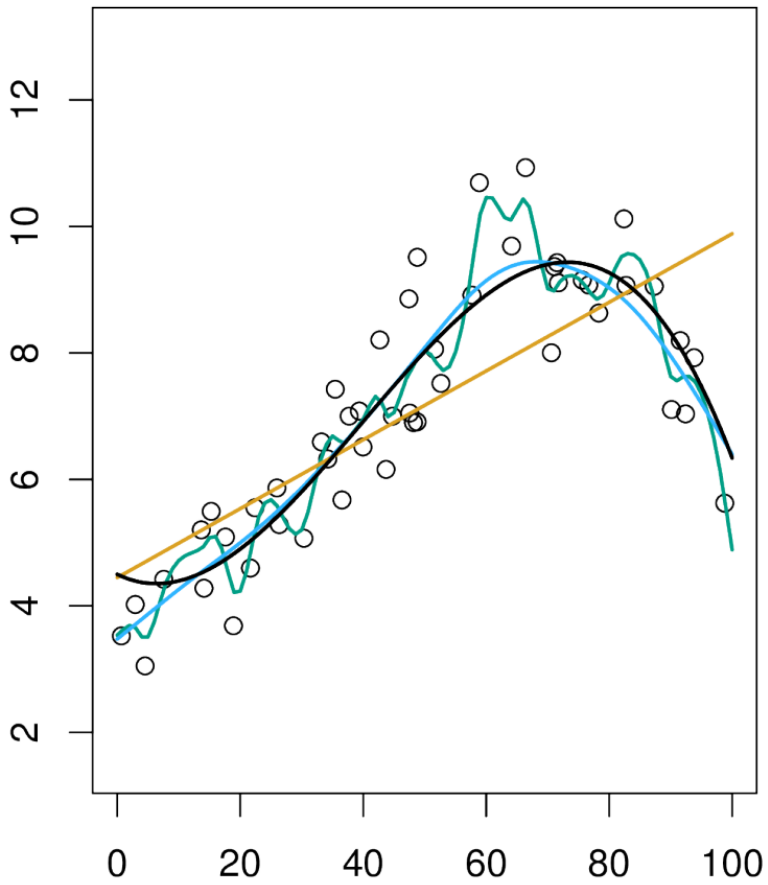
Want to **simultaneously** achieve **low variance** and **low bias**

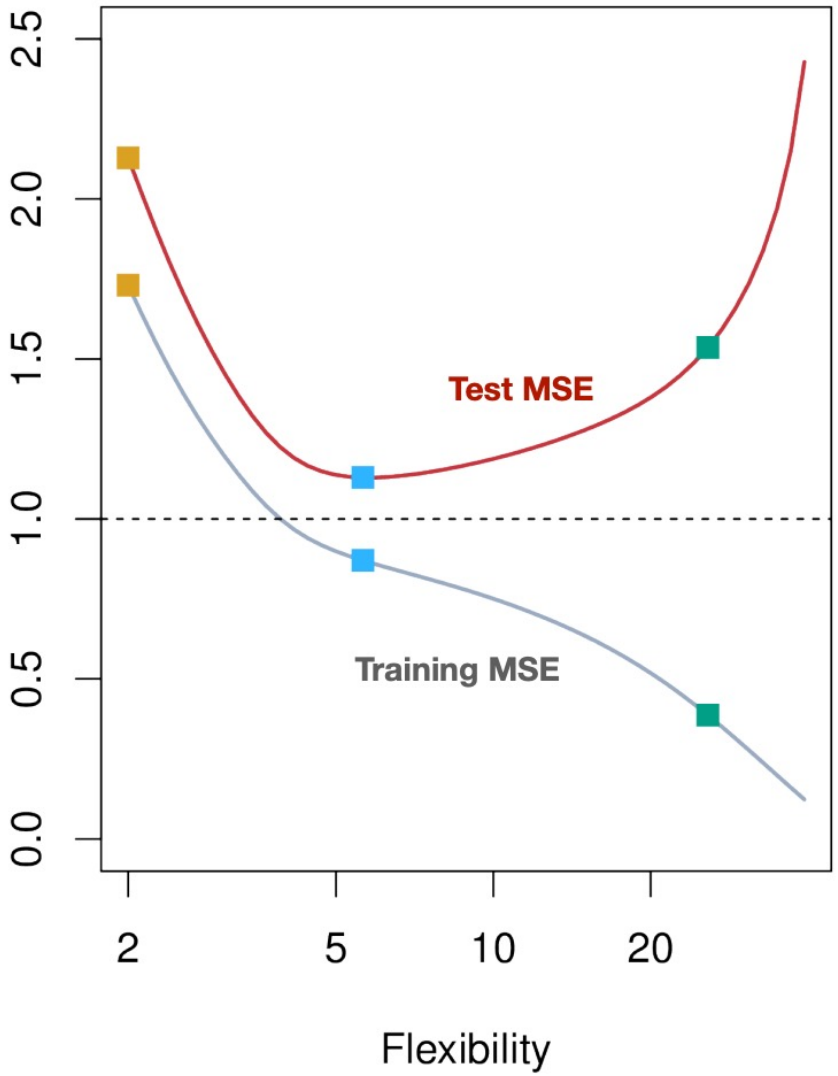


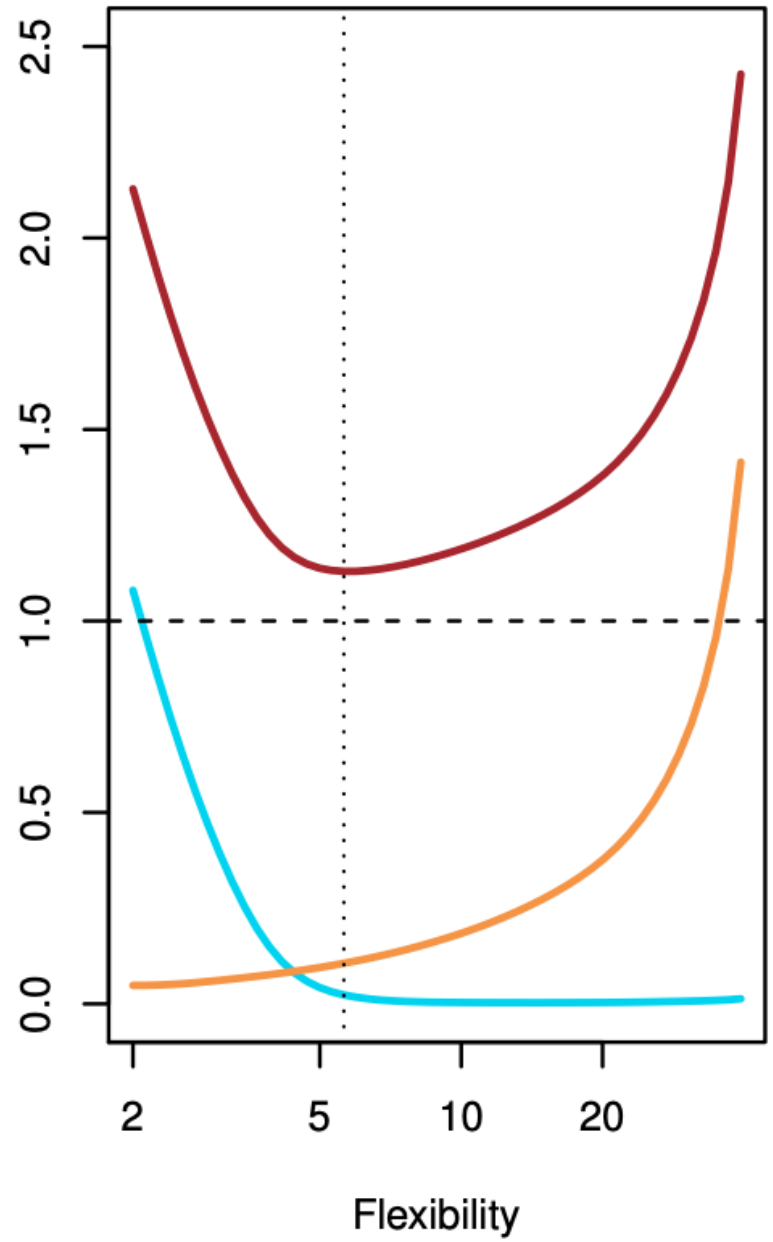
A simple model → high bias + low variance

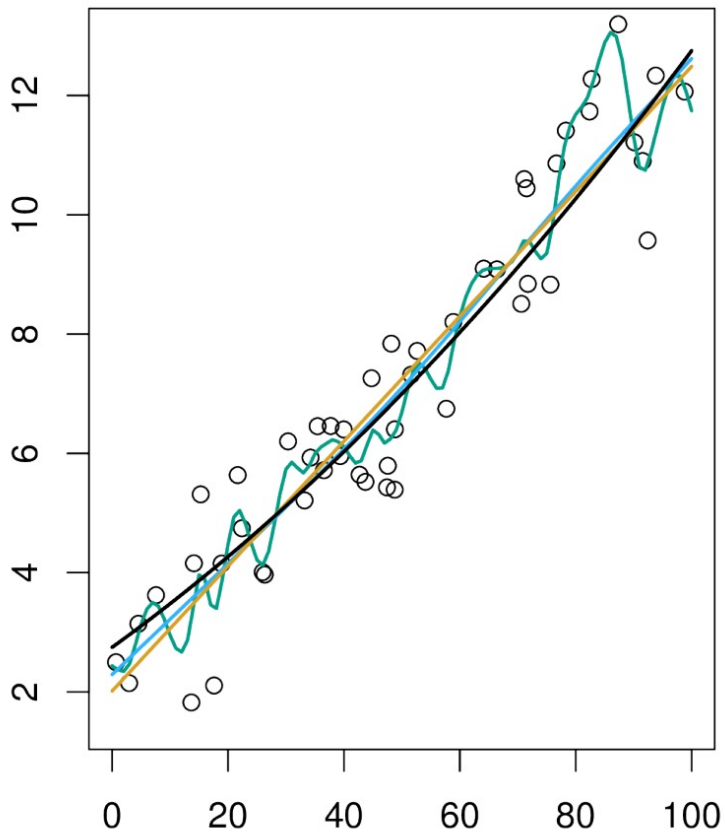
A flexible model → low bias + high variance

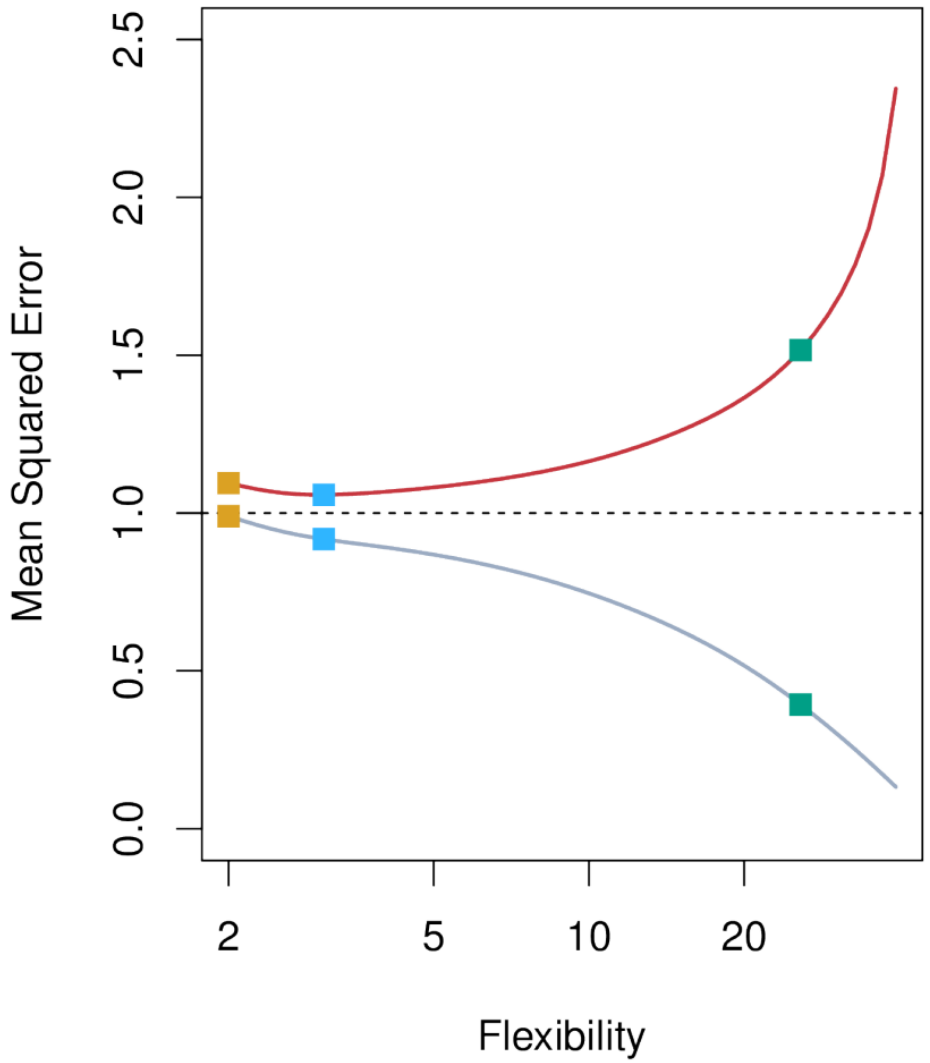
Challenge: find a method for which both the variance and (squared) bias are low

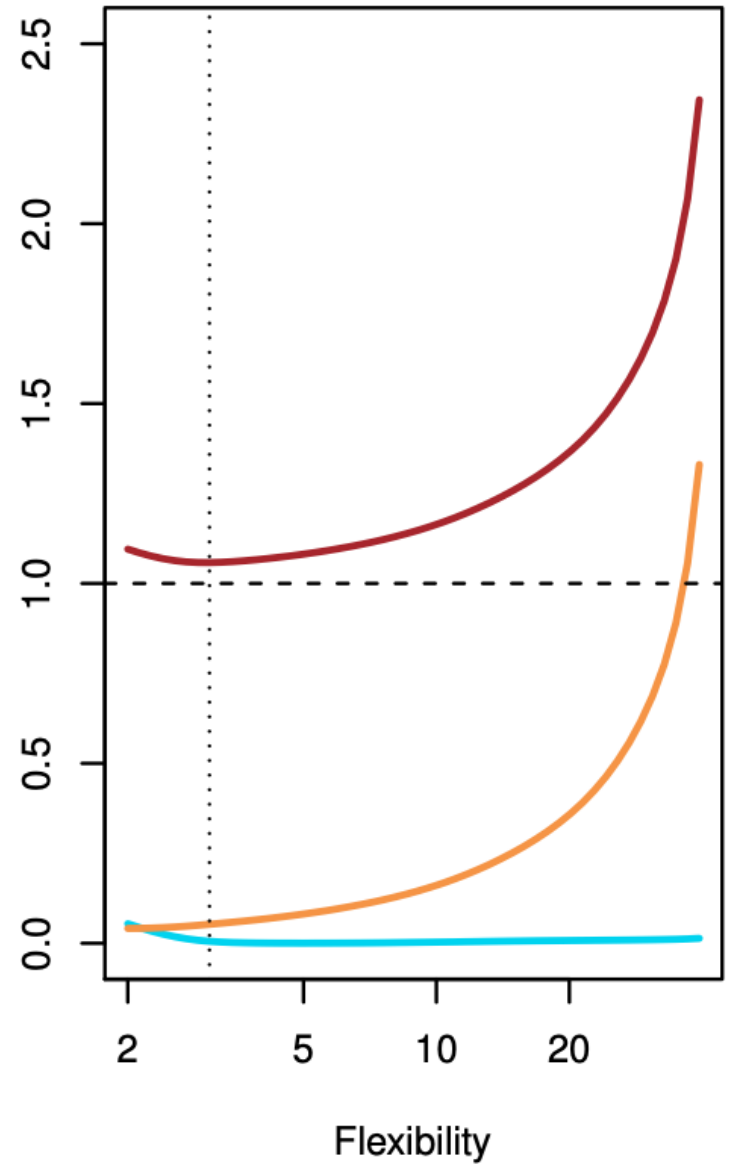

2
4
6
8
10
12
0
20
40
60
80
100

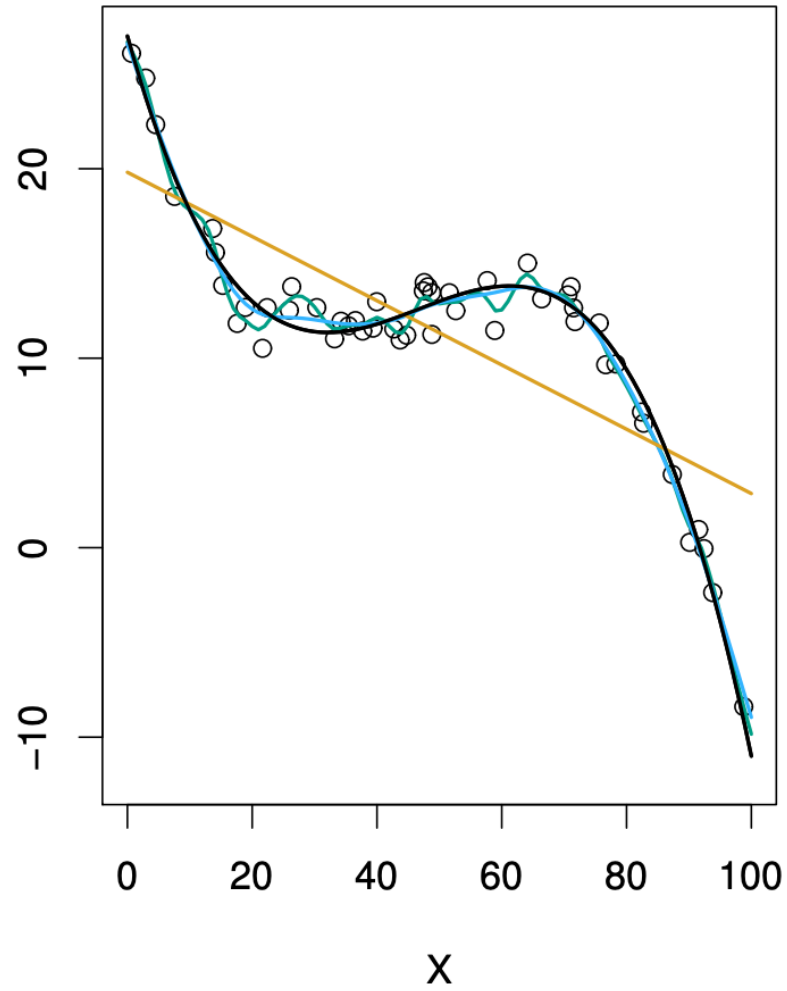

Test MSE
Training MSE
0.0
0.5
1.0
1.5
2.0
2.5
2
5
10
20
Flexibility

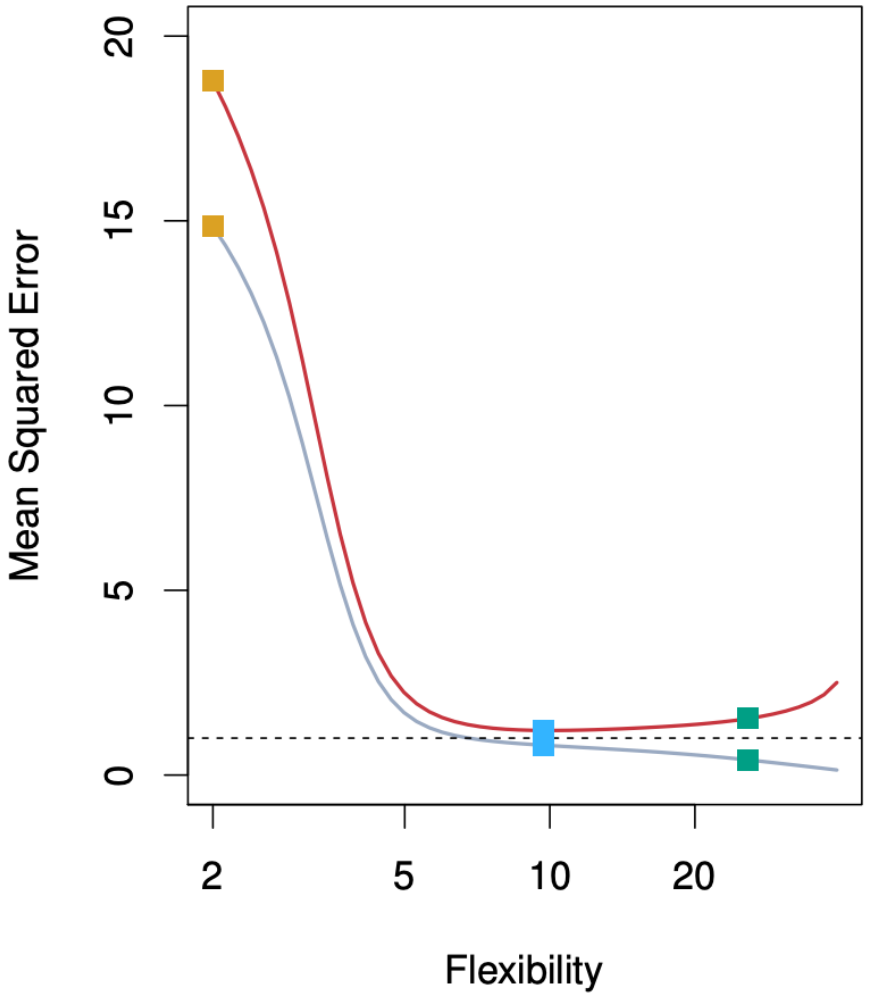

0.0
0.5
1.0
1.5
2.0
2.5
2
5
10
20
Flexibility

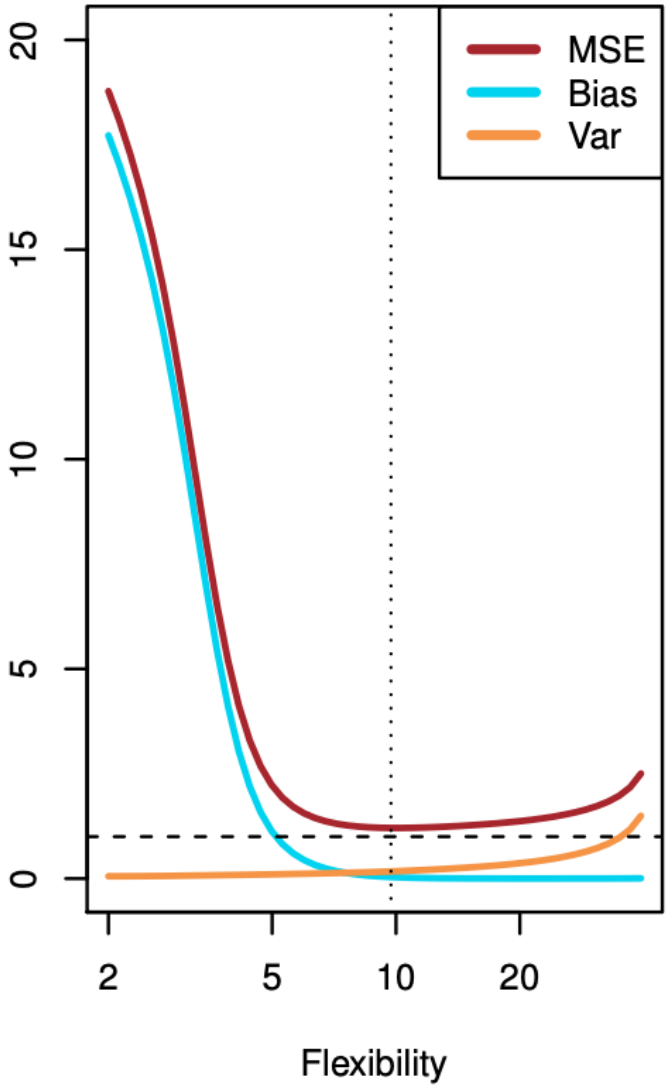

2
4
6
8
10
12
0
20
40
60
80
100


Mean Squared Error
0.0
0.5
1.0
1.5
2.0
2.5
2
5
10
20
Flexibility


0.0
0.5
1.0
1.5
2.0
2.5
2
5
10
20
Flexibility


20
10
0
−10
0
20
40
60
80
100
X


Mean Squared Error
20
15
10
5
0
2
5
10
20
Flexibility


MSE
Bias
Var
20
15
10
5
0
2
5
10
20
Flexibility

Bias-variance tradeoff

$$Y = f(X) + \varepsilon$$

non-random zero-mean noise

$$E\left[\left(\tilde{y}_0 - \hat{f}(\tilde{\mathbf{x}}_0)\right)^2\right] = Var\left(\hat{f}(\tilde{\mathbf{x}}_0)\right) + \left(Bias(\hat{f}(\tilde{\mathbf{x}}_0)\right)^2 + Var(\varepsilon)$$

Expected test MSE = Variance + Bias^2 + Irreducible error

$$= \mathrm{E}\left[\left(\hat{f}(\mathbf{x}_0) - \mathrm{E}\hat{f}(\mathbf{x}_0)\right)^2\right] + \left[\mathrm{E}\left[\hat{f}(\mathbf{x}_0)\right] - f(\mathbf{x}_0)\right]^2 + \mathrm{Var}(\varepsilon)$$

If we take

$$\hat{f}(\mathbf{x}_0) = \mathrm{E}\left[Y | X = \mathbf{x}_0\right]$$

then

$$\mathrm{E}\left[\left(\hat{f}(\mathbf{x}_0) - \mathrm{E}\hat{f}(\mathbf{x}_0)\right)^2\right] = 0 \quad \textbf{and} \quad \left[\mathrm{E}\left[\hat{f}(\mathbf{x}_0)\right] - f(\mathbf{x}_0)\right]^2 = 0$$

Bias-variance tradeoff

$$E\left[\left(y_0 - \hat{f}(\boldsymbol{x_0})\right)^2\right] = Var\left(\hat{f}(\boldsymbol{x_0})\right) + \left(Bias(\hat{f}(\boldsymbol{x_0})\right)^2 + Var(\varepsilon)$$

Expected test MSE = **Variance** + **Bias**^{2} + **Irreducible error**

$$= \underbrace{\mathrm{E}\left[\left(\hat{f}(\mathbf{x}_0) - \mathrm{E}\hat{f}(\mathbf{x}_0)\right)^2\right] + \left[\mathrm{E}\left[\hat{f}(\mathbf{x}_0)\right] - f(\mathbf{x}_0)\right]^2}_{\text{minimized!}} + \mathrm{Var}(\varepsilon)$$

$$= \mathrm{Var}(\varepsilon)$$

Irreducible error

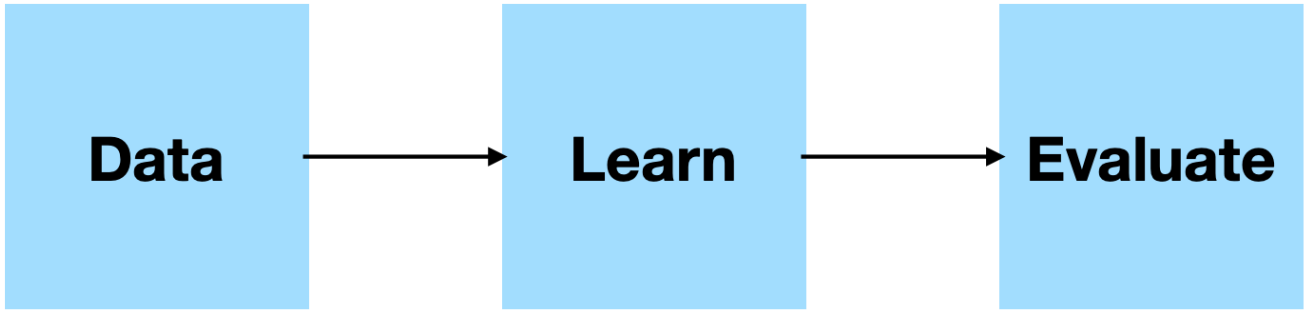
Bias-variance tradeoff

- The “best” we can do:

$$\hat{f}(\mathbf{x}_0) = \mathrm{E}\left[Y \mid X = \mathbf{x}_0\right]$$

- but this is unknown in practice!
- Because the actual population joint distribution of X and Y is unknown

Classification setting



Training set

$\{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \ldots, (\mathbf{x}_n, y_n)\}$

$\hat{f}$

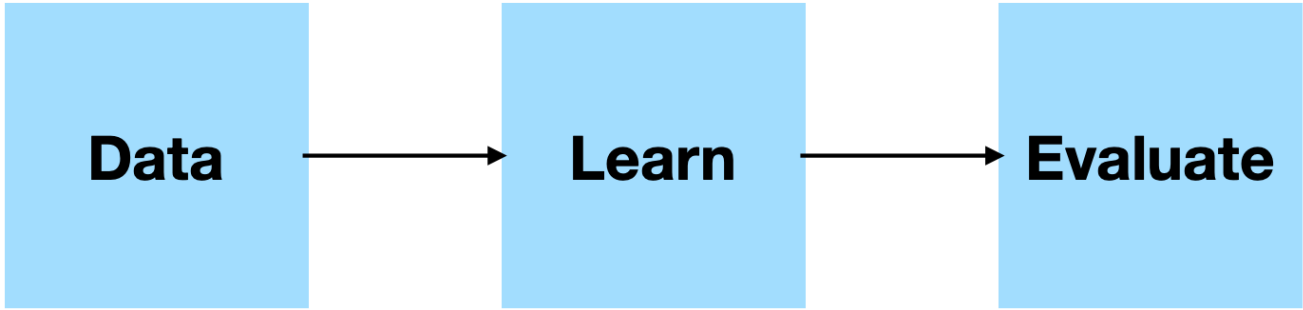
$y_1, \ldots, y_n$: **class label (qualitative), i.e.,**

$y_1 = \text{dog}, y_2 = \text{cat}, y_3 = \text{cat}, \ldots, y_n = \text{dog}$

$\hat{y}_1, \ldots, \hat{y}_n$: **predicted class label using $\hat{f}$, i.e.,**

$\hat{y}_1 = \text{dog}, \hat{y}_2 = \text{dog}, \hat{y}_3 = \text{cat}, \ldots, \hat{y}_n = \text{cat}$

$\mathbf{x}_i = (x_{i1}, x_{i2}, \ldots, x_{ip})$



Training set

$\{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \ldots, (\mathbf{x}_n, y_n)\}$

$\hat{f}$

how well does $\hat{f}$ learn?

Indicator function

$$I\left(y_i \neq \hat{y}_i\right) = \begin{cases} 1 & \textbf{if} \quad y_i \neq \hat{y}_i \\ 0 & \textbf{if} \quad y_i = \hat{y}_i \end{cases}$$

$$\text{Training error rate} = \frac{1}{n}\sum_{i=1}^{n} I\left(y_i \neq \hat{y}_i\right)$$

predicted class label that $\hat{f}$ gives for the ith training observation

Training Error Rate

fraction of **incorrect** classifications in **training** set

$$\text{Training error rate} = \frac{1}{n}\sum_{i=1}^{n} I\left(y_i \neq \hat{y}_i\right)$$

predicted class label that $\hat{f}$ gives for the ith training observation

Indicator function

$$I\left(y_i \neq \hat{y}_i\right) = \begin{cases} 1 & \textbf{if} \quad y_i \neq \hat{y}_i \\ 0 & \textbf{if} \quad y_i = \hat{y}_i \end{cases}$$

$y_1, \ldots, y_n$: **class label (qualitative), i.e.,**

$y_1 = \text{dog}, y_2 = \text{cat}, y_3 = \text{cat}, \ldots, y_n = \text{dog}$

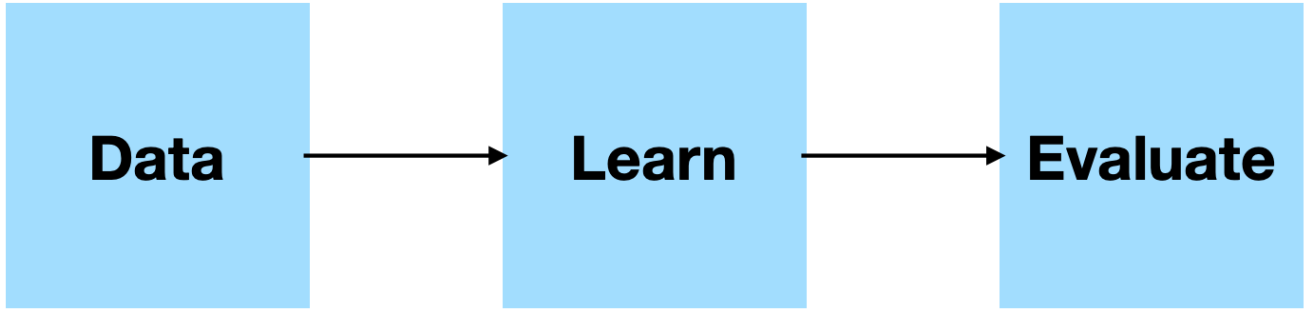
$\hat{y}_1, \ldots, \hat{y}_n$: **predicted class label using $\hat{f}$, i.e.,**

$\hat{y}_1 = \text{dog}, \hat{y}_2 = \text{dog}, \hat{y}_3 = \text{cat}, \ldots, \hat{y}_n = \text{cat}$

Training error rate: $\frac{\sum_{i=1}^{n} I(y_i \neq \hat{y}_i)}{n} = \frac{2}{5} = 0.40$

Training accuracy: $1 - \frac{\sum_{i=1}^{n} I(y_i \neq \hat{y}_i)}{n} = \frac{3}{5} = 0.60$

y_i	$\hat{y}_i$	$I(y_i \neq \hat{y}_i)$
dog	cat	1
cat	cat	0
bird	dog	1
dog	dog	0
dog	dog	0



Training set

$\{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \ldots, (\mathbf{x}_n, y_n)\}$ $\hat{f}$

Again, we are less interested in how $\hat{f}$ performs on training set

Consider test set $\{(\tilde{\mathbf{x}}_1, \tilde{y}_1), (\tilde{\mathbf{x}}_2, \tilde{y}_2), \ldots, (\tilde{\mathbf{x}}_m, \tilde{y}_m)\}$, not seen or used to train $\hat{f}$

$$\text{Test error rate} = \frac{1}{n}\sum_{i=1}^{n} I\left(\tilde{y}_i \neq \hat{\tilde{y}}_i\right)$$

predicted class label that $\hat{f}$ gives for the ith test observation

Test Error Rate

fraction of **incorrect** classifications in **test** set

Training set

$$\{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \ldots, (\mathbf{x}_n, y_n)\}$$

Test set

$$\{(\tilde{\mathbf{x}}_1, \tilde{y}_1), (\tilde{\mathbf{x}}_2, \tilde{y}_2), \ldots, (\tilde{\mathbf{x}}_m, y_m)\}$$

Training error rate

$$\frac{1}{n}\sum_{i=1}^{n} I\left(y_i \neq \hat{y}_i\right)$$

predicted class label that $\hat{f}$ gives for the ith training observation

Test error rate

$$\frac{1}{n}\sum_{i=1}^{n} I\left(\tilde{y}_i \neq \hat{\tilde{y}}_i\right)$$

predicted class label that $\hat{f}$ gives for the ith test observation

$\hat{f}$ is obtained on the training set!

Same discussion as in regression setting

- Having the **lowest training error rate** does not necessarily imply that a method also has the **lowest test error rate**

Higher model flexibility → lower training error rate

Higher model flexibility ↛ lower test error rate

Overfitting = Large test error rate + small training error rate

Estimating test error rate – same as regression:

- Computing training error is easy – test error, not so much
  - Either testing data is not available or it is, but we don't want to train on it
  - It would just become another training set!

- What do we do?

Cross-validation:

a method for estimating test MSE using only training data!

Next week!

Bias-variance tradeoff

Still want to **simultaneously** achieve **low variance** and **low bias**

A simple model → high bias + low variance

A flexible model → low bias + high variance

Challenge: find a method for which both the variance and bias are low

In classification setting, the "best" we can do

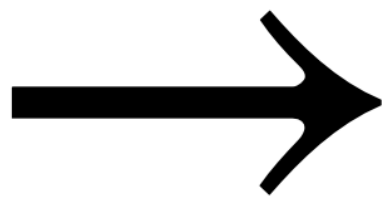
$$\hat{y}_0 = \text{argmax}_j \left\{ \text{Prob}\left(Y = j \,|\, X = \mathbf{x}_0\right) \right\}$$

conditional probability of $Y = j$,
given the observed predictor $\mathbf{x}_0$

Bayes classifier: assigns each observation to the most likely class, given its predictor values

For example, we want to classify species of an animal (dog, cat, bird) given its picture

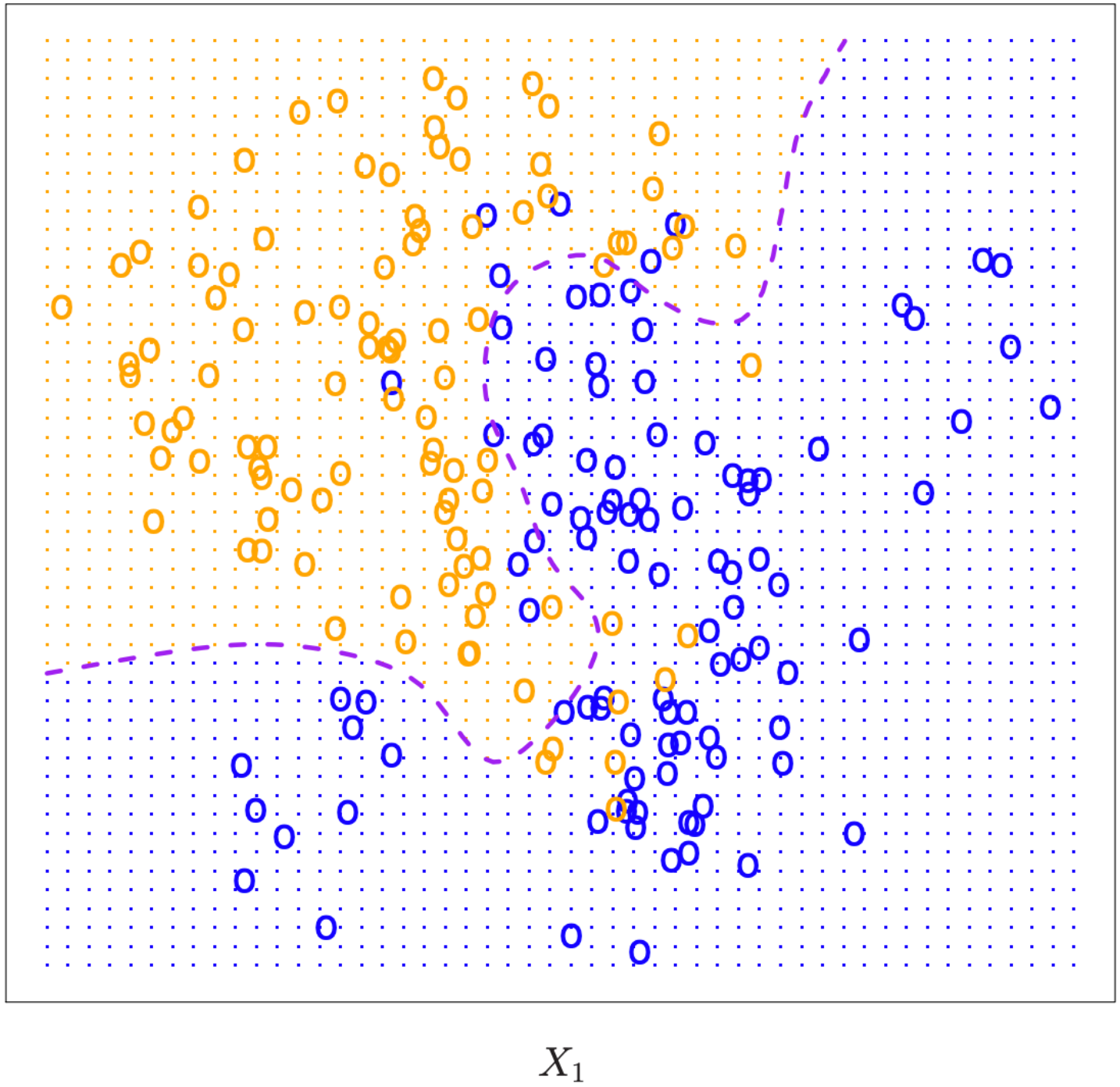
$$\text{Prob}\left(Y = \textbf{dog} \,|\, X = \mathbf{x}_0\right) = 0.5$$
$$\text{Prob}\left(Y = \textbf{cat} \,|\, X = \mathbf{x}_0\right) = 0.3$$
$$\text{Prob}\left(Y = \textbf{bird} \,|\, X = \mathbf{x}_0\right) = 0.2$$



$$\hat{y}_0 = \textbf{dog}$$

In a two-class problem:

$$\hat{y}_0 = \begin{cases} 1 & \textbf{if Prob}\left(Y = 1 \,|\, X = \mathbf{x}_0\right) > 0.5 \\ 2 & \textbf{if Prob}\left(Y = 1 \,|\, X = \mathbf{x}_0\right) \leq 0.5 \end{cases}$$



Each circle is an observation

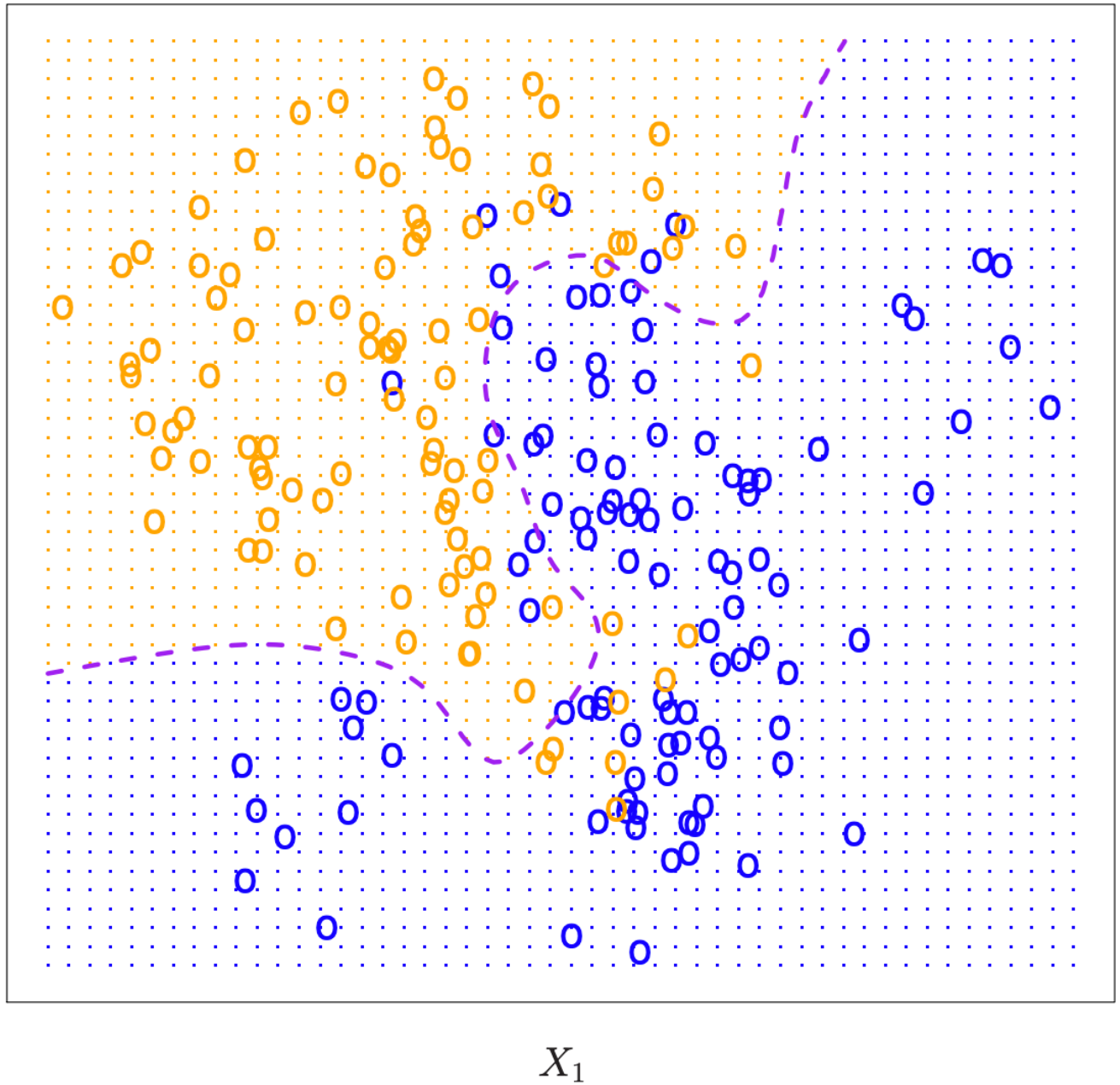
location: predictor X

color: class label Y, $Y = 1$ **vs.** $Y = 2$

Areas:

$$\text{Prob}\left(Y = 1 \,|\, X = \mathbf{x}_0\right) > 50\,\%$$

$$\text{Prob}\left(Y = 2 \,|\, X = \mathbf{x}_0\right) > 50\,\%$$

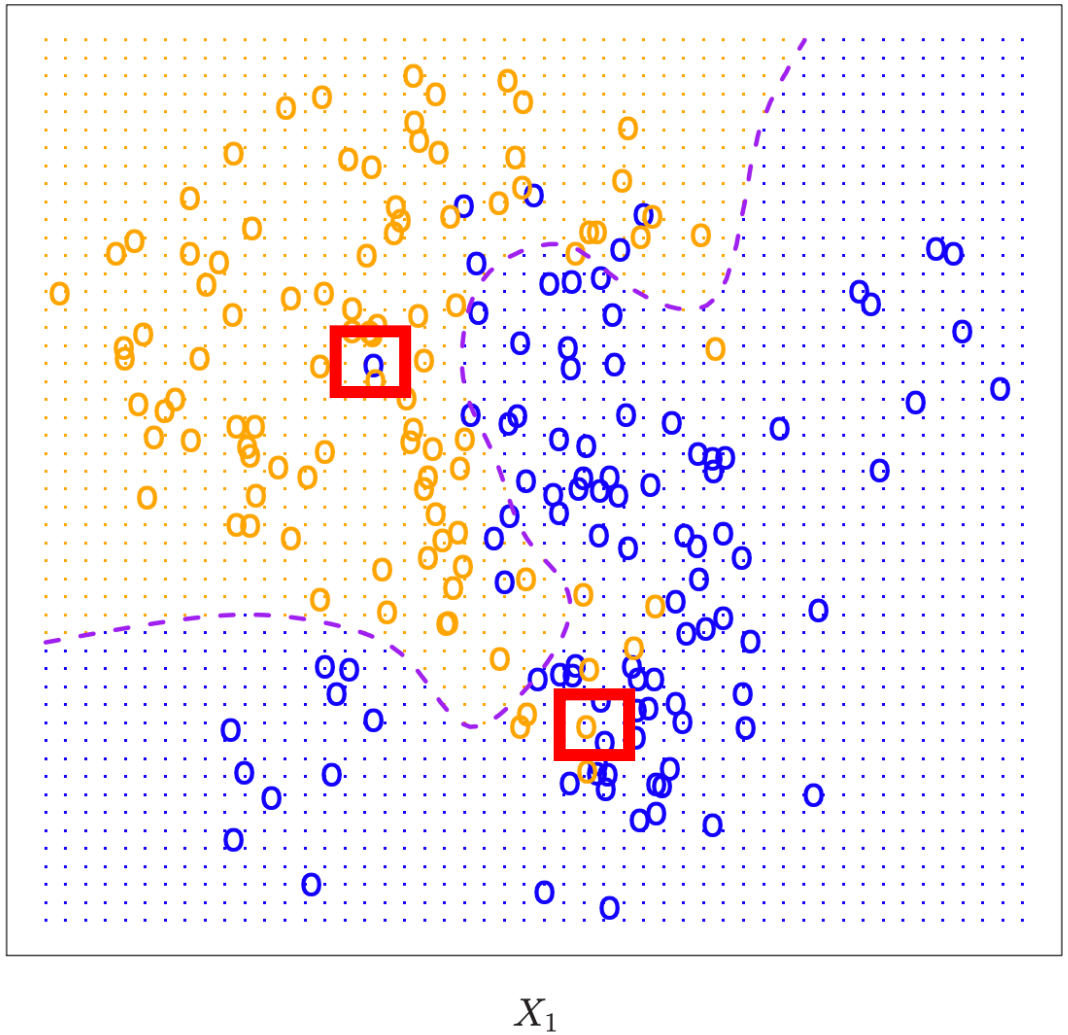


$$\hat{y}_0 = \text{argmax}_j \left\{ \text{Prob}\left(Y = j \,|\, X = \mathbf{x}_0\right) \right\}$$

conditional probability of $Y = j$,
given the observed predictor $\mathbf{x}_0$

Bayes error rate

$$1 - \text{E}\left[\max_j \text{Prob}\left(Y = j \,|\, X\right)\right]$$



Regression

X

quantitative Y

Training/Test

Mean Squared Error (MSE)

$$\hat{f}(\mathbf{x}_0) = \mathrm{E}\left[Y \mid X = \mathbf{x}_0\right]$$

$$\mathrm{Var}(\varepsilon)$$

Classification

X

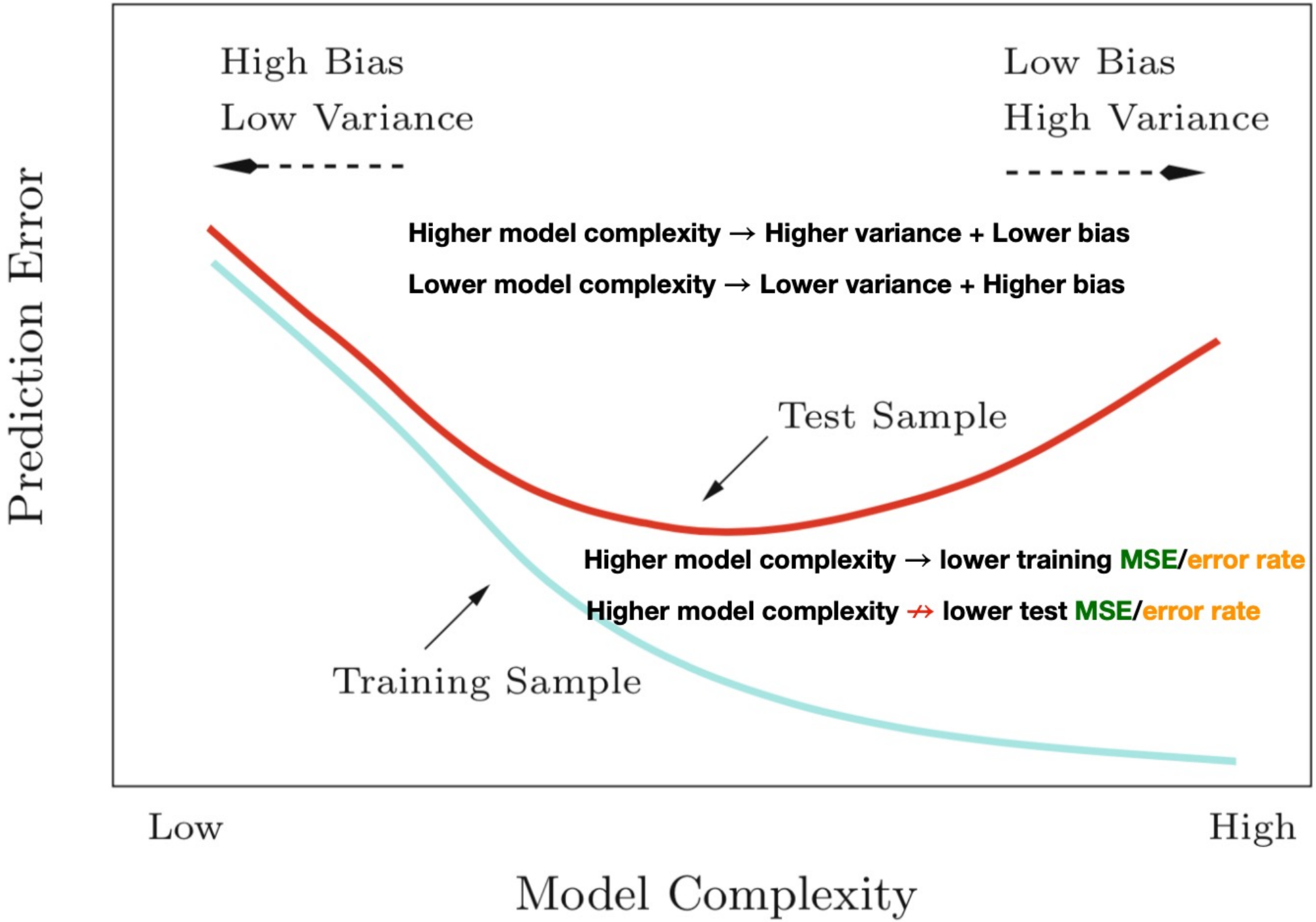
qualitative Y

Training/Test

Error Rate

$$\hat{y}_0 = \mathrm{argmax}_j \left\{ \mathrm{Prob}\left(Y = j \mid X = \mathbf{x}_0\right) \right\}$$

$$1 - \mathrm{E}\left[\max_j \mathrm{Prob}\left(Y = j \mid X\right)\right]$$


High Bias
Low Variance
Low Bias
High Variance
Higher model complexity → Higher variance + Lower bias
Lower model complexity → Lower variance + Higher bias
Test Sample
Higher model complexity → lower training MSE/error rate
Higher model complexity ↛ lower test MSE/error rate
Training Sample
Prediction Error
Low
High
Model Complexity

k-nearest neighbors (KNN)

A starter machine learning model

K-nearest neighbors

Regression

$$\hat{f}(\mathbf{x}_0) = \mathrm{E}\left[Y \,|\, X = \mathbf{x}_0\right]$$

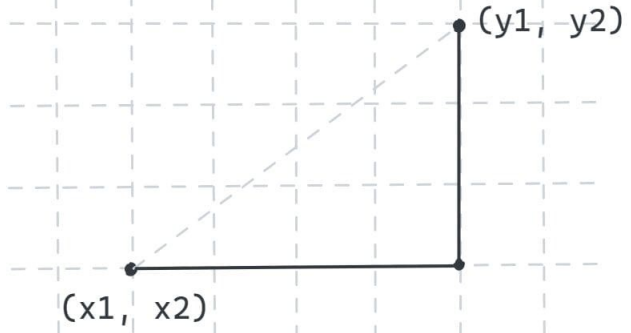
Classification

$$\hat{y}_0 = \mathrm{argmax}_j \left\{ \mathrm{Prob}\left(Y = j \,|\, X = \mathbf{x}_0\right) \right\}$$

In practice, we never know the conditional distribution of Y given X

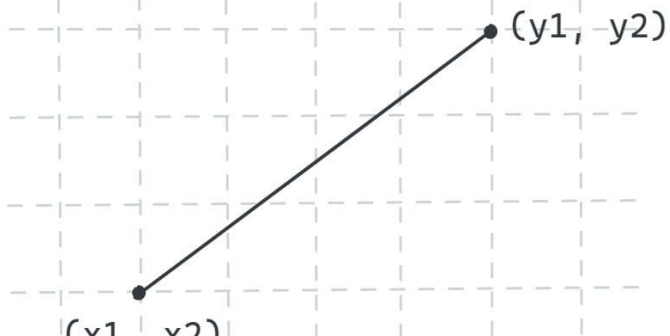
Estimate!!

Idea: look at the K-nearest neighbors of $\mathbf{x}_0$



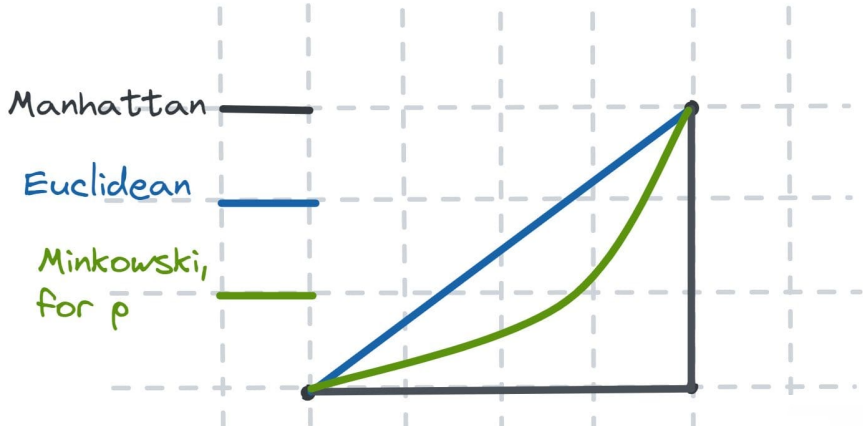
Manhattan Distance in 2D Plane

d(x,y) = |x1 – y1| + |x2 – y2|



Euclidean Distance in 2D Plane

$$d(\mathbf{x}, \mathbf{y}) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$$



Minkowski when p = 1 --> Manhattan
Minkowski when p = 2 --> Euclidean

$$d(\mathbf{x}, \mathbf{y}) = \left(\sum_{i=1}^{n} |x_i - y_i|^p \right)^{1/p} \quad for\ p \geq 1$$

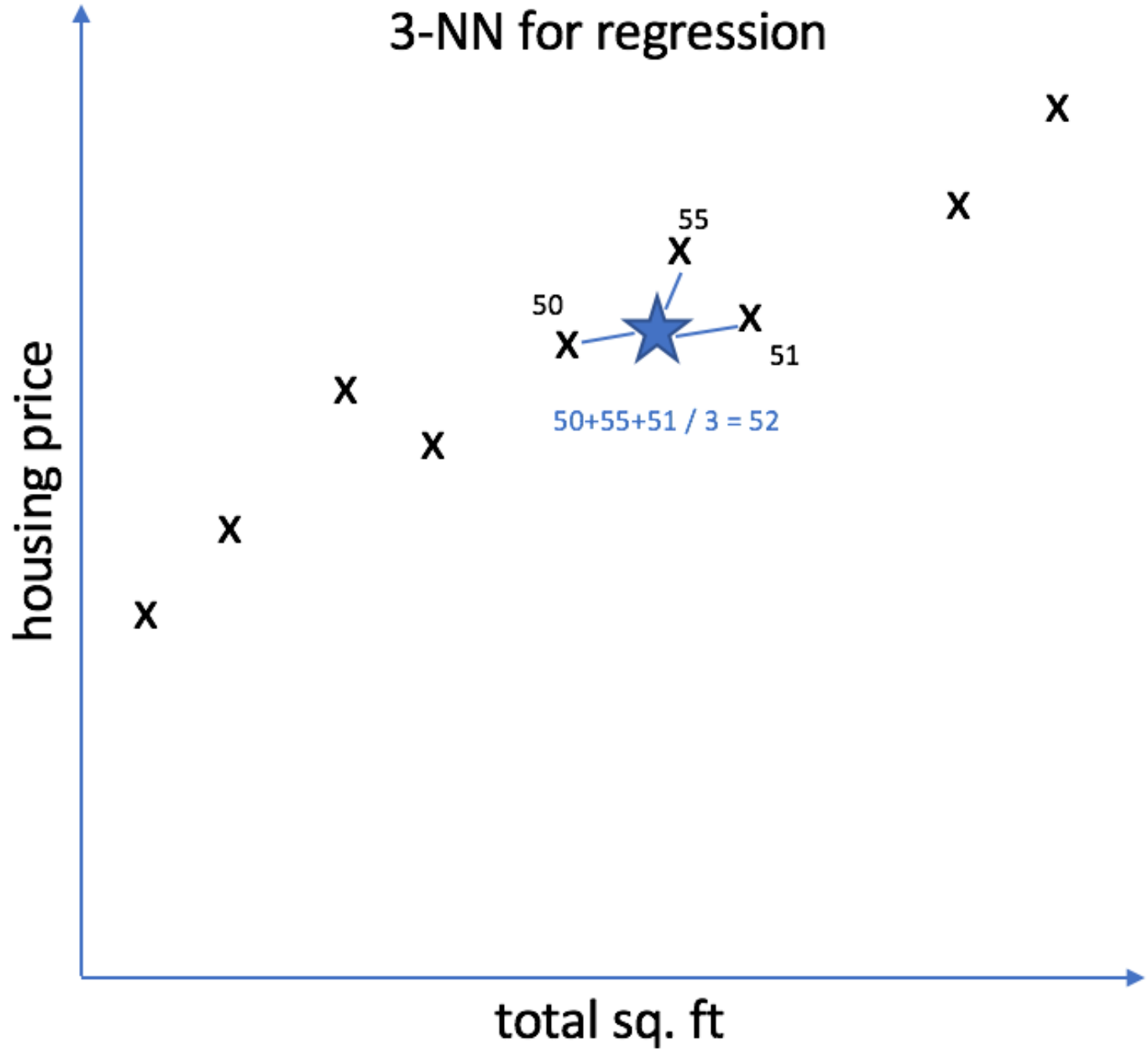
K-nearest neighbors

$$\hat{f}(\mathbf{x}_0) = \mathrm{E}\left[Y \mid X = \mathbf{x}_0\right]$$

Estimation idea: look at the K-nearest neighbors of $\mathbf{x}_0$

$$\mathcal{N}_0 = \left\{i : \mathbf{x}_i \textbf{ is among the K nearest neighbors of } \mathbf{x}_0\right\}$$

$$\hat{f}_K(\mathbf{x}_0) = \frac{1}{K} \sum_{i \in \mathcal{N}_0} y_i$$


3-NN for regression
housing price
total sq. ft
55
50
51
50+55+51 / 3 = 52

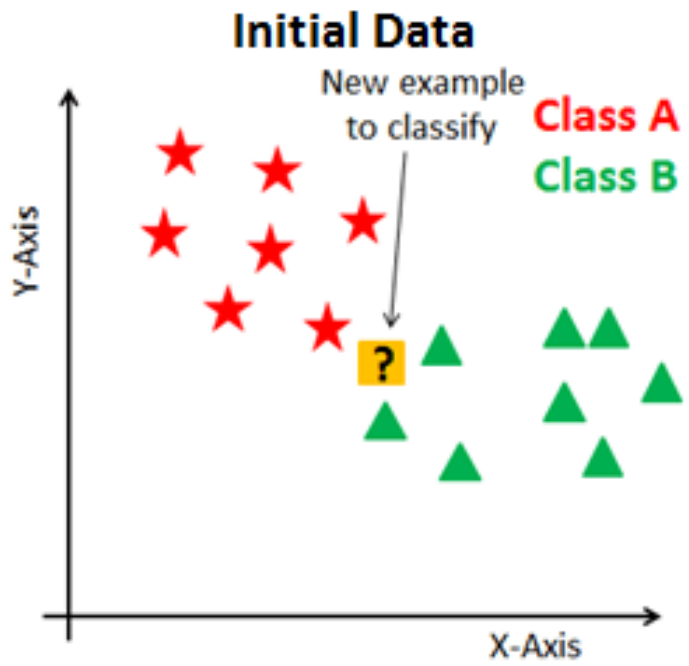
K-nearest neighbors

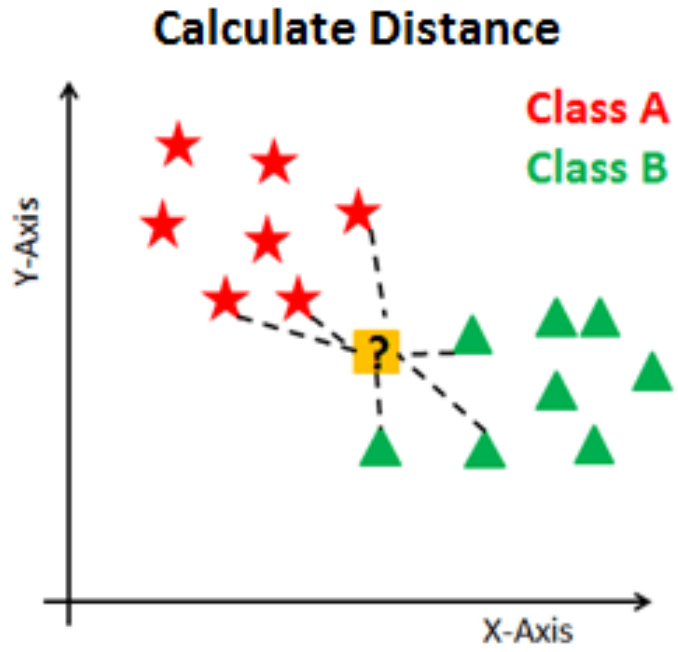
$$\hat{y}_0 = \text{argmax}_j \left\{ \text{Prob}\left(Y = j \,|\, X = \mathbf{x}_0\right) \right\}$$

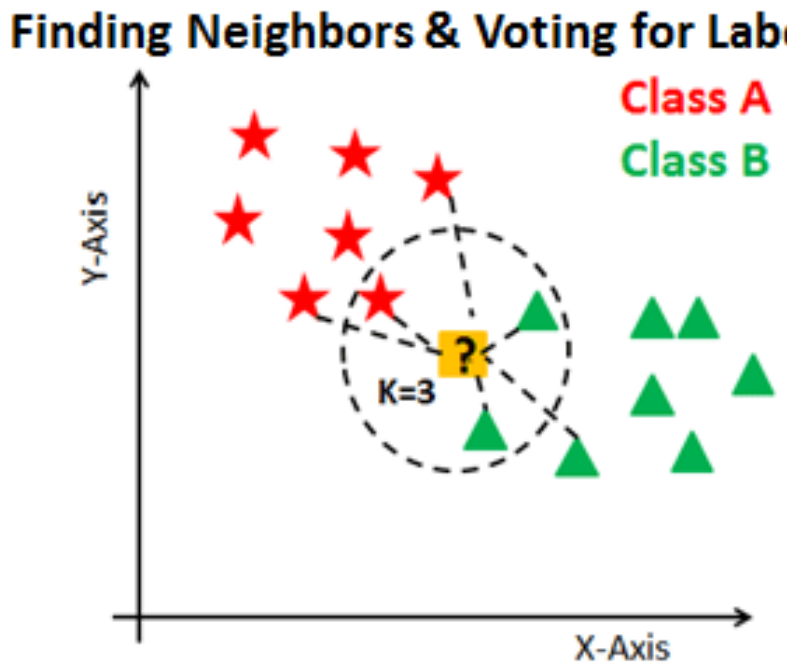
Estimation idea: look at the K-nearest neighbors of $\mathbf{x}_0$

$$\mathcal{N}_0 = \left\{ i : \mathbf{x}_i \textbf{ is among the K nearest neighbors of } \mathbf{x}_0 \right\}$$

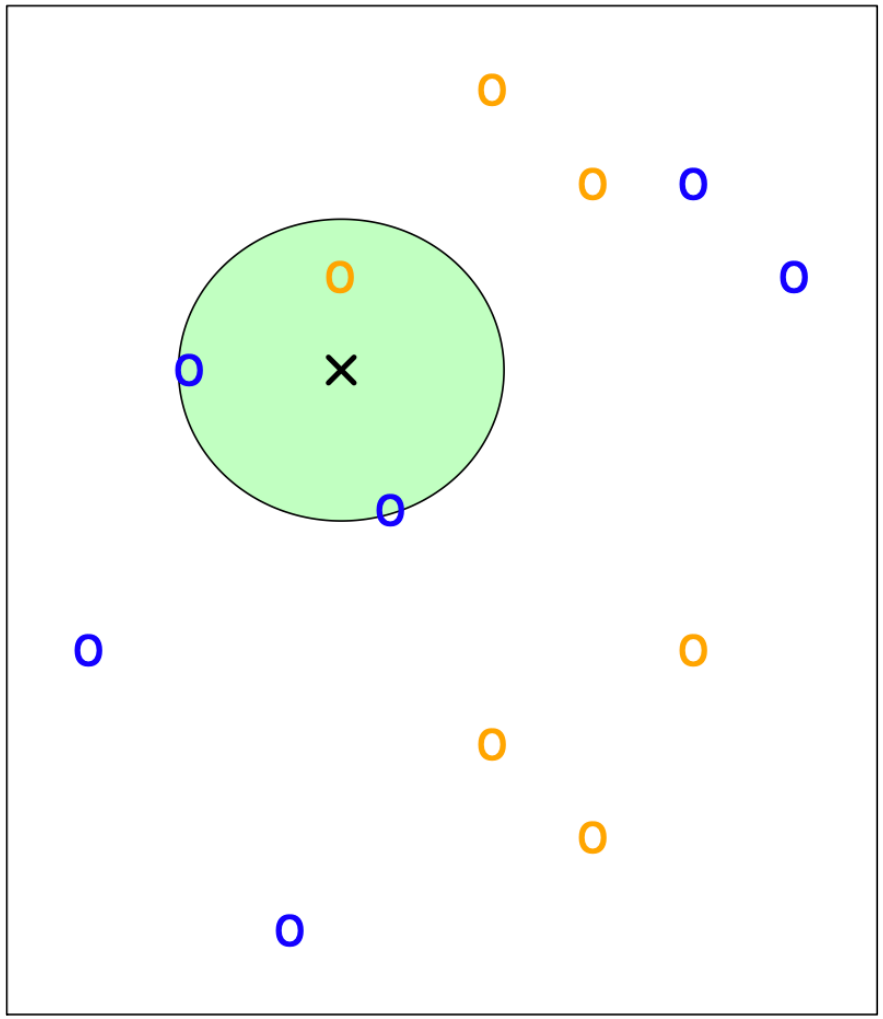
$$\hat{y}_K = \text{argmax}_j \left\{ \frac{1}{K} \sum_{i \in \mathcal{N}_0} I(y_i = j) \right\}$$

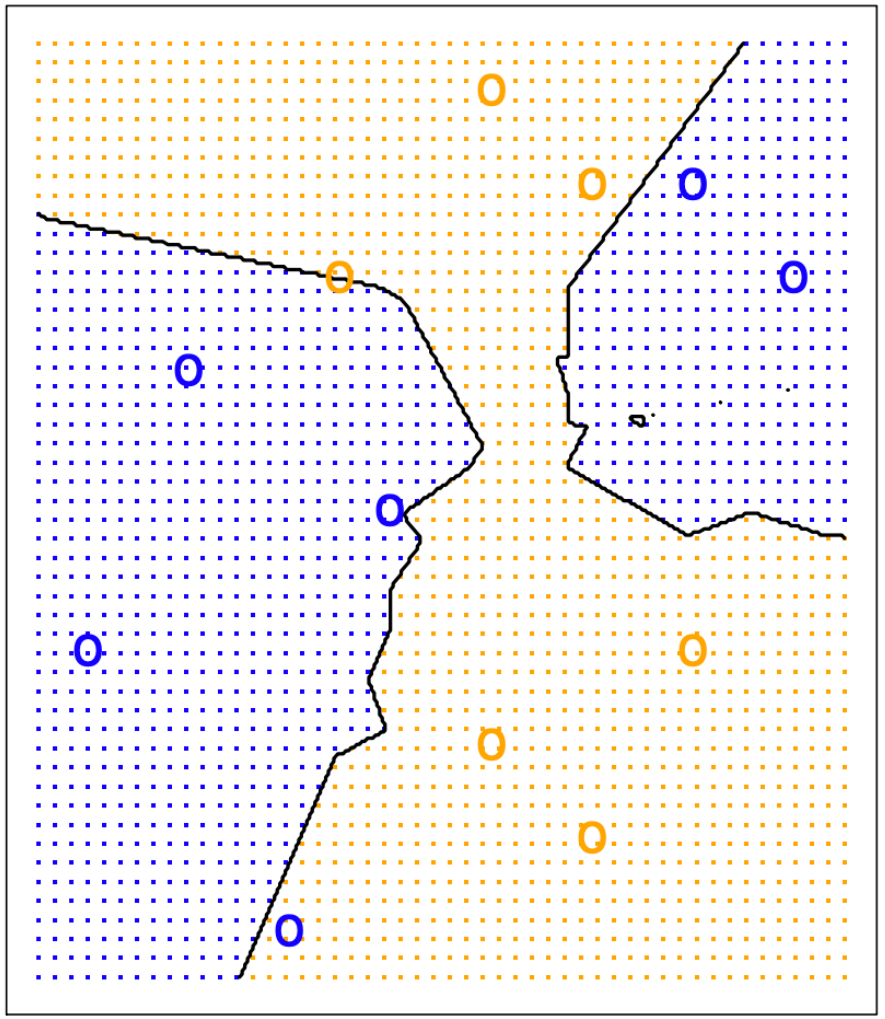

Initial Data
New example
to classify
Class A
Class B
?
Y-Axis
X-Axis


Calculate Distance
Class A
Class B
?
Y-Axis
X-Axis


Finding Neighbors & Voting for Labels
Class A
Class B
?
K=3
Y-Axis
X-Axis

K-nearest neighbors





Ties in KNN

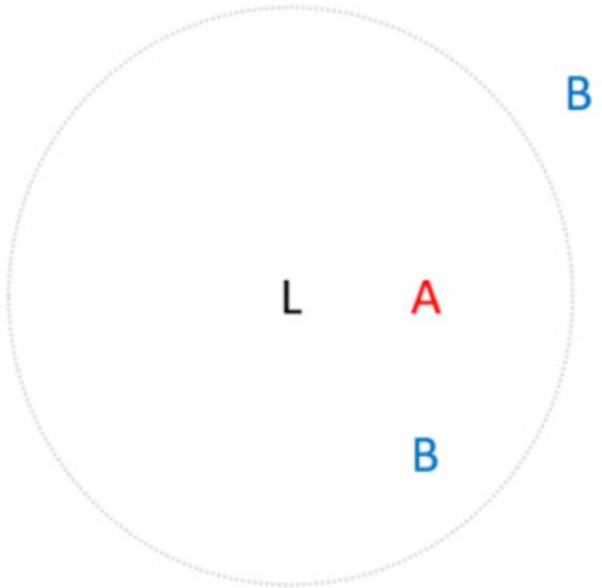


Figure i. Of the three neighbors shown, the two nearest are obvious.

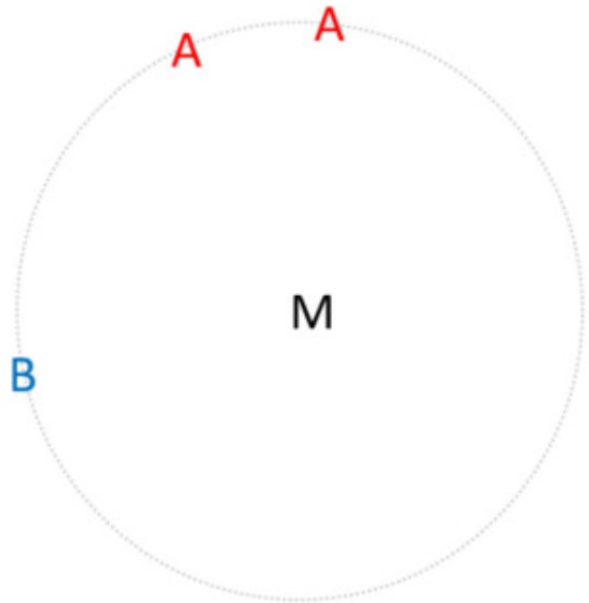


Figure ii. Three neighbors are equidistant. Which two should be chosen?

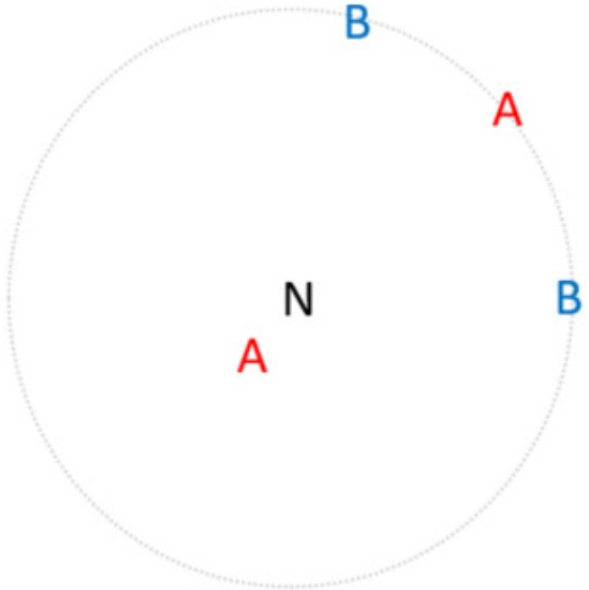


Figure iii. Of the three neighbors shown, one is obvious, but which of the other two should be chosen?

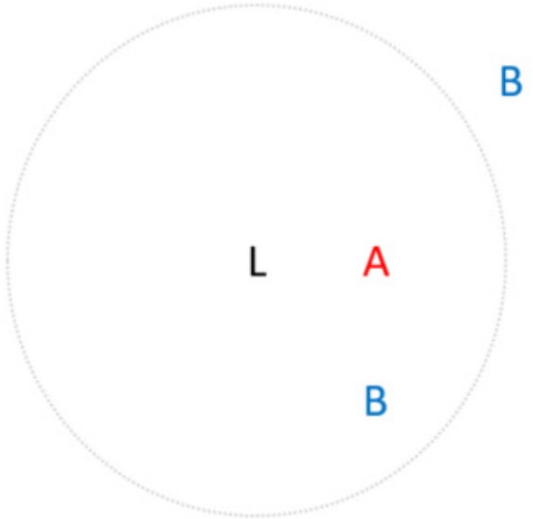


Figure i. Of the three neighbors shown, the two nearest are obvious.

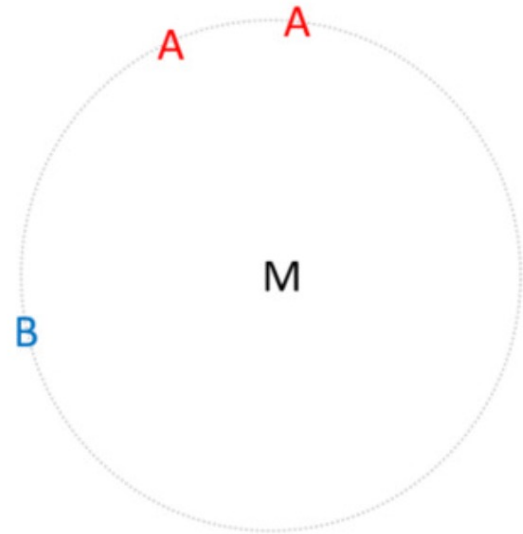


Figure ii. Three neighbors are equidistant. Which two should be chosen?

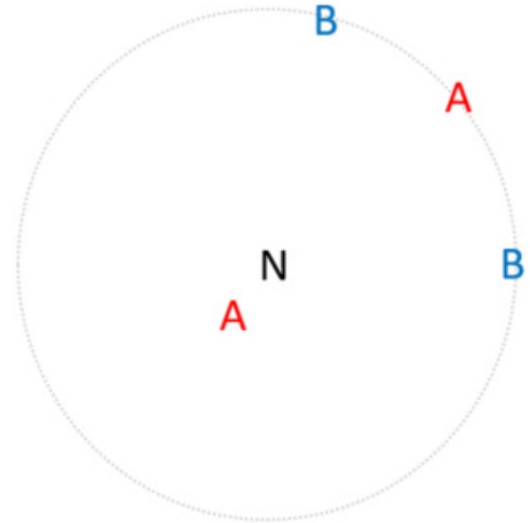
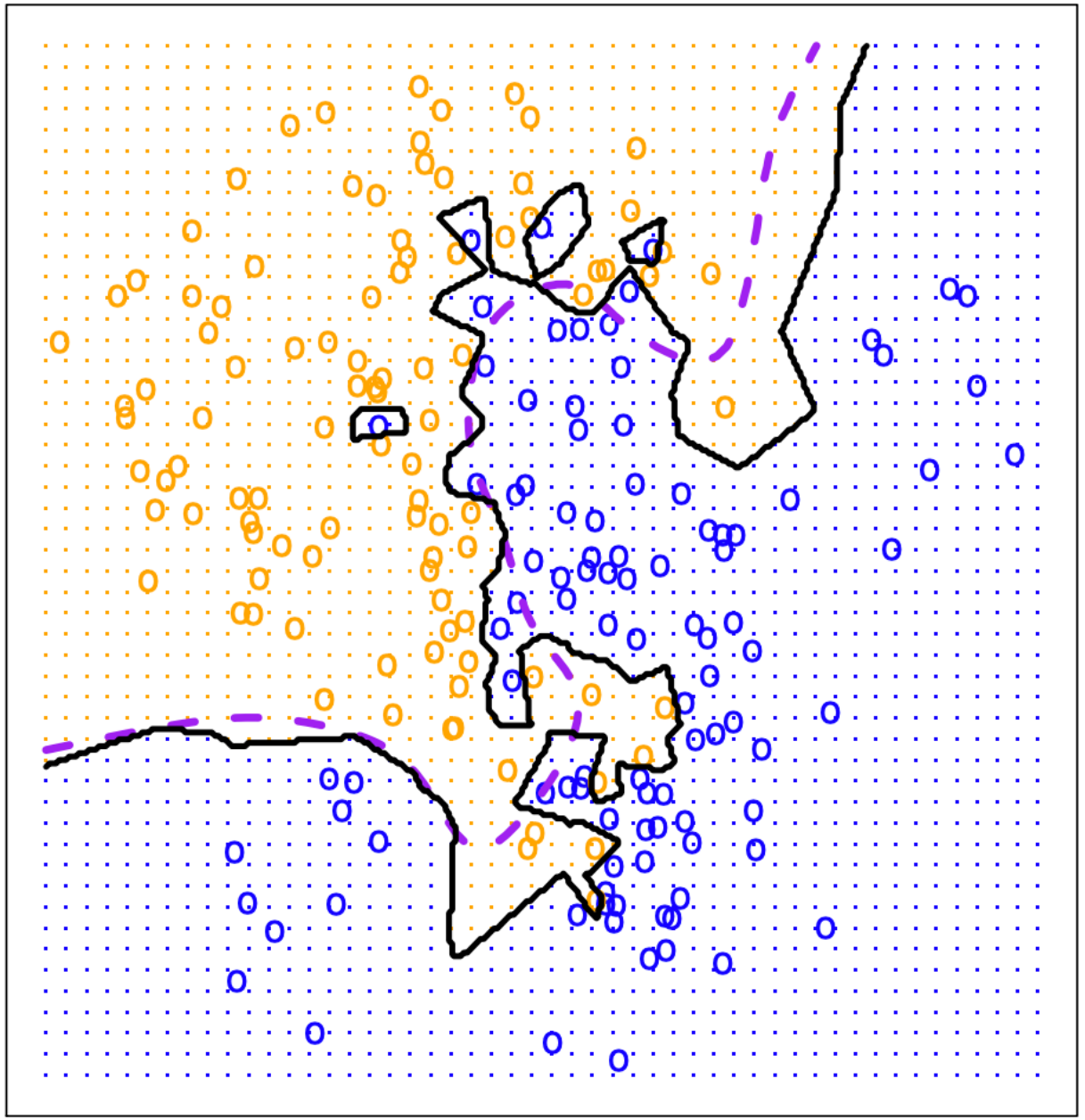


Figure iii. Of the three neighbors shown, one is obvious, but which of the other two should be chosen?

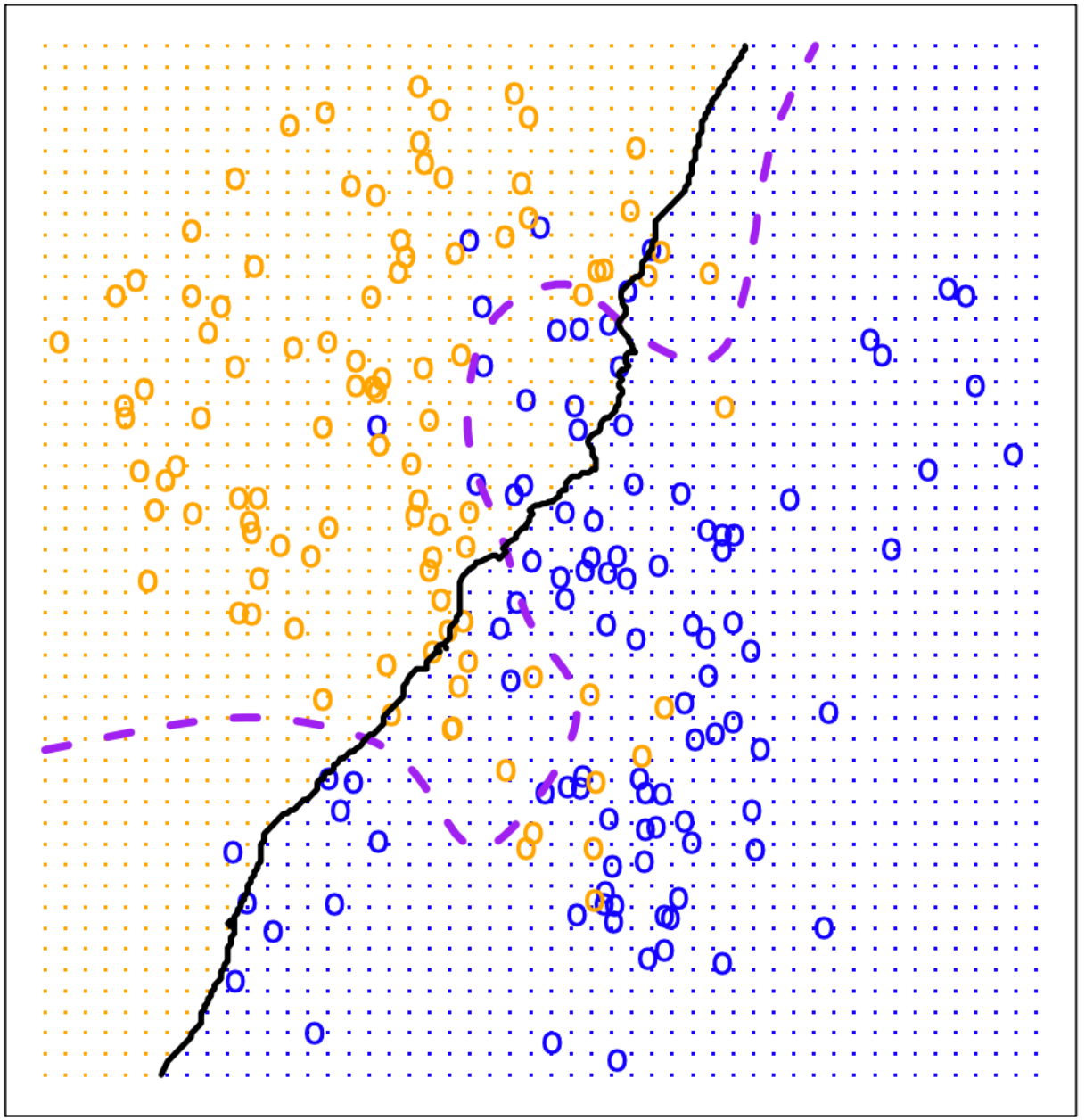
Unclassified	Neighbors
L	A B
M	A A B
N	A A B B

Table i. The selected nearest neighbors for each point.

KNN: K=1



KNN: K=100



K-nearest neighbors bias-variance tradeoff

$$\hat{f}_K(\mathbf{x}_0) = \frac{1}{K} \sum_{i \in \mathcal{N}_0} y_i$$

$$Y = f(X) + \varepsilon$$

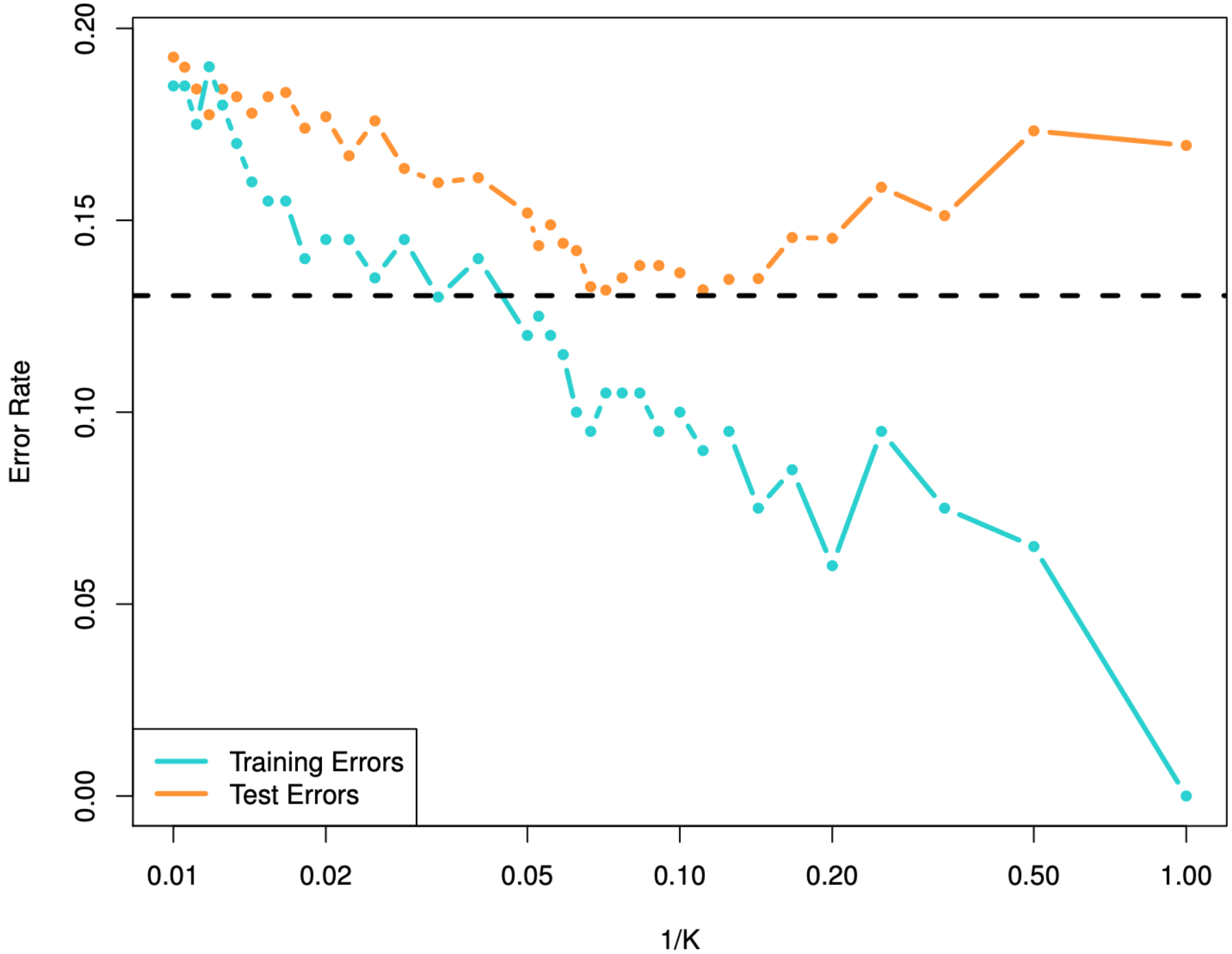
non-random zero-mean noise

For fixed $(\mathbf{x}_0, y_0)$

$$\mathrm{E}\left[\left(y_0 - \hat{f}(\mathbf{x}_0)\right)^2\right] = \mathrm{Var}(\hat{f}(\mathbf{x}_0)) + \left[\mathrm{Bias}(\hat{f}(\mathbf{x}_0))\right]^2 + \mathrm{Var}(\varepsilon),$$

$$= \frac{\mathrm{Var}(\varepsilon)}{k} + \left[f(\mathbf{x}_0) - \frac{1}{k} \sum_{i \in \mathcal{N}_0} f(\mathbf{x}_i)\right]^2 + \mathrm{Var}(\varepsilon),$$

k increase	**decrease**	**increase**	**beyond our control**
k decrease	**increase**	**decrease**	


Error Rate
0.00
0.05
0.10
0.15
0.20
0.01
0.02
0.05
0.10
0.20
0.50
1.00
1/K
Training Errors
Test Errors

Limitations of K-nearest neighbors

K-nearest neighbors methods can be pretty good for small p (i.e., $p \leq 4$) and large n

Many more sophisticated versions of non-parametric methods, e.g., kernel estimators…

Fail when p is very large: there are very few (almost no) data points in the neighbors

Curse of Dimensionality!